

Nov. 12, 2025
SRF Frontiers Workshop

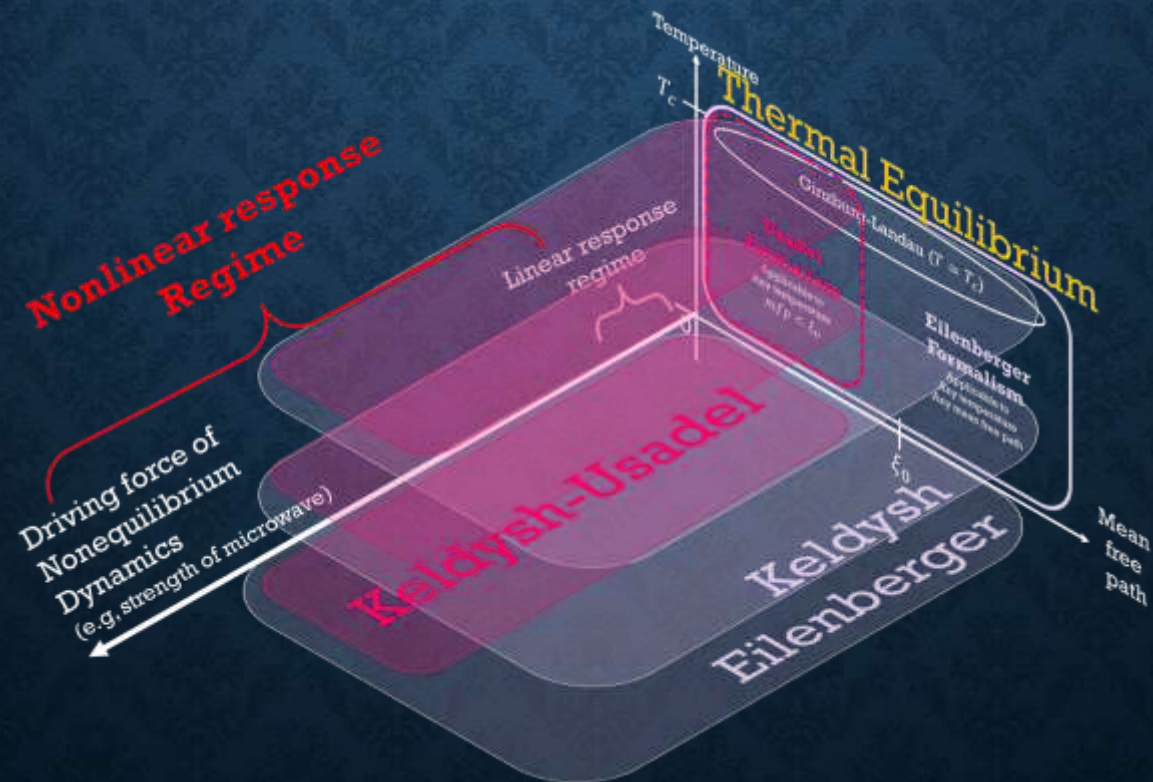
THEORETICAL PERSPECTIVE OF SRF PERFORMANCE



High energy Accelerator Research Organization (KEK)
Innovation Center for Applied Superconducting Accelerators (iCASA)

Takayuki Kubo

Overview of the theoretical frameworks



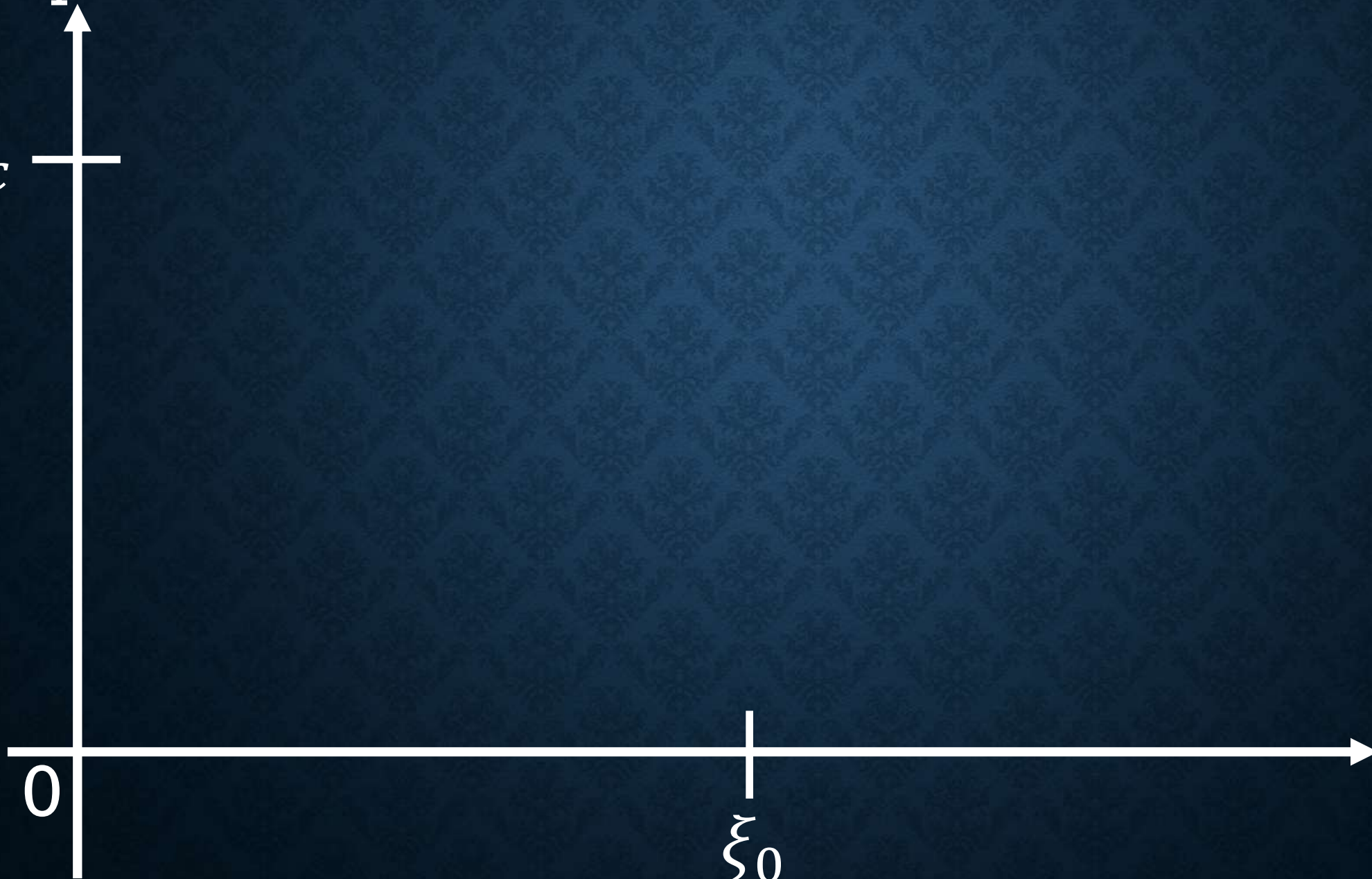
Temperature

T_c

0

ξ_0

Mean
free
path



Temperature

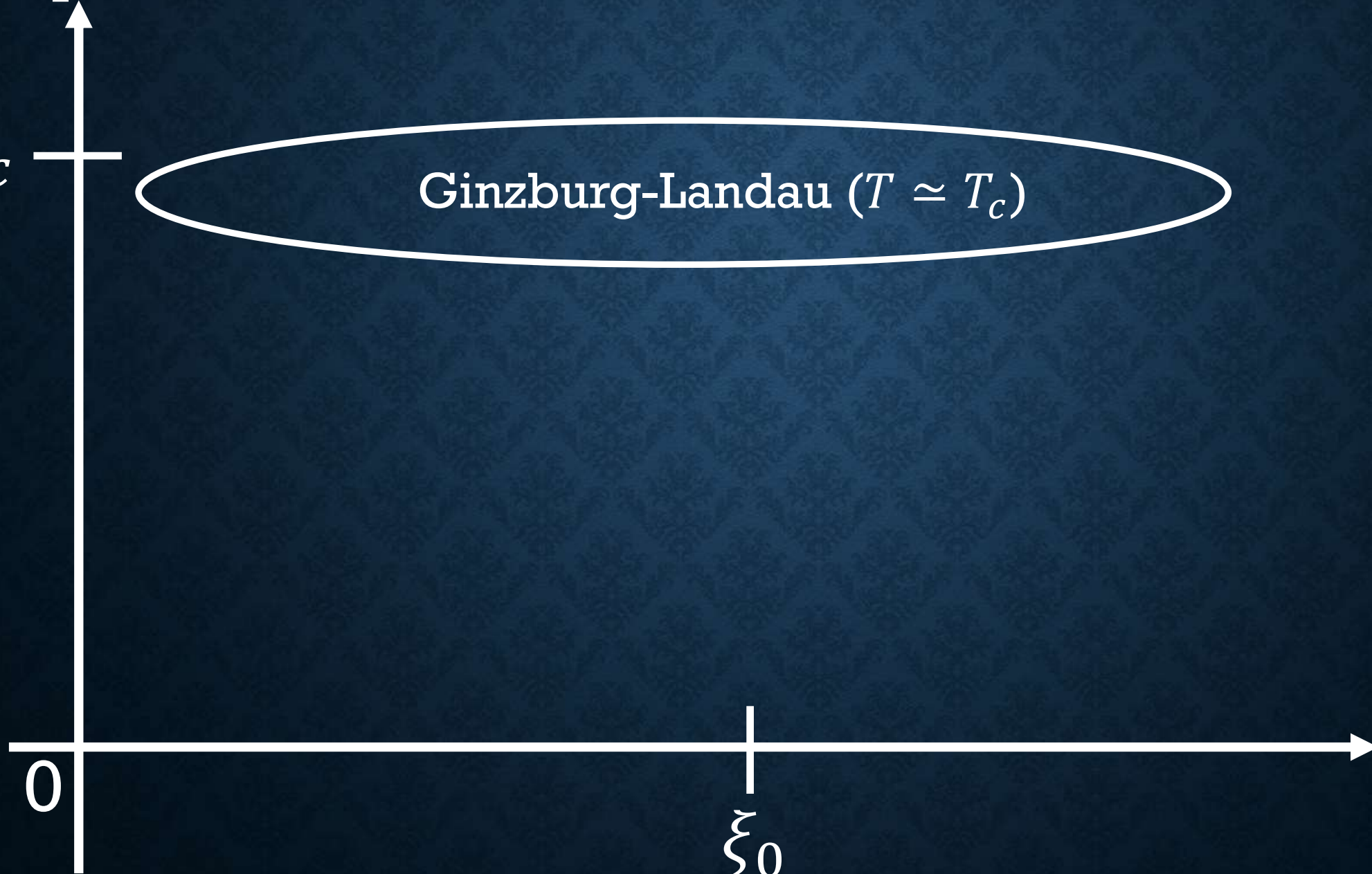
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Ginzburg-Landau ($T \simeq T_c$)

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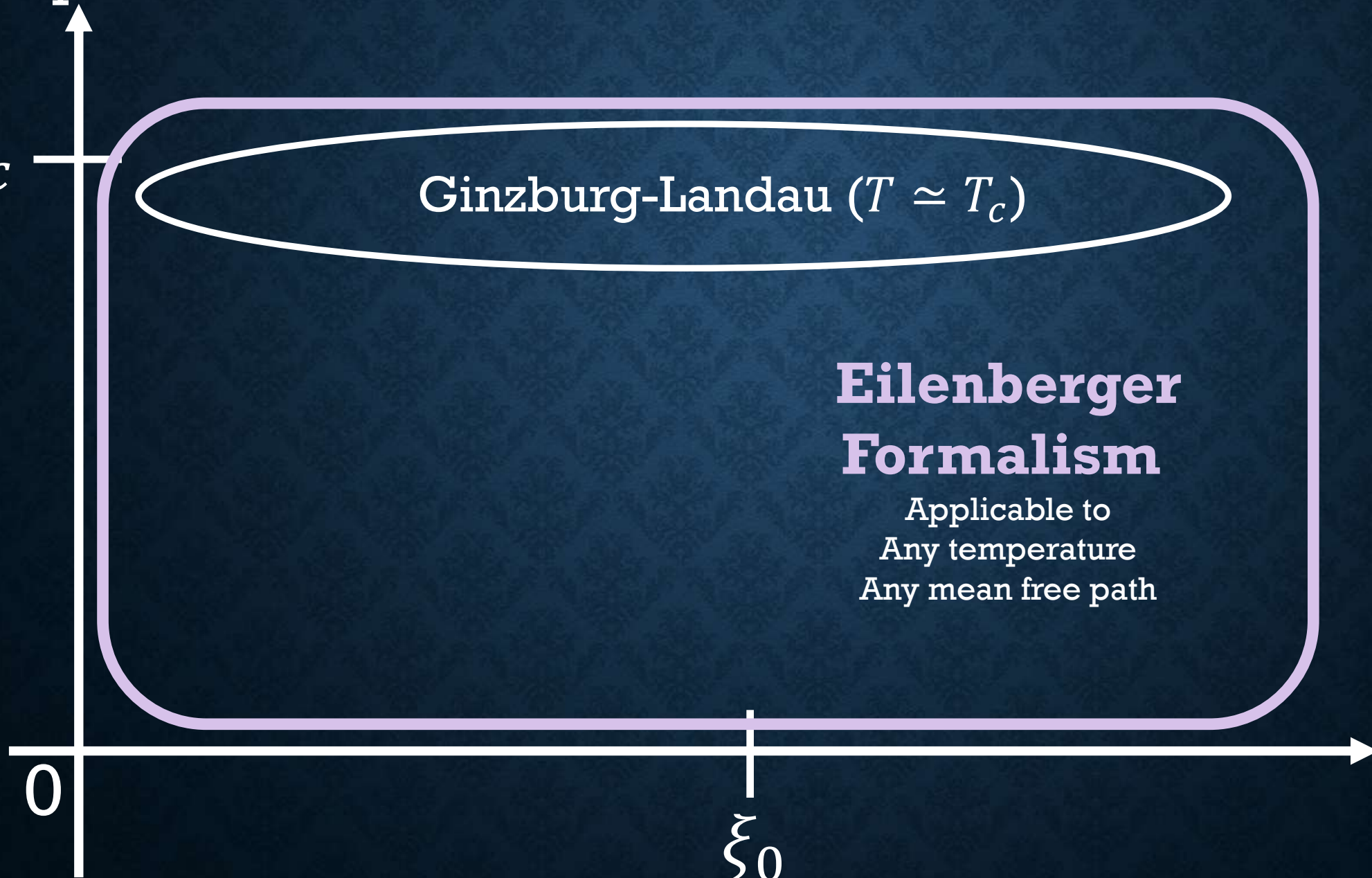
**Eilenberger
Formalism**

Applicable to
Any temperature
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Applicable to
Any temperature

$$mfp < \xi_0$$

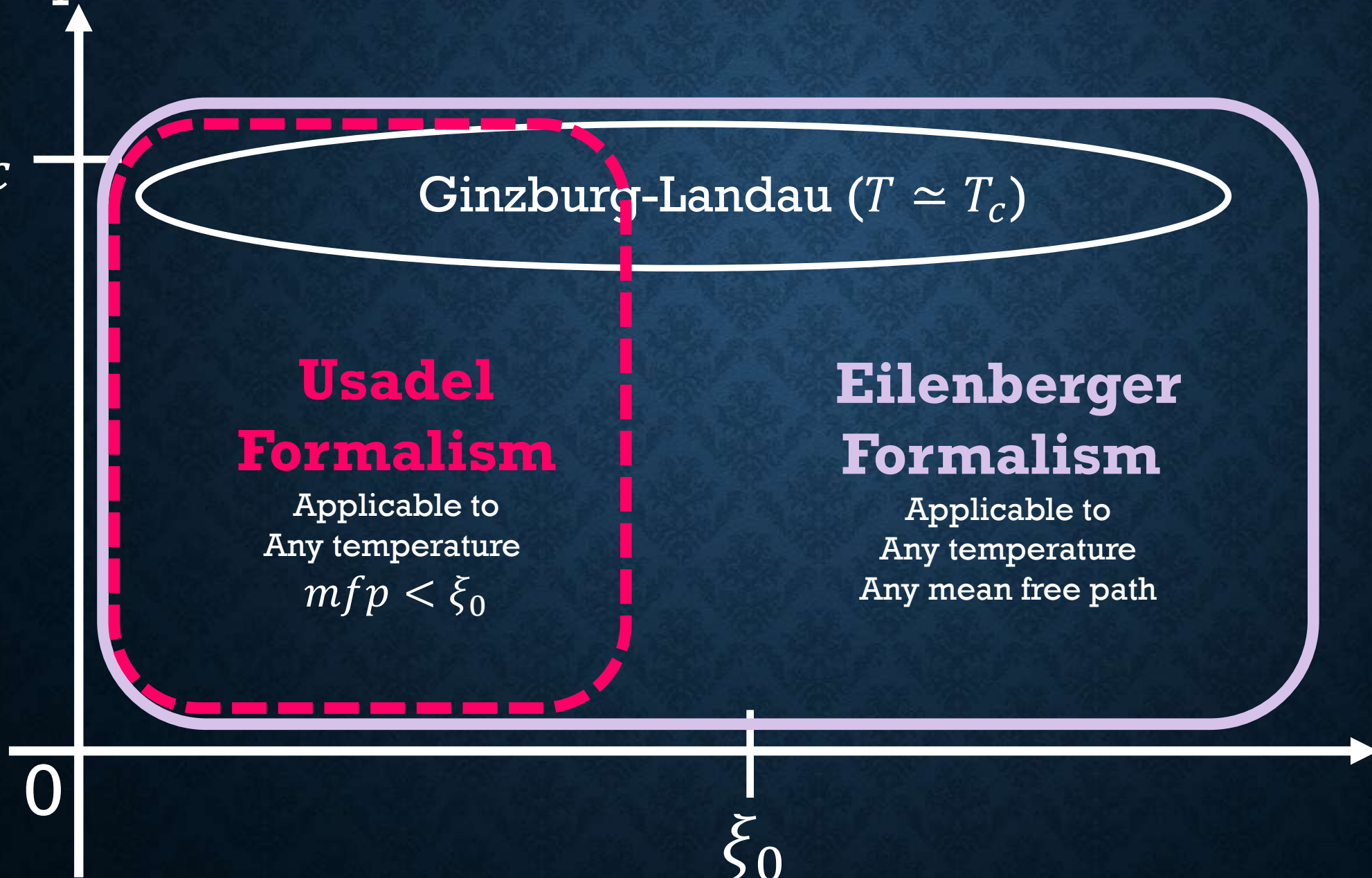
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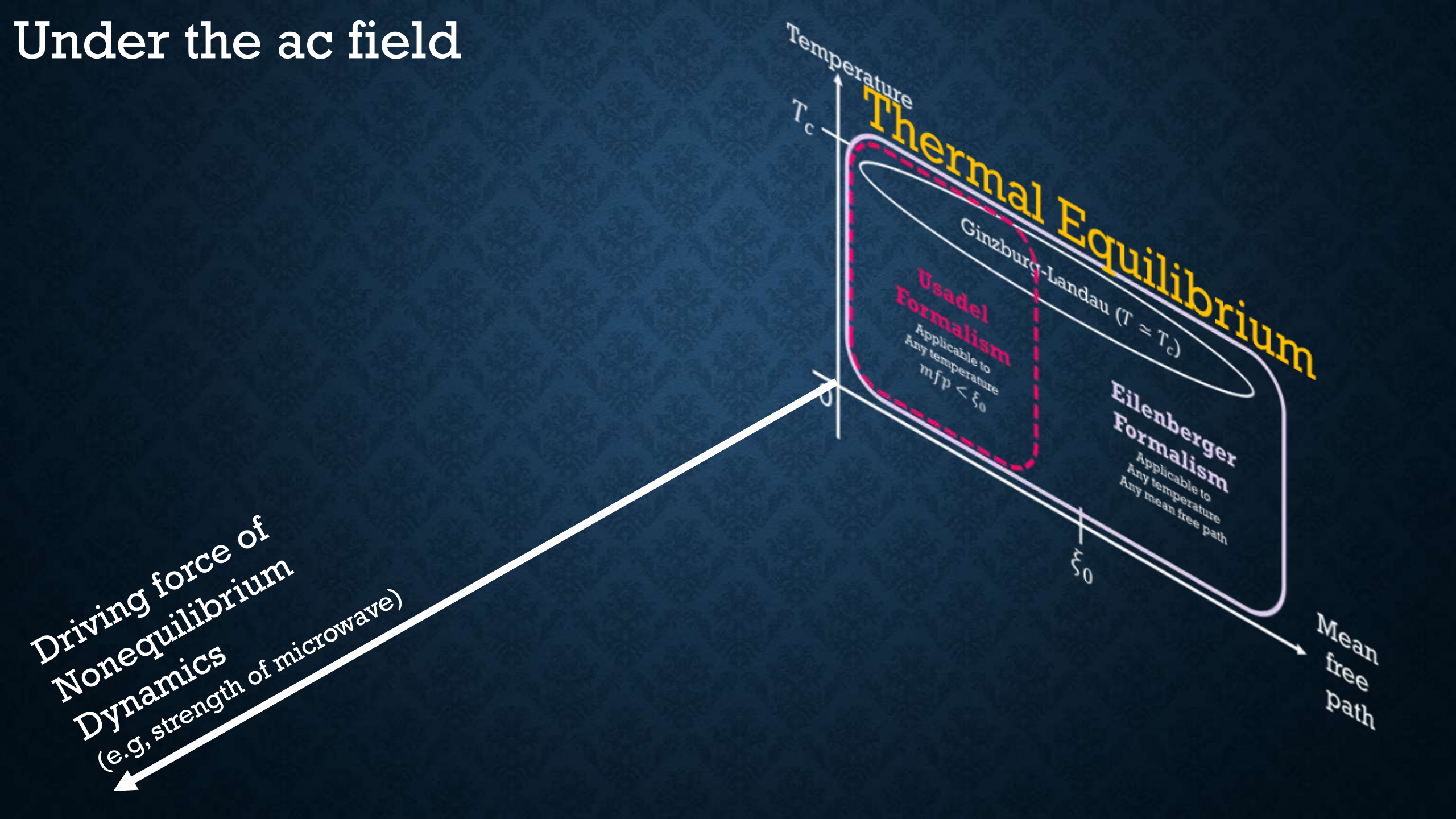
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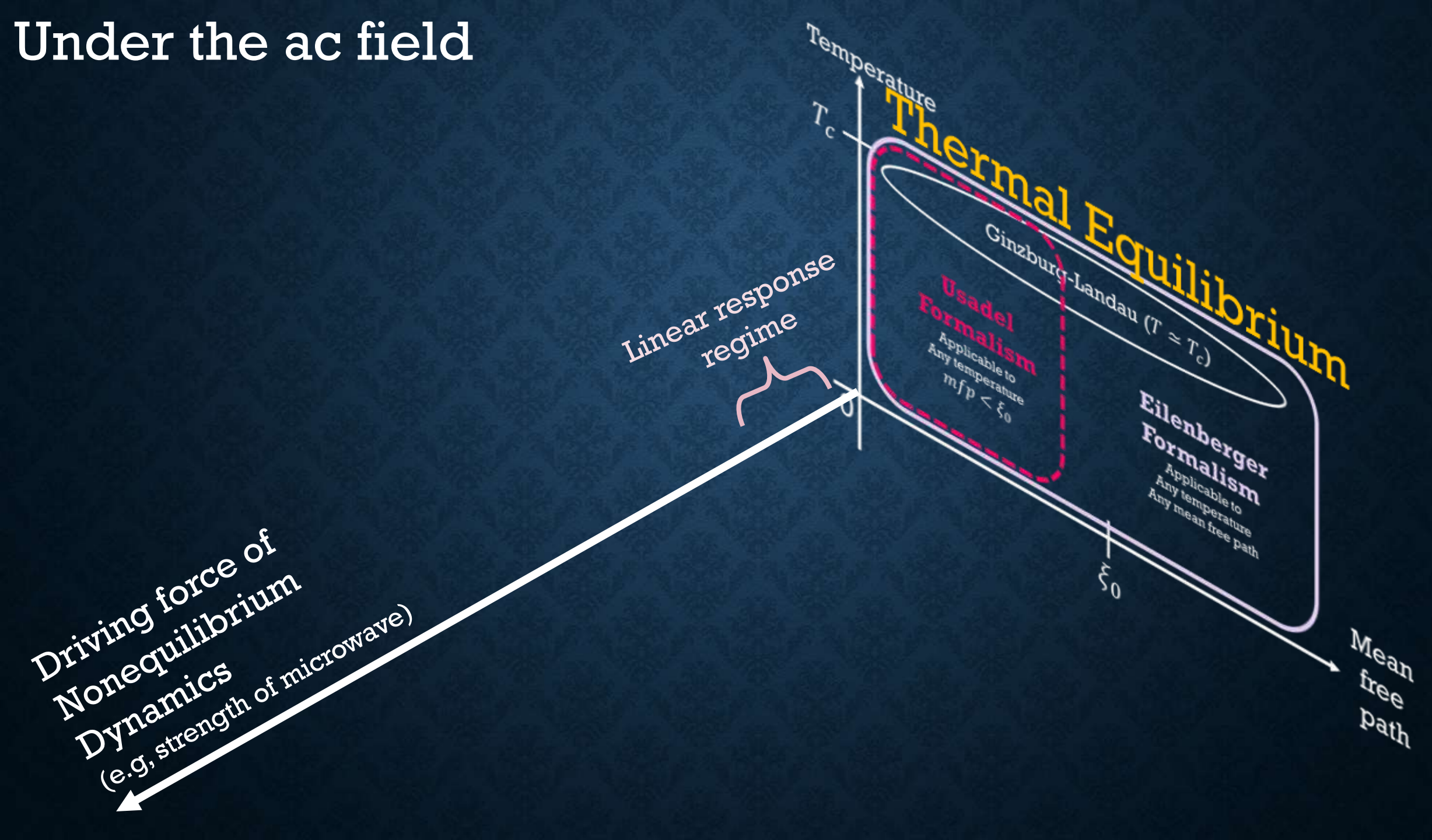
Mean
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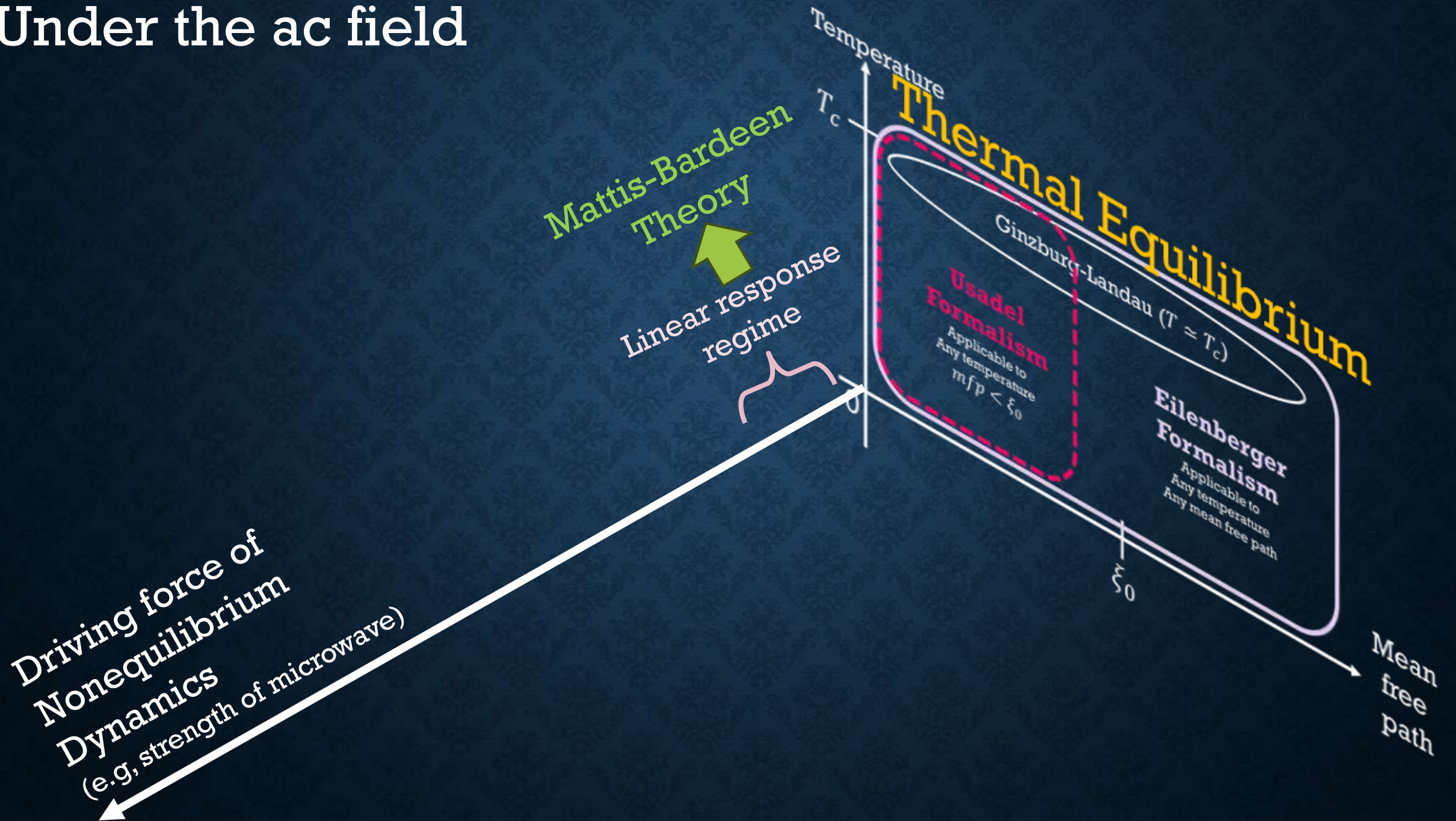
Under the ac field



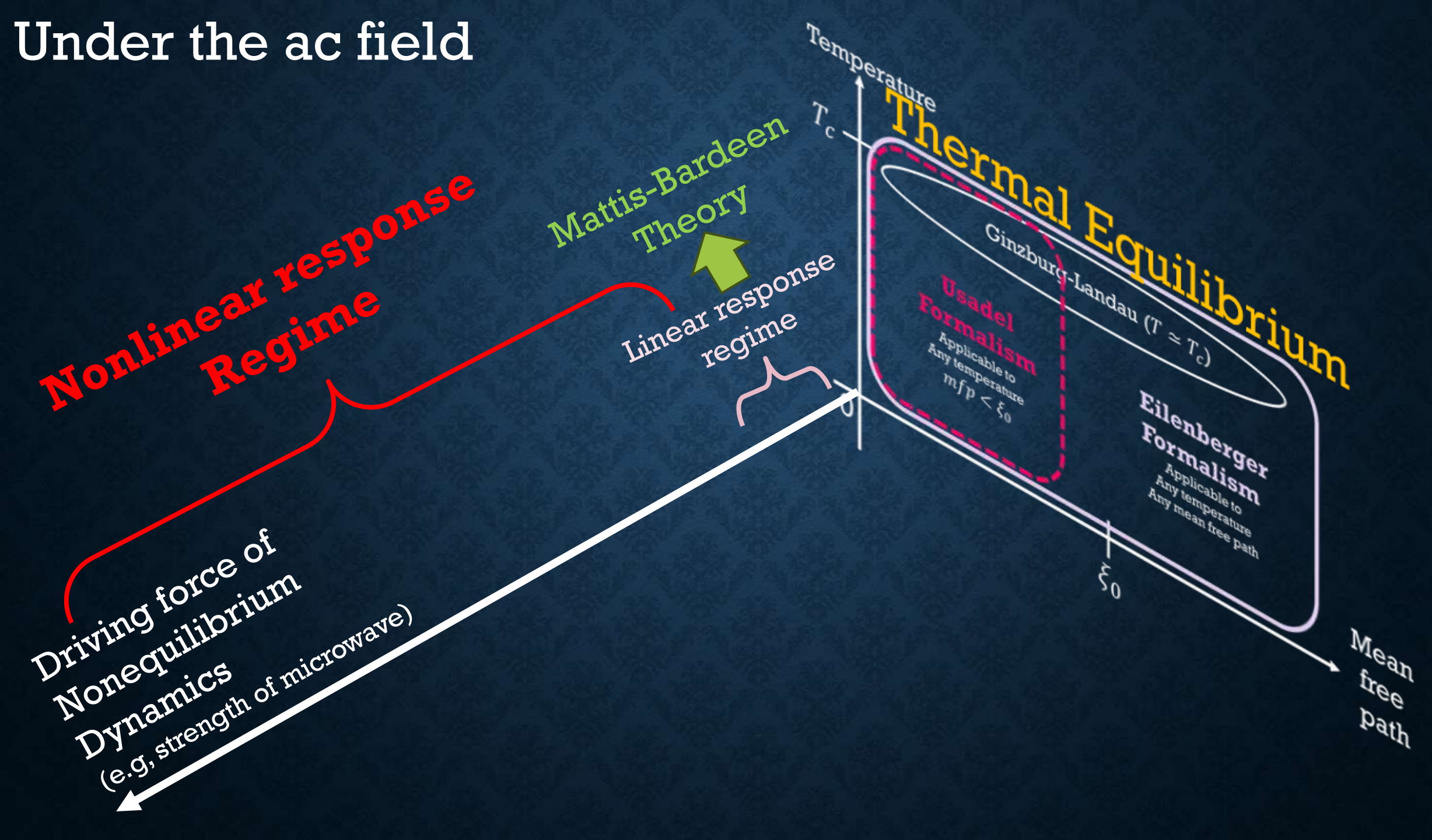
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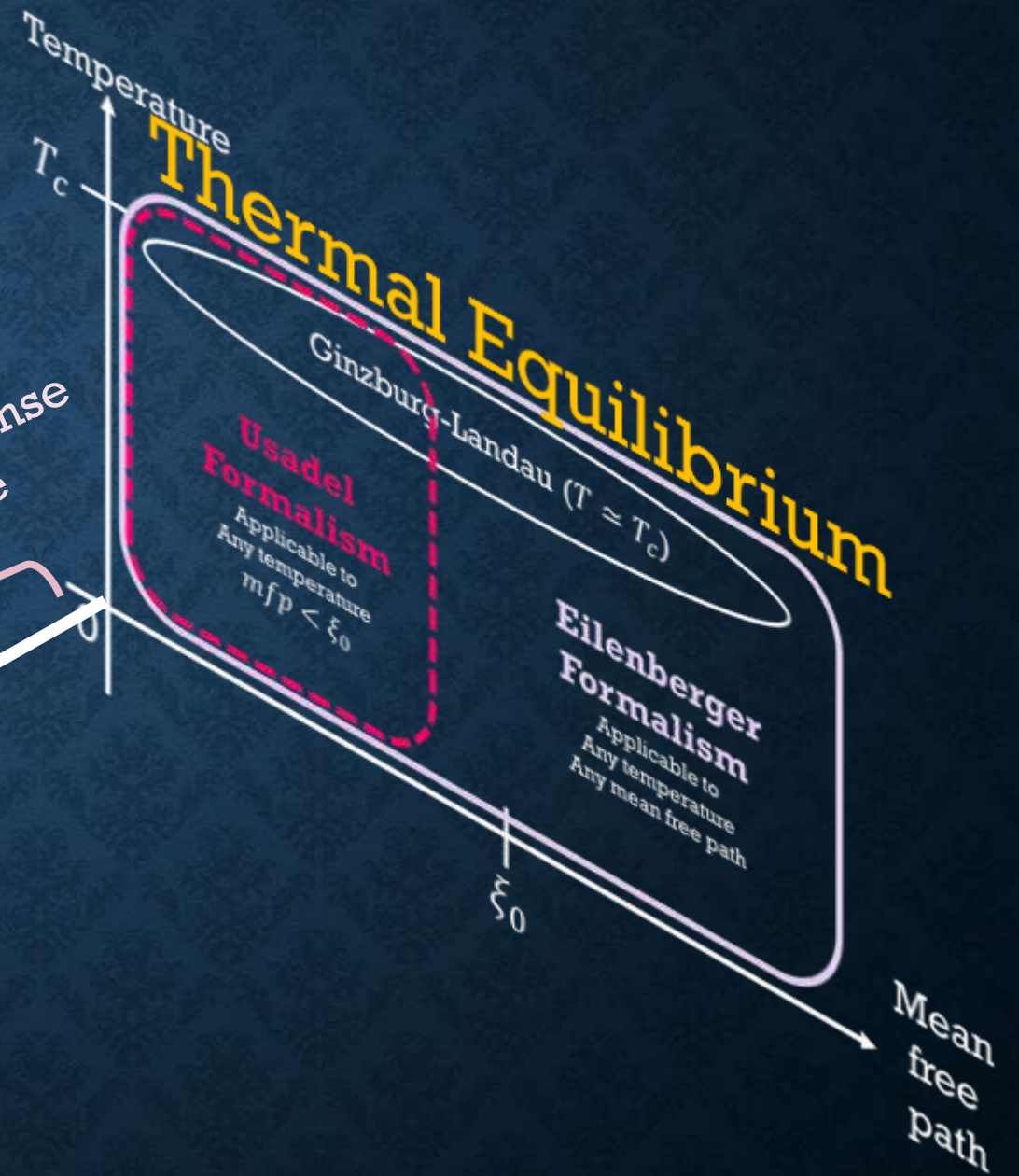
Unfortunately, this is
the **SRF** accelerator regime

**Nonlinear response
Regime**

Driving force of
Nonequilibrium
Dynamics
(e.g, strength of microwave)

Mattis-Bardeen
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Linear response
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Quality factor Q_0

$$Q_0 = \frac{\omega U}{P} \sim \frac{U}{P_1}$$

P_1 : dissipation per
RF cycle

Q_0 : how many cycles it
takes for the stored
energy to completely
dissipate



The **SRF** accelerator regime
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Keldysh
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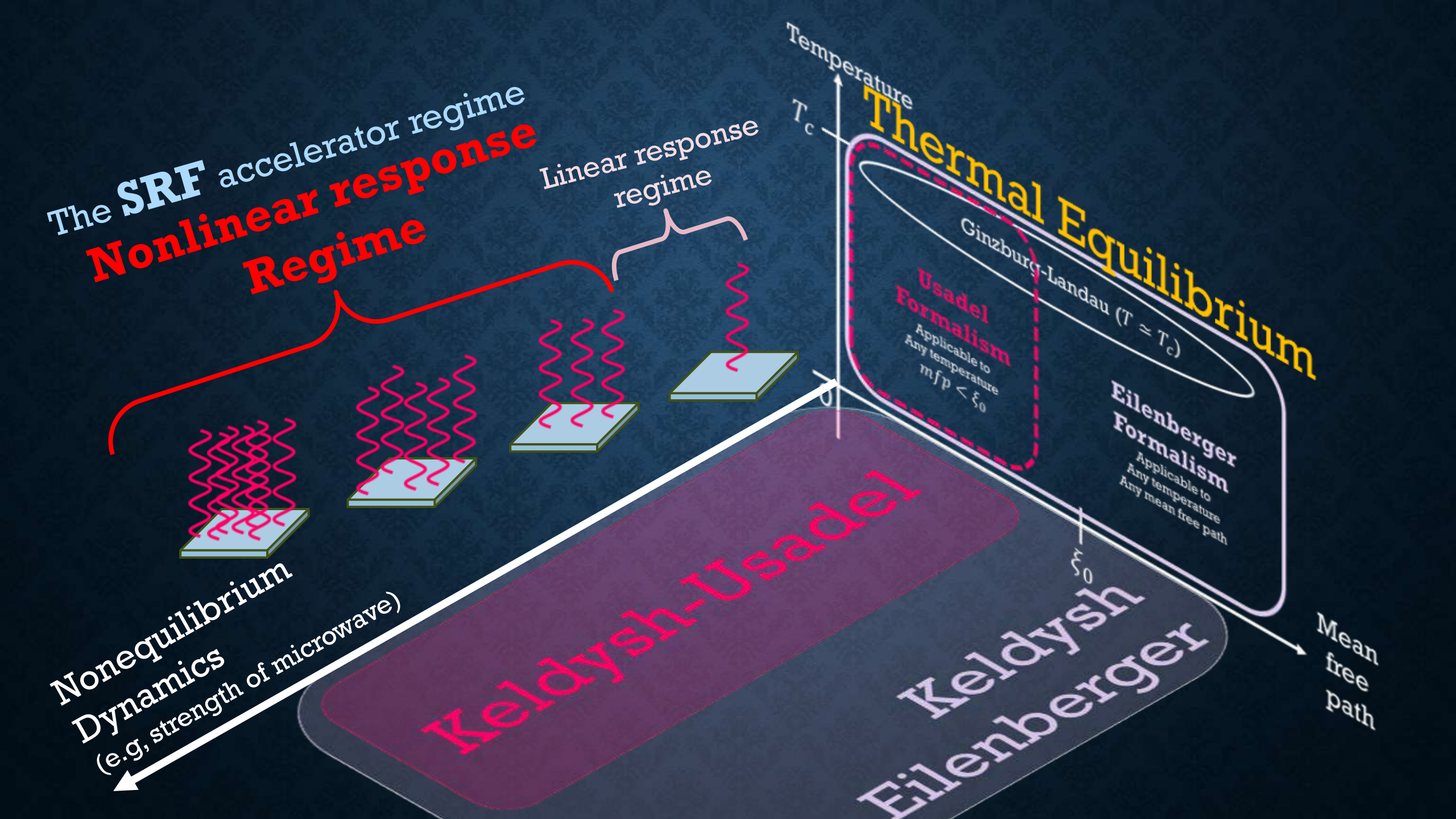
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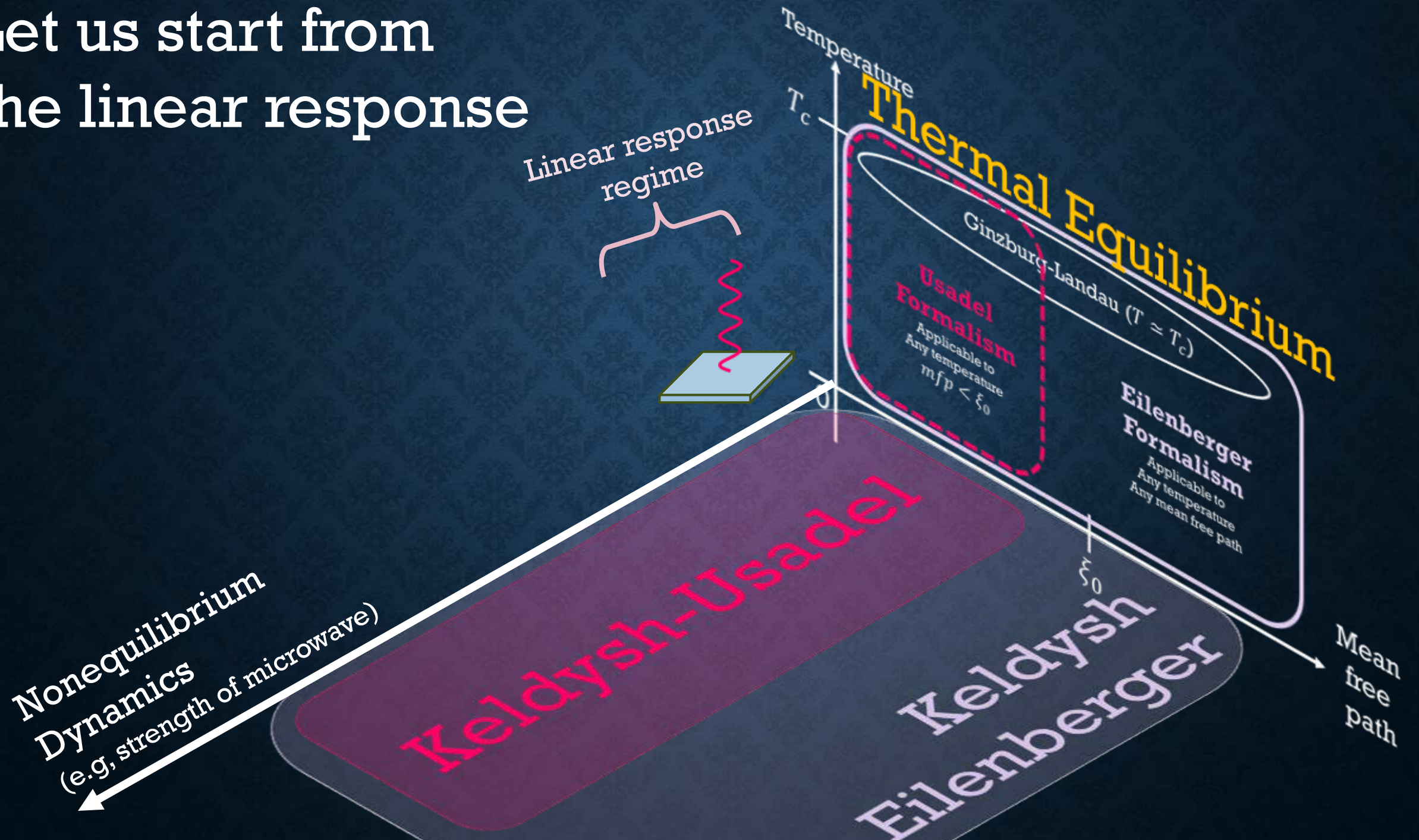
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Let us start from the linear response



Start from the Keldysh-Eilenberger equations (a set of equations for Green's functions and pair potential Δ)

D. Rainer and J. A. Sauls,
Strong coupling theory of superconductivity, in Superconductivity:
From Basic Physics to the Latest Developments, edited by P. N.
Butcher and Y. Lu (World Scientific, Singapore, 1995), p. 45.

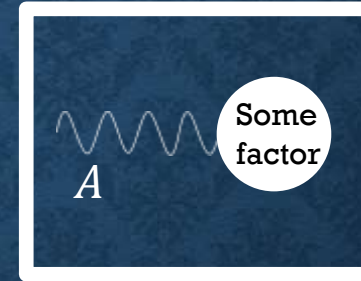
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We apply a small ac electromagnetic field and calculate the first-order perturbation to the current.

$$J \sim A \times \bullet \sim$$



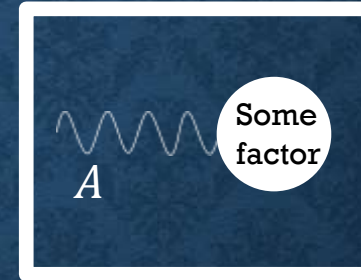
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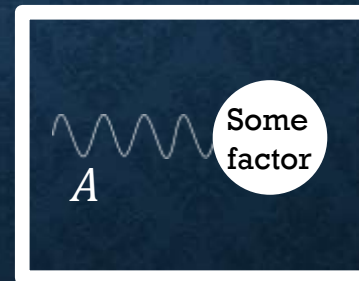
We apply a small ac electromagnetic field and calculate the first-order perturbation to the current.

$$J \sim A \times \bullet \sim$$



Then,
the complex conductivity is given by

$$\sigma = \frac{J}{E} \sim \frac{\text{[Diagram: wavy line A, Some factor]}}{A}$$



Complex conductivity formula for arbitrary MFP (for $\lambda \gg \xi$)

$$\begin{aligned}\sigma(\ell, T, \omega) &= \sigma_1 + i\sigma_2 \\ &= \frac{-3i\sigma_n}{4\omega\tau} \int_{-\infty}^{\infty} d\epsilon [\kappa\mathcal{T}_- + \kappa^*\mathcal{T}_+ + \kappa^a(\mathcal{T}_+ - \mathcal{T}_-)], \\ \kappa &= \frac{1}{3} \frac{g_+g_- + f_+f_- - 1}{d_+ + d_-}, \\ \kappa^a &= \frac{1}{3} \frac{-g_+g_-^* - f_+f_-^* - 1}{d_+ - d_-^*}.\end{aligned}$$

$\tau = \ell/v_f$ is the electron relaxation time. We define $\epsilon_{\pm} := \epsilon \pm \hbar\omega/2$ and set $g_{\pm} := g(\epsilon_{\pm})$, $f_{\pm} := f(\epsilon_{\pm})$, $d_{\pm} := d(\epsilon_{\pm})$, and $\mathcal{T}_{\pm} := \mathcal{T}(\epsilon_{\pm})$, where $g(\epsilon) = -i(\epsilon + i0)/\sqrt{\Delta^2 - (\epsilon + i0)^2}$, $f(\epsilon) = -i\Delta/\sqrt{\Delta^2 - (\epsilon + i0)^2}$, $d(\epsilon) = i\gamma + i\sqrt{\Delta^2 - (\epsilon + i0)^2}$, and $\mathcal{T}(\epsilon) := \tanh(\epsilon/2k_B T) = 1 - 2f_{\text{FD}}(\epsilon)$, with $f_{\text{FD}}(\epsilon)$ the

- W. Zimmermann, E. Brandt, M. Bauer, E. Seider, and L. Genzel, Optical conductivity of BCS superconductors with arbitrary purity, *Physica C* **183**, 99 (1991).
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- T. Kubo, Effects of Nonmagnetic Impurities and Subgap States on the Kinetic Inductance, Complex Conductivity, Quality Factor, and Depairing Current Density, *Physical Review Applied* **17**, 014018 (2022)

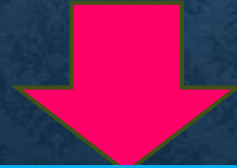
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**In the dirty limit
($\text{mfp} \ll \xi_0$)**



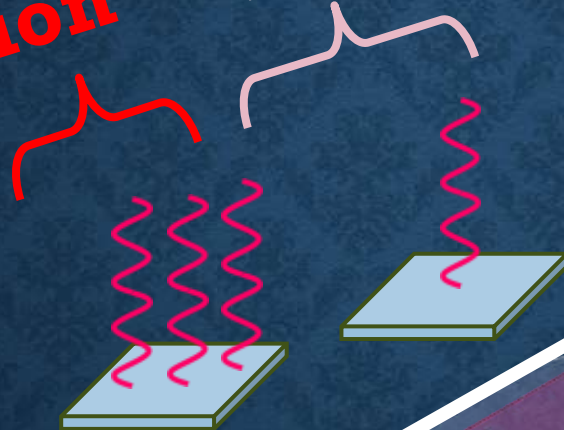
$$\begin{aligned}\sigma_1(T, \omega) &= \frac{\sigma_n}{\hbar\omega} \int_{-\infty}^{\infty} d\epsilon [f_{\text{FD}}(\epsilon) - f_{\text{FD}}(\epsilon')] M_1(\epsilon, \omega), \\ M_1(\epsilon, \omega) &= \text{Reg}(\epsilon)\text{Reg}(\epsilon') + \text{Re}f(\epsilon)\text{Re}f(\epsilon'), \\ \sigma_2(T, \omega) &= \frac{\sigma_n}{\hbar\omega} \int_{-\infty}^{\infty} d\epsilon \tanh \frac{\epsilon}{2k_B T} M_2(\epsilon, \omega), \\ M_2(\epsilon, \omega) &= \text{Reg}(\epsilon)\text{Im}g(\epsilon') + \text{Re}f(\epsilon)\text{Im}f(\epsilon').\end{aligned}$$

$$\epsilon' := \epsilon + \hbar\omega$$

The well-known
dirty-limit Mattis-Bardeen formula

**Leading order
nonlinear
correction**

Linear response
regime



Nonequilibrium
Dynamics
(e.g. strength of microwave)

Keldysh-Usadel

**Keldysh
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Temperature
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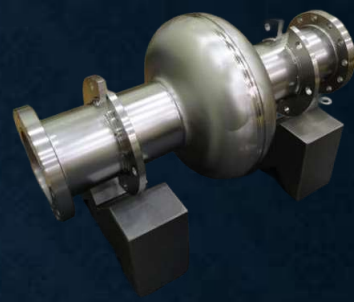
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Keldysh–Usadel Equation for SC under a strong ac field



arXiv:2509.09766

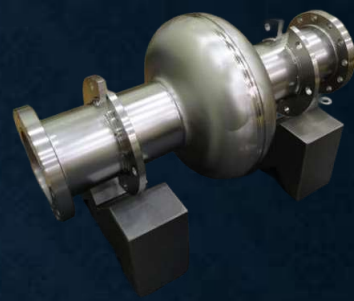
$$\begin{aligned}
 & -i\frac{s}{8} \sum_{\eta=\pm\omega_{ac}} \sum_{\eta'=\pm\omega_{ac}} 2\pi\delta(\omega + \eta + \eta') \times \\
 & \left[\hat{\tau}_3 \hat{g}_e^r \left(\epsilon - \frac{\hbar\eta}{2} + \frac{\hbar\eta'}{2} \right) \hat{\tau}_3 \hat{g}_e^r \left(\epsilon + \frac{\hbar\eta}{2} + \frac{\hbar\eta'}{2} \right) \right. \\
 & \left. - \hat{g}_e^r \left(\epsilon - \frac{\hbar\eta}{2} - \frac{\hbar\eta'}{2} \right) \hat{\tau}_3 \hat{g}_e^r \left(\epsilon + \frac{\hbar\eta}{2} - \frac{\hbar\eta'}{2} \right) \hat{\tau}_3 \right] \\
 & = \left(\epsilon + \frac{\hbar\omega}{2} \right) \hat{\tau}_3 \delta \hat{g}^r(\epsilon, \omega) - \left(\epsilon - \frac{\hbar\omega}{2} \right) \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \\
 & + \delta \hat{\Delta}(\omega) \hat{g}_e^r \left(\epsilon - \frac{\hbar\omega}{2} \right) - \hat{g}_e^r \left(\epsilon + \frac{\hbar\omega}{2} \right) \delta \hat{\Delta}(\omega) \\
 & + [\hat{\Delta}_e, \delta \hat{g}^r(\epsilon, \omega)]. \quad (11)
 \end{aligned}$$

$$\delta \Delta(\omega) = -\frac{\mathcal{G}}{8} \int d\epsilon \text{Tr} [(-i\tau_2) \delta \hat{g}^K(\epsilon, \omega)]. \quad (15)$$

$$\begin{aligned}
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 & - \hat{g}_e^R \left(\epsilon - \frac{\hbar\eta}{2} - \frac{\hbar\eta'}{2} \right) \hat{\tau}_3 \hat{g}_e^K \left(\epsilon + \frac{\hbar\eta}{2} - \frac{\hbar\eta'}{2} \right) \hat{\tau}_3 \\
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We solve these equations to obtain the **nonequilibrium corrections** ($\delta \hat{g}^{R,A,K}$ and $\delta \Delta$).

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 \end{aligned}$$

We solve these equations to obtain the **nonequilibrium corrections** ($\delta \hat{g}^{R,A,K}$ and $\delta \Delta$).

To obtain the current response,
we substitute the solutions
($\delta \hat{g}^{R,A,K}$, $\delta \Delta$) into

$$\begin{aligned}
 \mathbf{J}(t) &= \mathbf{J}^{(1)}(t) + \mathbf{J}^{(3)}(t) \\
 &= \frac{\sigma_n}{8e} \int \left\{ \mathbf{S}^{(1)}(\epsilon, t) + \mathbf{S}^{(3)}(\epsilon, t) \right\} d\epsilon, \quad (49)
 \end{aligned}$$

$$\mathbf{S}^{(1)}(\epsilon, t) = \text{Tr} \left[\hat{\tau}_3 \left\{ \hat{g}_e^R \circ (\hat{\partial} \circ \hat{g}_e^K) + \hat{g}_e^K \circ (\hat{\partial} \circ \hat{g}_e^A) \right\} \right] \quad (50)$$

$$\begin{aligned}
 \mathbf{S}^{(3)}(\epsilon, t) &= \text{Tr} \left[\hat{\tau}_3 \left\{ \hat{g}_e^R \circ (\hat{\partial} \circ \delta \hat{g}^K) + \hat{g}_e^K \circ (\hat{\partial} \circ \delta \hat{g}^A) \right. \right. \\
 & \quad \left. \left. + \delta \hat{g}^R \circ (\hat{\partial} \circ \hat{g}_e^K) + \delta \hat{g}^K \circ (\hat{\partial} \circ \hat{g}_e^A) \right\} \right]. \quad (51)
 \end{aligned}$$

← Linear response

← **Nonlinear** correction
of $\mathcal{O}(A^3)$

Schematically, the results can be illustrated as

$$J \sim Agg + Ag\delta g$$

Linear
response

Nonlinear corrections



$$\begin{aligned} \mathbf{J}(t) &= \mathbf{J}^{(1)}(t) + \mathbf{J}^{(3)}(t) \\ &= \frac{\sigma_n}{8e} \int \left\{ \mathbf{S}^{(1)}(\epsilon, t) + \mathbf{S}^{(3)}(\epsilon, t) \right\} d\epsilon, \end{aligned} \quad (49)$$

$$\mathbf{S}^{(1)}(\epsilon, t) = \text{Tr} \left[\hat{\tau}_3 \left\{ \hat{g}_e^R \circ (\hat{\partial} \circ \hat{g}_e^K) + \hat{g}_e^K \circ (\hat{\partial} \circ \hat{g}_e^A) \right\} \right] \quad (50)$$

$$\begin{aligned} \mathbf{S}^{(3)}(\epsilon, t) &= \text{Tr} \left[\hat{\tau}_3 \left\{ \hat{g}_e^R \circ (\hat{\partial} \circ \delta \hat{g}^K) + \hat{g}_e^K \circ (\hat{\partial} \circ \delta \hat{g}^A) \right. \right. \\ &\quad \left. \left. + \delta \hat{g}^R \circ (\hat{\partial} \circ \hat{g}_e^K) + \delta \hat{g}^K \circ (\hat{\partial} \circ \hat{g}_e^A) \right\} \right]. \end{aligned} \quad (51)$$

← Linear response

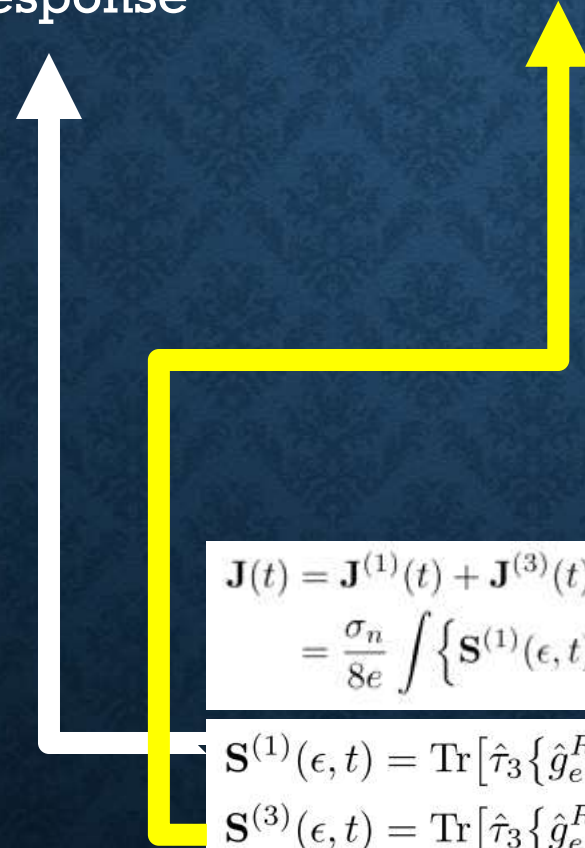
← **Nonlinear** correction
of $\mathcal{O}(A^3)$

Schematically, the results can be illustrated as

$$J \sim Agg + Ag\delta g$$

Linear
response

Nonlinear corrections



$$\begin{aligned} \mathbf{J}(t) &= \mathbf{J}^{(1)}(t) + \mathbf{J}^{(3)}(t) \\ &= \frac{\sigma_n}{8e} \int \left\{ \mathbf{S}^{(1)}(\epsilon, t) + \mathbf{S}^{(3)}(\epsilon, t) \right\} d\epsilon, \quad (49) \end{aligned}$$

$$\mathbf{S}^{(1)}(\epsilon, t) = \text{Tr} \left[\hat{\tau}_3 \left\{ \hat{g}_e^R \circ (\hat{\partial} \circ \hat{g}_e^K) + \hat{g}_e^K \circ (\hat{\partial} \circ \hat{g}_e^A) \right\} \right] \quad (50)$$

$$\begin{aligned} \mathbf{S}^{(3)}(\epsilon, t) &= \text{Tr} \left[\hat{\tau}_3 \left\{ \hat{g}_e^R \circ (\hat{\partial} \circ \delta \hat{g}^K) + \hat{g}_e^K \circ (\hat{\partial} \circ \delta \hat{g}^A) \right. \right. \\ &\quad \left. \left. + \delta \hat{g}^R \circ (\hat{\partial} \circ \hat{g}_e^K) + \delta \hat{g}^K \circ (\hat{\partial} \circ \hat{g}_e^A) \right\} \right]. \quad (51) \end{aligned}$$

← Linear response

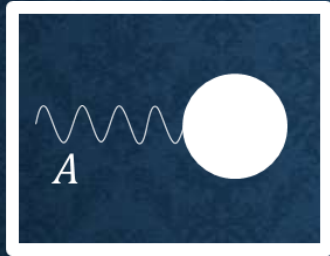
← **Nonlinear** correction
of $\mathcal{O}(A^3)$

Schematically, the results can be illustrated as

$$J \sim \boxed{Agg} + Ag\delta g$$

Linear
response

Nonlinear corrections

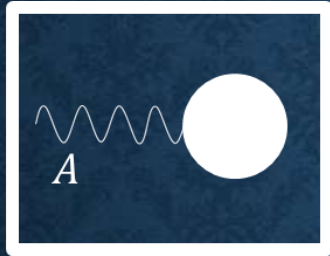


Schematically, the results can be illustrated as

$$J \sim \boxed{A g g} + \boxed{A g \delta g} \quad \delta g \sim A^2 + \delta\Delta \quad \text{and} \quad \delta\Delta \sim \Psi A^2$$

Linear
response

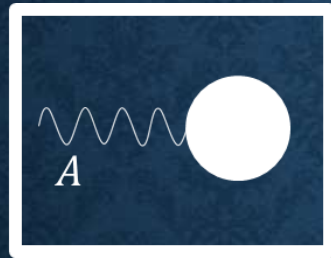
Nonlinear corrections



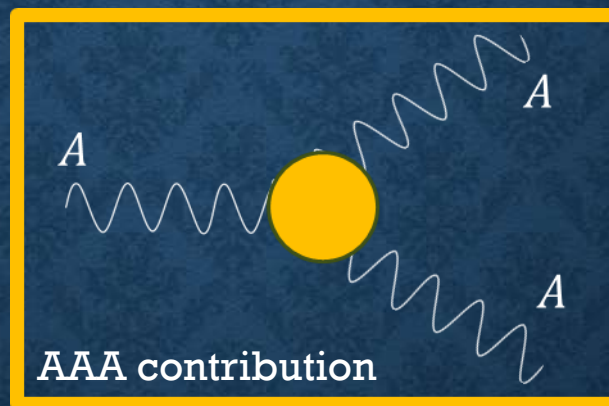
Schematically, the results can be illustrated as

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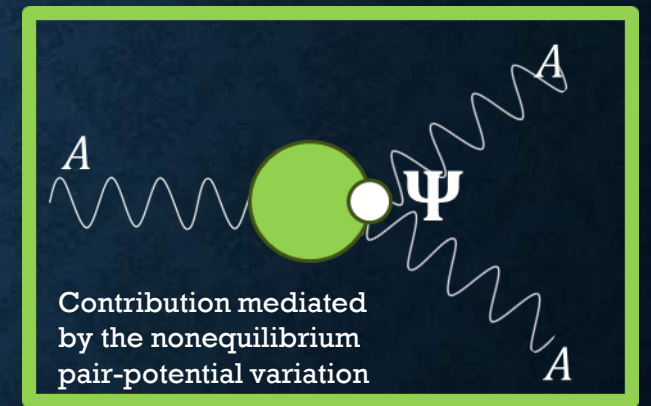
Linear
response



Nonlinear corrections



+

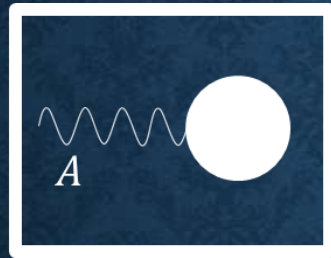


Contribution mediated
by the nonequilibrium
pair-potential variation

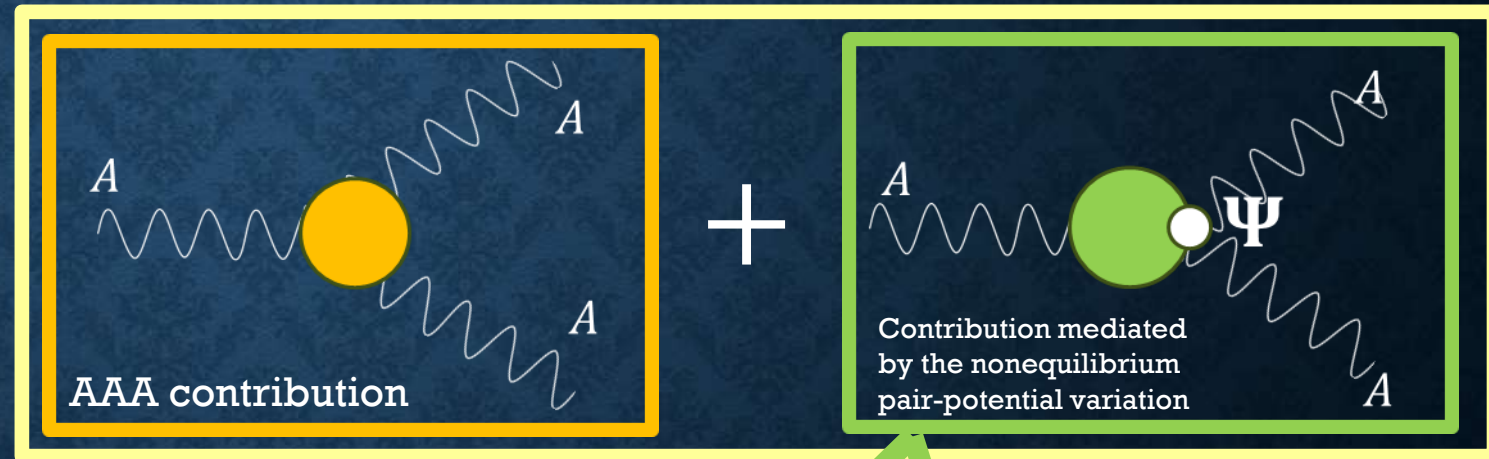
Schematically, the results can be illustrated as

$$J \sim \boxed{A g g} + \boxed{A g \delta g} \quad \delta g \sim A^2 + \delta\Delta \quad \text{and} \quad \delta\Delta \sim \Psi A^2$$

Linear
response



Nonlinear corrections



Note: This theory automatically includes the **Eliashberg effect** (microwave-induced gap enhancement) as well as the **Higgs-mode** contribution.

Time averaged $\delta\Delta$:
Eliashberg effect

Time dependent $\delta\Delta$:
Higgs

Eliashberg effect?

Higgs?

Eliashberg effect?

Higgs?

The variation of order parameter $\delta\Delta$ does not appear in standard linear-response theories such as Mattis–Bardeen theory because $\delta\Delta$ is an $\mathcal{O}(A^2)$ effect, but it does emerge in nonlinear response.

$$\delta\Delta \propto A_0^2 \cos^2 \omega t$$

Eliashberg effect?

Higgs?

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$$\delta\Delta \propto A_0^2 \cos^2 \omega t \propto 1 + \cos 2\omega t$$

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Time-averaged nonequilibrium gap variation

Eliashberg effect?

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Time-averaged nonequilibrium gap variation

Time dependent nonequilibrium gap variation

Eliashberg effect?

Higgs?

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$$\delta\Delta \propto A_0^2 \cos^2 \omega t \propto 1 + \cos 2\omega t$$

Time-averaged nonequilibrium gap variation

Time dependent nonequilibrium gap variation

quasiparticles



Δ

Eliashberg effect?

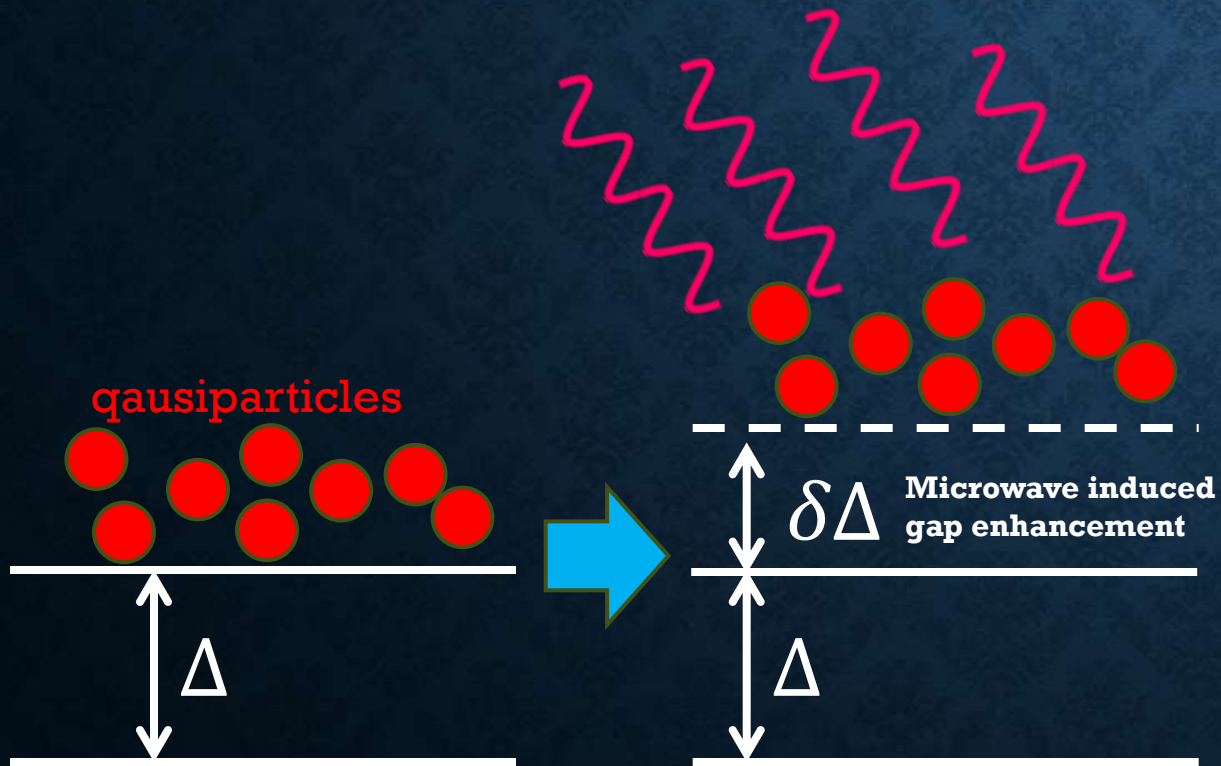
Higgs?

The variation of order parameter $\delta\Delta$ does not appear in standard linear-response theories such as Mattis–Bardeen theory because $\delta\Delta$ is an $\mathcal{O}(A^2)$ effect, but it does emerge in nonlinear response.

$$\delta\Delta \propto A_0^2 \cos^2 \omega t \propto 1 + \cos 2\omega t$$

Time-averaged nonequilibrium gap variation

Time dependent nonequilibrium gap variation



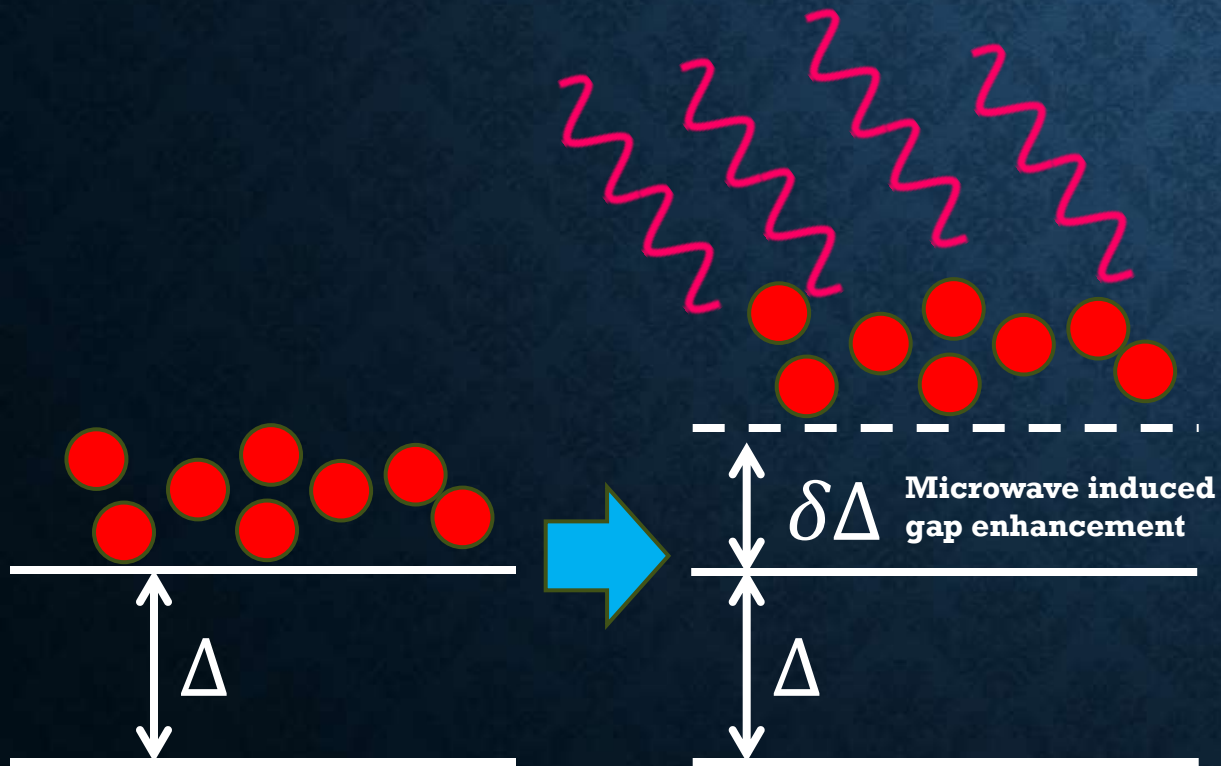
Eliashberg effect?

Higgs?

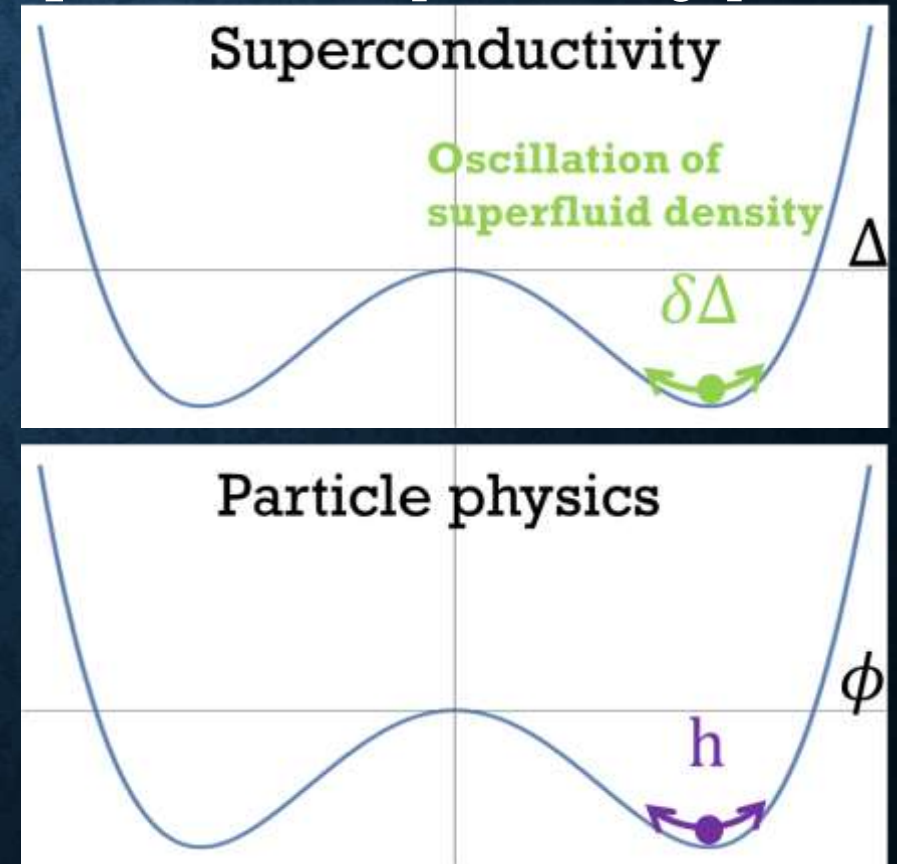
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$$\delta\Delta \propto A_0^2 \cos^2 \omega t \propto 1 + \cos 2\omega t$$

Time-averaged nonequilibrium gap variation



Time dependent nonequilibrium gap variation

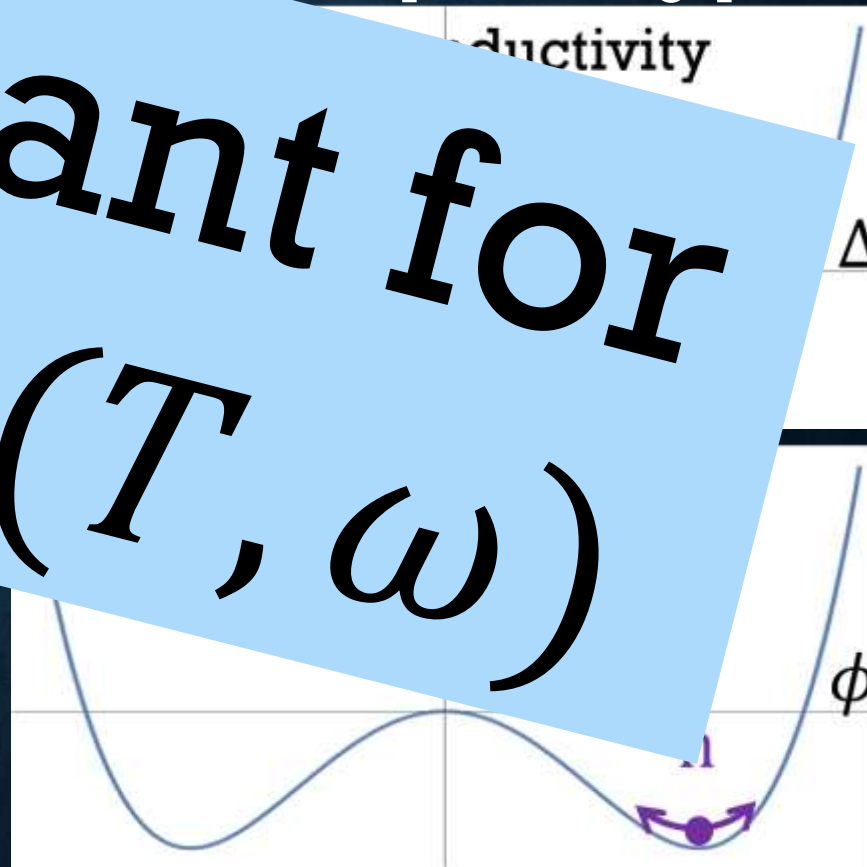
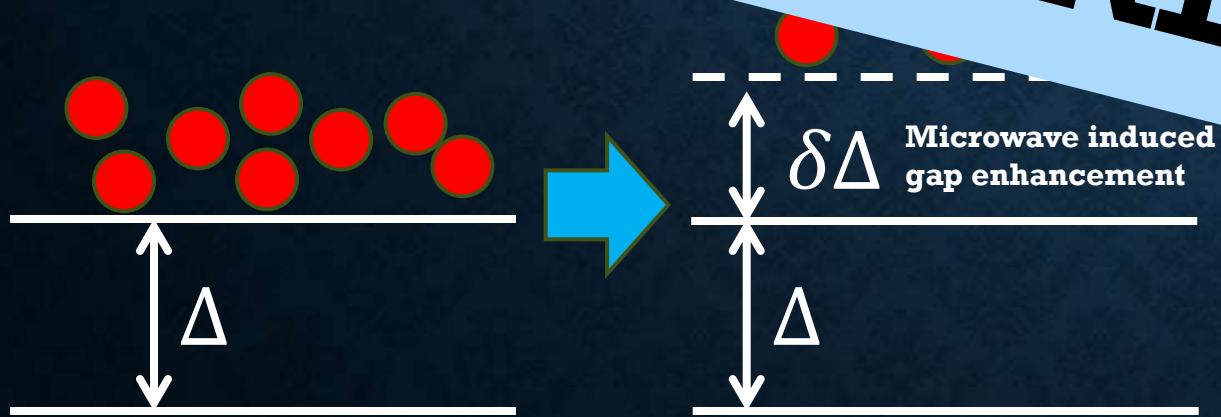


Eliashberg effect?

Higgs?

The variation of order parameter $\delta\Delta$ does not appear in standard linear-response theories such as Mattis–Bardarshan because $\delta\Delta$ is an $\mathcal{O}(A^2)$ effect, but it does emerge in nonlinear response.

Not important for σ_1 at SRF's (T, ω)



Nonlinear correction

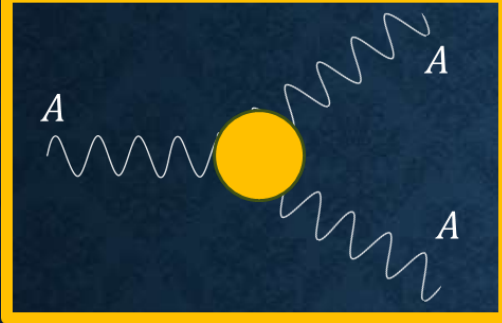
arXiv:2509.09766

$$\delta\sigma_1 = \delta\sigma_1^{qqq} + \delta\sigma_1^{\text{Higgs}} + \delta\sigma_1^{\text{Eliash}}$$

Nonlinear correction

arXiv:2509.09766

$$\delta\sigma_1 = \delta\sigma_1^{qqq} + \delta\sigma_1^{\text{Higgs}} + \delta\sigma_1^{\text{Eliash}}$$



$$\delta\sigma_1^{qqq} = 2\sqrt{\pi}\sigma_n \text{Re}[I_{1H}^{qqq}] \frac{s}{\hbar\omega_{ac}} \quad I_{1H}^{qqq} = \frac{-1}{16\sqrt{\pi}} \int K_{1H} d\epsilon,$$

$$K_{1H}(\epsilon, \omega_{ac}) = \sum_{i=1}^6 K_{1H,i}, \quad (61)$$

$$K_{1H,1} = i \frac{(F_1 - F_{-3})F_{-1} + (G_1 - G_{-3})G_{-1}}{4\hbar\omega_{ac}(F_1 + F_{-3})} \times \left[\left\{ (F_1 + F_{-3})G_{-1}^* + (G_1 + G_{-3})F_{-1}^* \right\} (\mathcal{T}_{-1} - \mathcal{T}_{-3}) + \left\{ (F_1 + F_{-3})G_{-1} + (G_1 + G_{-3})F_{-1} \right\} \mathcal{T}_{-1} \right], \quad (62)$$

$$K_{1H,2} = i \frac{(G_3 - G_{-1})G_1 + (F_3 - F_{-1})F_1}{4\hbar\omega_{ac}(F_3 + F_{-1})} \times \left\{ (F_3 + F_{-1})G_1 + (G_3 + G_{-1})F_1 \right\} \mathcal{T}_{-1}, \quad (63)$$

$$K_{1H,3} = -i \frac{(F_3 - F_{-1}^*)G_1 + (G_3 - G_{-1}^*)F_1}{4\hbar\omega_{ac}(F_3 - F_{-1}^*)} \times \left[\left\{ (G_3 + G_{-1}^*)G_1^* + (F_3 + F_{-1}^*)F_1^* \right\} (\mathcal{T}_3 - \mathcal{T}_1) + \left\{ (G_3 + G_{-1}^*)G_1 + (F_3 + F_{-1}^*)F_1 \right\} (\mathcal{T}_{-1} - \mathcal{T}_1) \right], \quad (64)$$

$$K_{1H,4} = -i \frac{F_1 G_{-1} + G_1 F_{-1}}{2(\epsilon_{-1} G_{-1} - F_{-1} \Delta)} \times \left\{ (F_1 + F_{-3})G_{-1} + (G_1 + G_{-3})F_{-1} \right\} \mathcal{T}_{-1}, \quad (65)$$

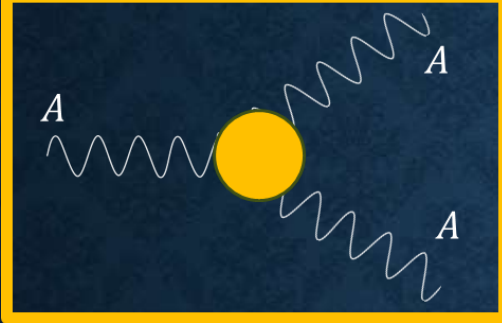
$$K_{1H,5} = -i \frac{(F_3 + F_{-1})G_1 + (G_3 + G_{-1})F_1}{2(\epsilon_1 G_1 - F_1 \Delta)} \times \left\{ (G_1 F_{-1}^* + F_1 G_{-1}^*) (\mathcal{T}_{-1} - \mathcal{T}_1) + (G_1 F_{-1} + G_{-1} F_1) \mathcal{T}_{-1} \right\}, \quad (66)$$

$$K_{1H,6} = -\frac{G_{-1}^* \text{Im} F_1 + F_{-1}^* \text{Im} G_1}{\epsilon_1 \text{Im} G_1 - \text{Im} F_1 \Delta} \times \left\{ (\text{Re} G_3 \text{Im} F_1 + \text{Im} G_1 \text{Re} F_3) (\mathcal{T}_3 - \mathcal{T}_1) + (\text{Re} G_{-1} \text{Im} F_1 + \text{Im} G_1 \text{Re} F_{-1}) (\mathcal{T}_{-1} - \mathcal{T}_1) \right\}, \quad (67)$$

Nonlinear correction

arXiv:2509.09766

$$\delta\sigma_1 = \delta\sigma_1^{qqq} + \delta\sigma_1^{\text{Higgs}} + \delta\sigma_1^{\text{Eliash}}$$



$$\delta\sigma_1^{qqq} = 2\sqrt{\pi}\sigma_n \text{Re}[I_{1H}^{qqq}] \frac{s}{\hbar\omega_{ac}} \quad I_{1H}^{qqq} = \frac{-1}{16\sqrt{\pi}} \int K_{1H} d\epsilon,$$

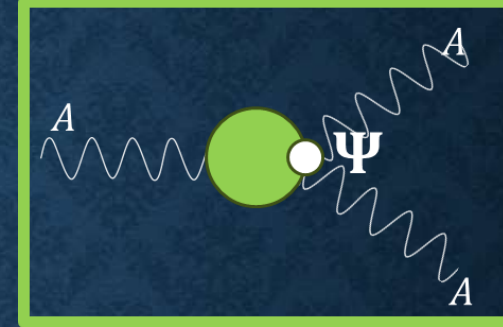
$$K_{1H}(\epsilon, \omega_{ac}) = \sum_{i=1}^6 K_{1H,i}, \quad (61)$$

$$\begin{aligned} K_{1H,1} &= i \frac{(F_1 - F_{-3})F_{-1} + (G_1 - G_{-3})G_{-1}}{4\hbar\omega_{ac}(F_1 + F_{-3})} \\ &\times \left[\left\{ (F_1 + F_{-3})G_{-1}^* + (G_1 + G_{-3})F_{-1}^* \right\} (\mathcal{T}_{-1} - \mathcal{T}_{-3}) \right. \\ &\left. + \left\{ (F_1 + F_{-3})G_{-1} + (G_1 + G_{-3})F_{-1} \right\} \mathcal{T}_{-1} \right], \quad (62) \\ K_{1H,2} &= i \frac{(G_3 - G_{-1})G_1 + (F_3 - F_{-1})F_1}{4\hbar\omega_{ac}(F_3 + F_{-1})} \\ &\times \left\{ (F_3 + F_{-1})G_1 + (G_3 + G_{-1})F_1 \right\} \mathcal{T}_{-1}, \quad (63) \\ K_{1H,3} &= -i \frac{(F_3 - F_{-1}^*)G_1 + (G_3 - G_{-1}^*)F_1}{4\hbar\omega_{ac}(F_3 - F_{-1}^*)} \\ &\times \left[\left\{ (G_3 + G_{-1}^*)G_1^* + (F_3 + F_{-1}^*)F_1^* \right\} (\mathcal{T}_3 - \mathcal{T}_1) \right. \\ &\left. + \left\{ (G_3 + G_{-1}^*)G_1 + (F_3 + F_{-1}^*)F_1 \right\} (\mathcal{T}_{-1} - \mathcal{T}_1) \right], \quad (64) \end{aligned}$$

$$K_{1H,4} = -i \frac{F_1 G_{-1} + G_1 F_{-1}}{2(\epsilon_{-1} G_{-1} - F_{-1} \Delta)} \times \left\{ (F_1 + F_{-3})G_{-1} + (G_1 + G_{-3})F_{-1} \right\} \mathcal{T}_{-1}, \quad (65)$$

$$K_{1H,5} = -i \frac{(F_3 + F_{-1})G_1 + (G_3 + G_{-1})F_1}{2(\epsilon_1 G_1 - F_1 \Delta)} \times \left\{ (G_1 F_{-1}^* + F_1 G_{-1}^*)(\mathcal{T}_{-1} - \mathcal{T}_1) \right. \\ \left. + (G_1 F_{-1} + G_{-1} F_1) \mathcal{T}_{-1} \right\}, \quad (66)$$

$$K_{1H,6} = -\frac{G_{-1}^* \text{Im} F_1 + F_{-1}^* \text{Im} G_1}{\epsilon_1 \text{Im} G_1 - \text{Im} F_1 \Delta} \times \left\{ (\text{Re} G_3 \text{Im} F_1 + \text{Im} G_1 \text{Re} F_3)(\mathcal{T}_3 - \mathcal{T}_1) \right. \\ \left. + (\text{Re} G_{-1} \text{Im} F_1 + \text{Im} G_1 \text{Re} F_{-1})(\mathcal{T}_{-1} - \mathcal{T}_1) \right\}, \quad (67)$$



Higgs
Contribution:
Time dependent $\delta\Delta$

$$\delta\sigma_1^{\text{Higgs}} = 2\sqrt{\pi}\sigma_n \text{Re}[I_{1H}^{\text{Higgs}}] \frac{s}{\hbar\omega_{ac}}$$

$$I_{1H}^{\text{Higgs}} = \frac{-1}{16\sqrt{\pi}} \Psi_{2H} \int Z_{1H}^{\text{Higgs}} d\epsilon$$

$$Z_{1H}^{\text{Higgs}}(\epsilon, \omega_{ac}) = \sum_{i=1,2,3} Z_{1H,i}^{\text{Higgs}}, \quad (71)$$

$$Z_{1H,1}^{\text{Higgs}} = -\frac{2(F_1 - F_{-3})}{\hbar\omega_{ac}(F_1 + F_{-3})} \times \left[\left\{ (F_1 + F_{-3})G_{-1}^* + (G_1 + G_{-3})F_{-1}^* \right\} (\mathcal{T}_{-1} - \mathcal{T}_{-3}) \right. \\ \left. + \left\{ (F_1 + F_{-3})G_{-1} + (G_1 + G_{-3})F_{-1} \right\} \mathcal{T}_{-1} \right], \quad (72)$$

$$Z_{1H,2}^{\text{Higgs}} = -\frac{2(F_3 - F_{-1})}{\hbar\omega_{ac}(F_3 + F_{-1})} \times \left\{ (F_3 + F_{-1})G_1 + (G_3 + G_{-1})F_1 \right\} \mathcal{T}_{-1}, \quad (73)$$

$$Z_{1H,3}^{\text{Higgs}} = -\frac{2(F_3 + F_{-1}^*)}{\hbar\omega_{ac}(F_3 - F_{-1}^*)} \times \left\{ (F_3 - F_{-1}^*)G_1 + (G_3 - G_{-1}^*)F_1 \right\} (\mathcal{T}_3 - \mathcal{T}_{-1}) \quad (74)$$

Eliashberg
Contribution:
Time averaged $\delta\Delta$

$$\delta\sigma_1^{\text{Eliash}} = 2\sqrt{\pi}\sigma_n \text{Re}[I_{1H}^{\text{Eliash}}] \frac{s}{\hbar\omega_{ac}}$$

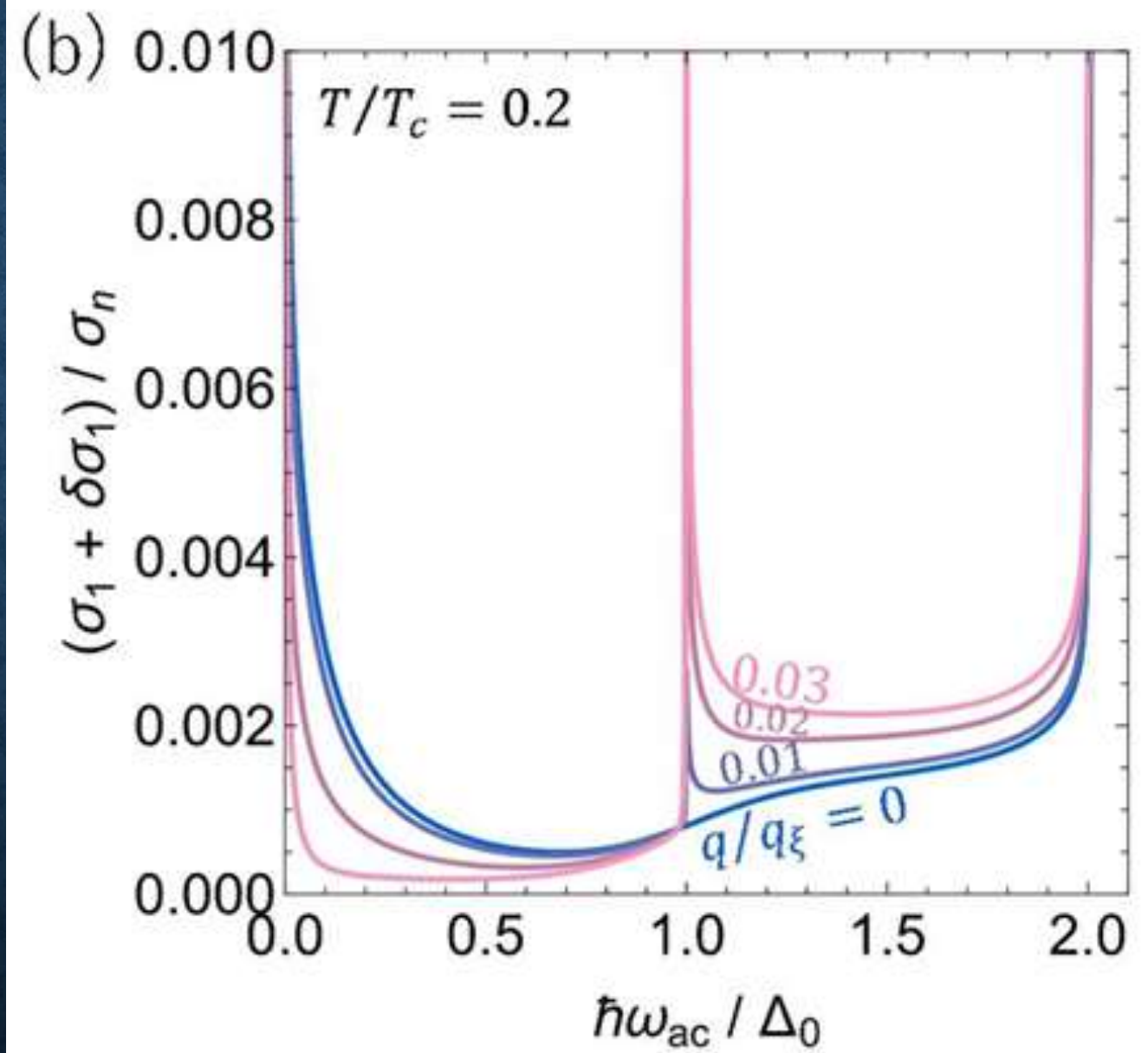
$$I_{1H}^{\text{Eliash}} = \frac{-1}{16\sqrt{\pi}} \Psi_{0H} \int Z_{1H}^{\text{Eliash}} d\epsilon$$

$$Z_{1H}^{\text{Eliash}}(\epsilon, \omega_{ac}) = \sum_{i=1,2} Z_{1H,i}^{\text{Eliash}}, \quad (68)$$

$$Z_{1H,1}^{\text{Eliash}} = \frac{4G_{-1}(G_1 F_{-1} + F_1 G_{-1})}{\epsilon_{-1} G_{-1} - F_{-1} \Delta} \mathcal{T}_{-1}, \quad (69)$$

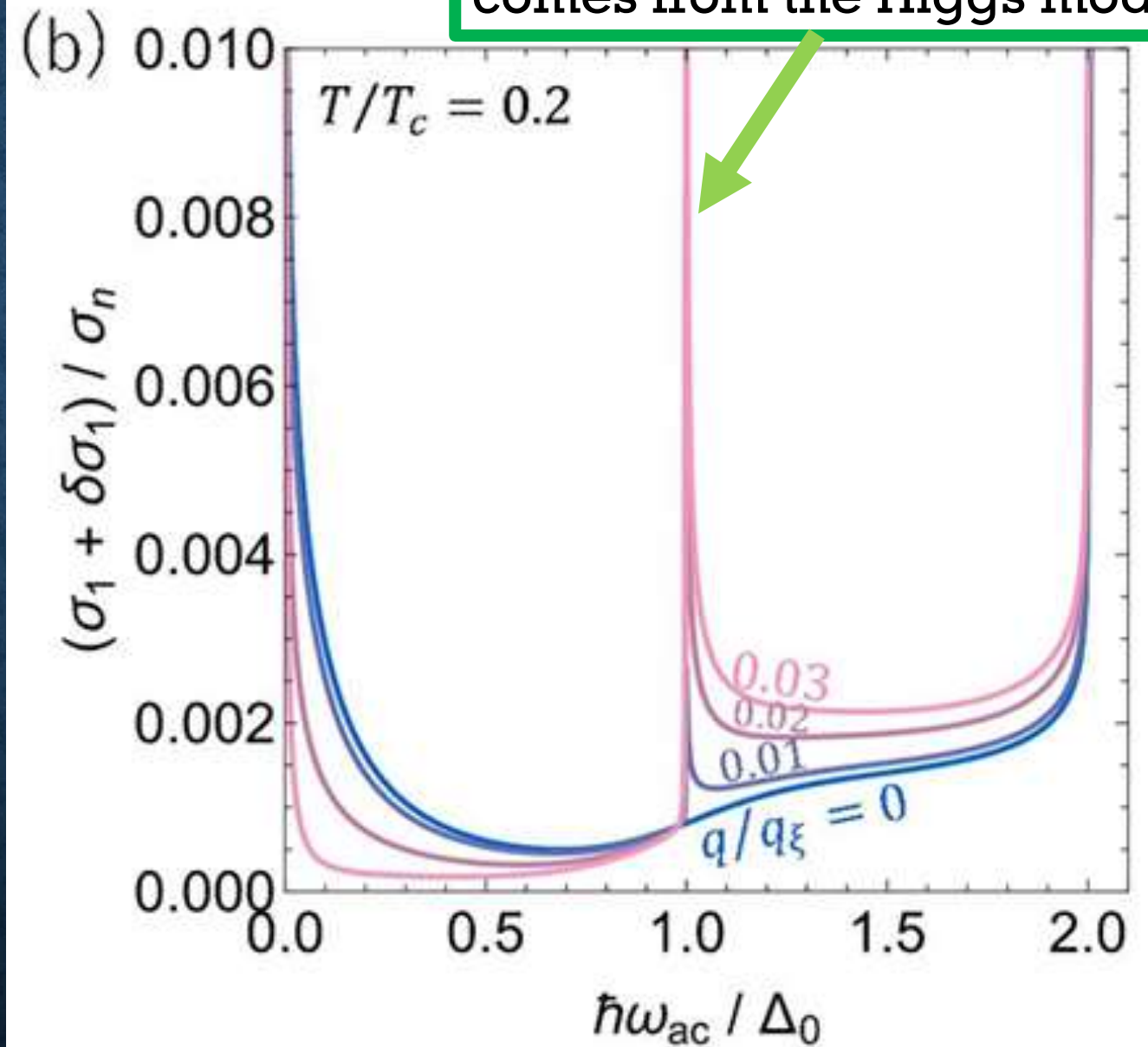
$$Z_{1H,2}^{\text{Eliash}} = \frac{4G_1}{\epsilon_1 G_1 - F_1 \Delta} \left\{ (G_1 F_{-1} + G_{-1} F_1) \mathcal{T}_{-1} \right. \\ \left. + (G_1 F_{-1}^* + F_1 G_{-1}^*)(\mathcal{T}_{-1} - \mathcal{T}_1) \right\}, \quad (70)$$

Results



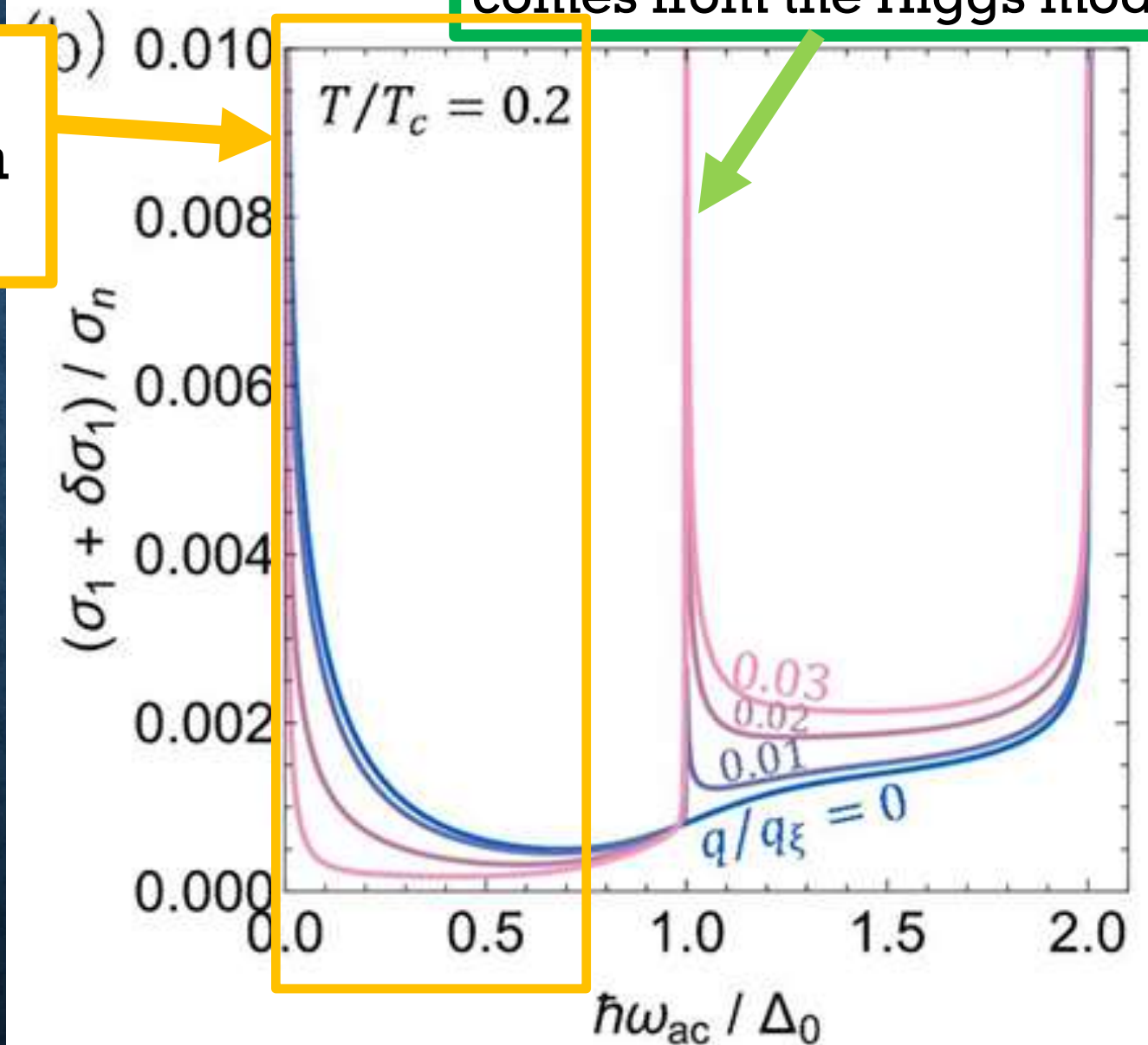
Results

The resonance peak at Δ comes from the Higgs mode



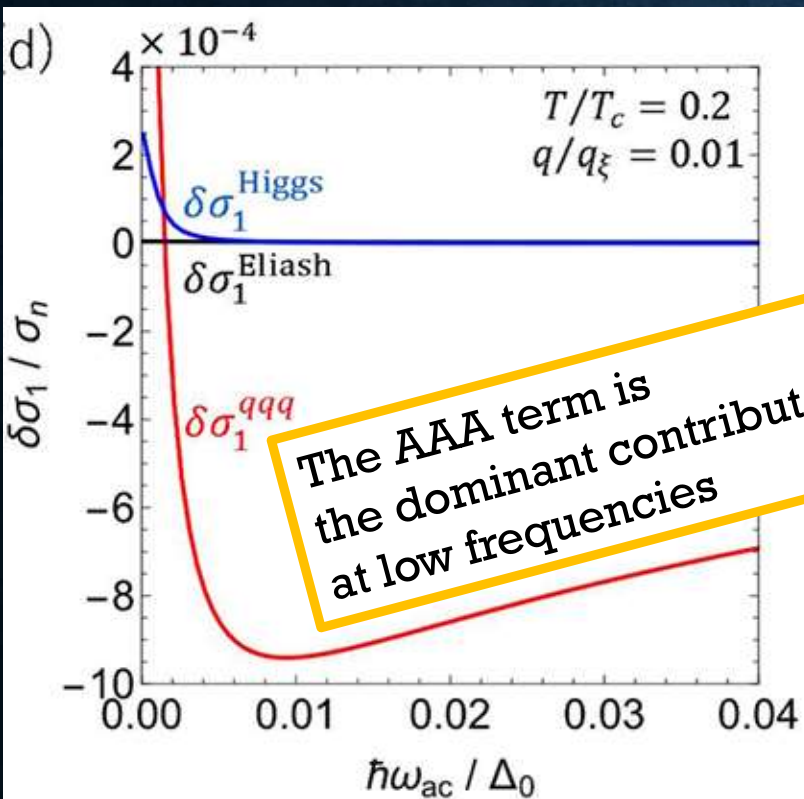
Results

At lower frequencies, $\sigma_1 + \delta\sigma_1$ is a decreasing function of the ac amplitude (q).



Results

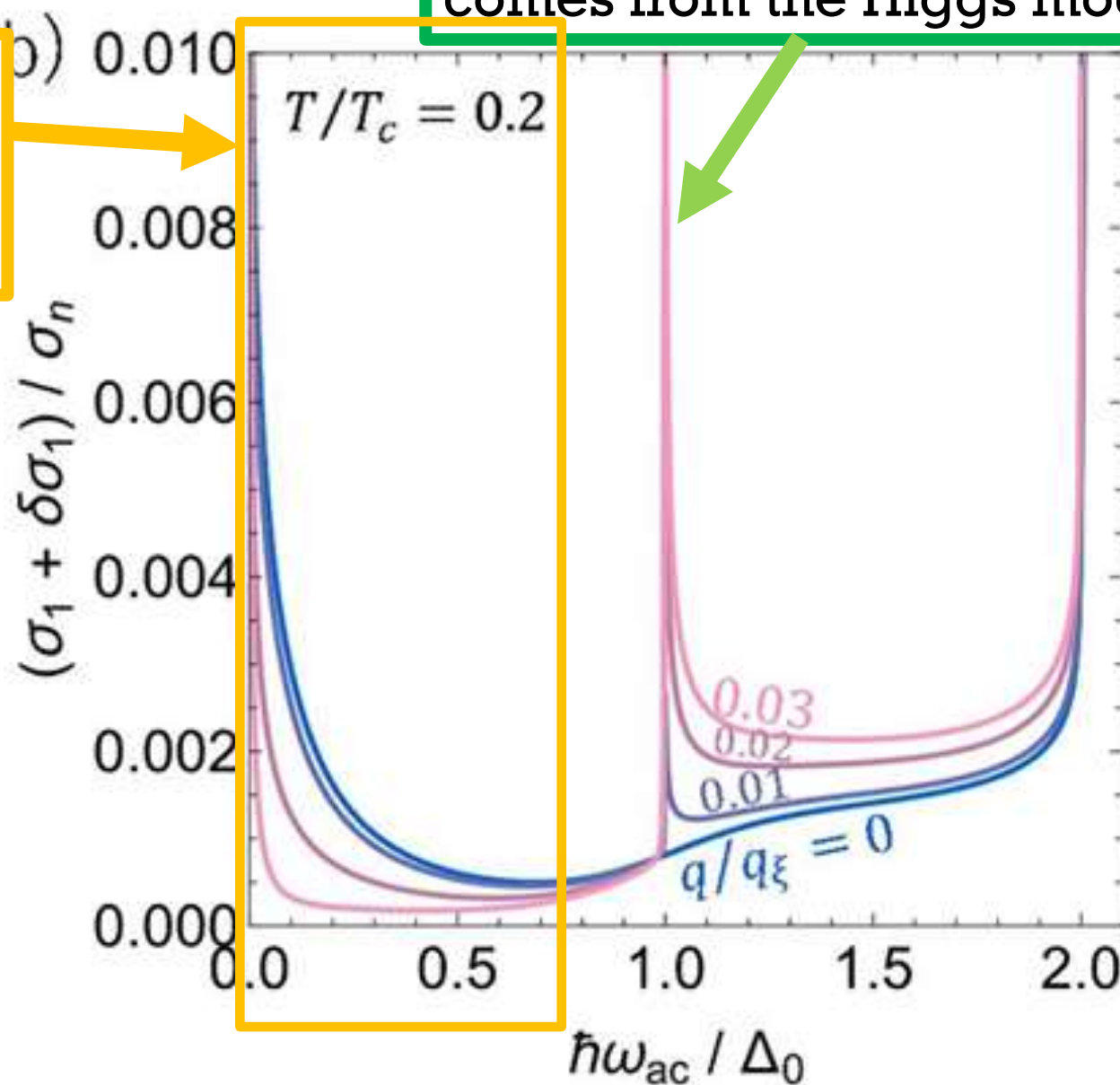
At lower frequencies, $\sigma_1 + \delta\sigma_1$ is a decreasing function of the ac amplitude (q).



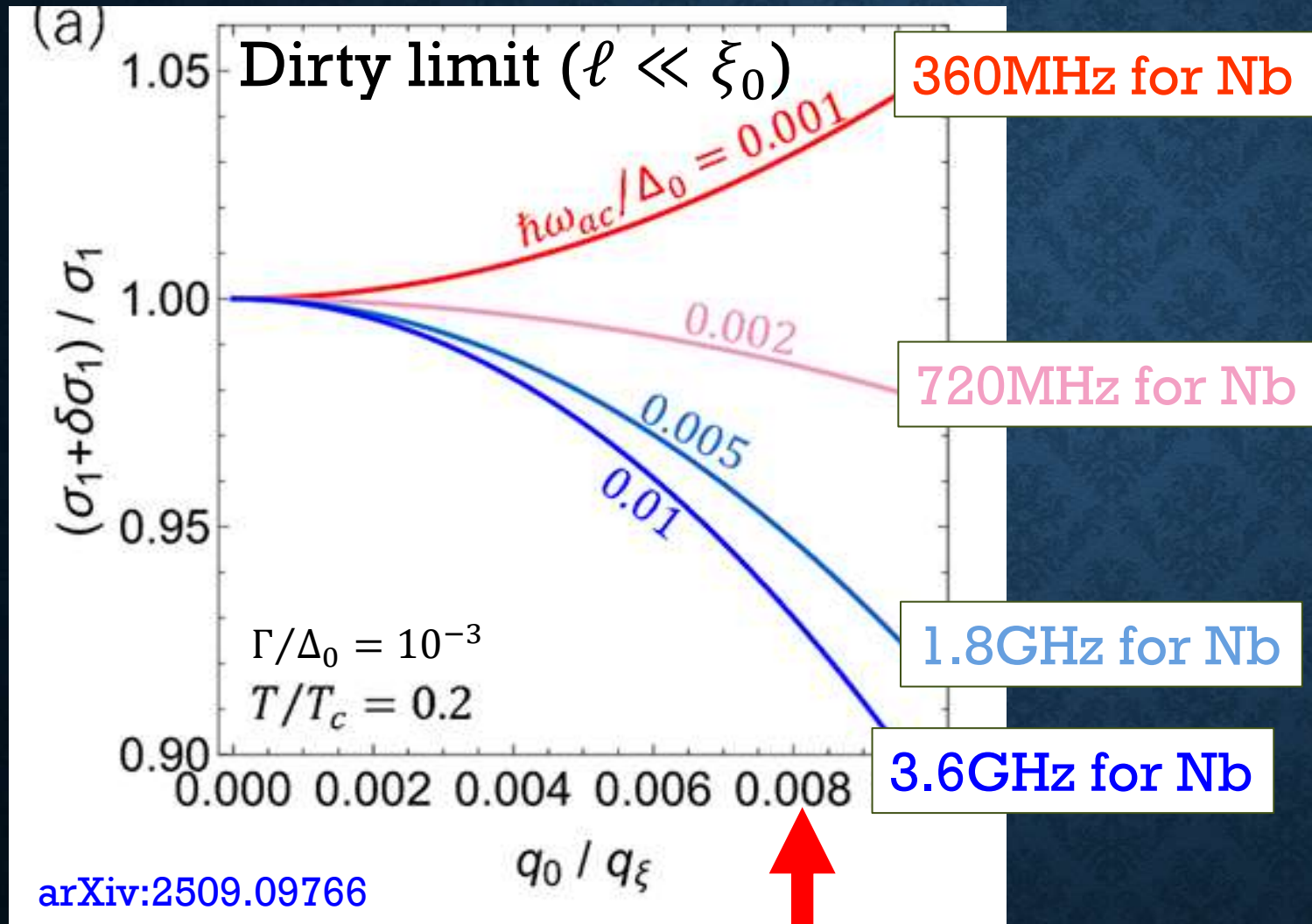
The AAA term is the dominant contributor at low frequencies



The resonance peak at Δ comes from the Higgs mode

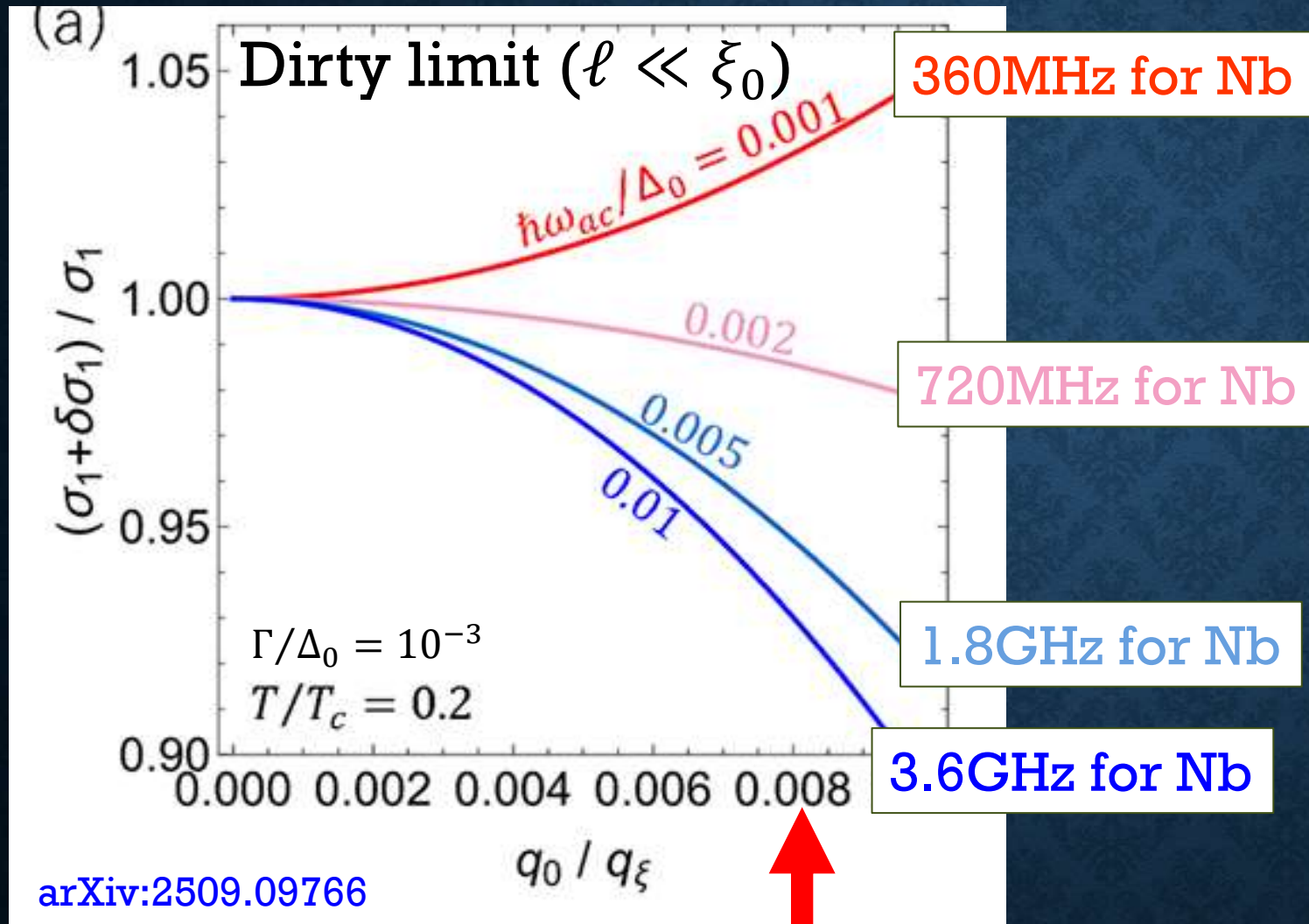


ac amplitude dependent dissipative conductivity



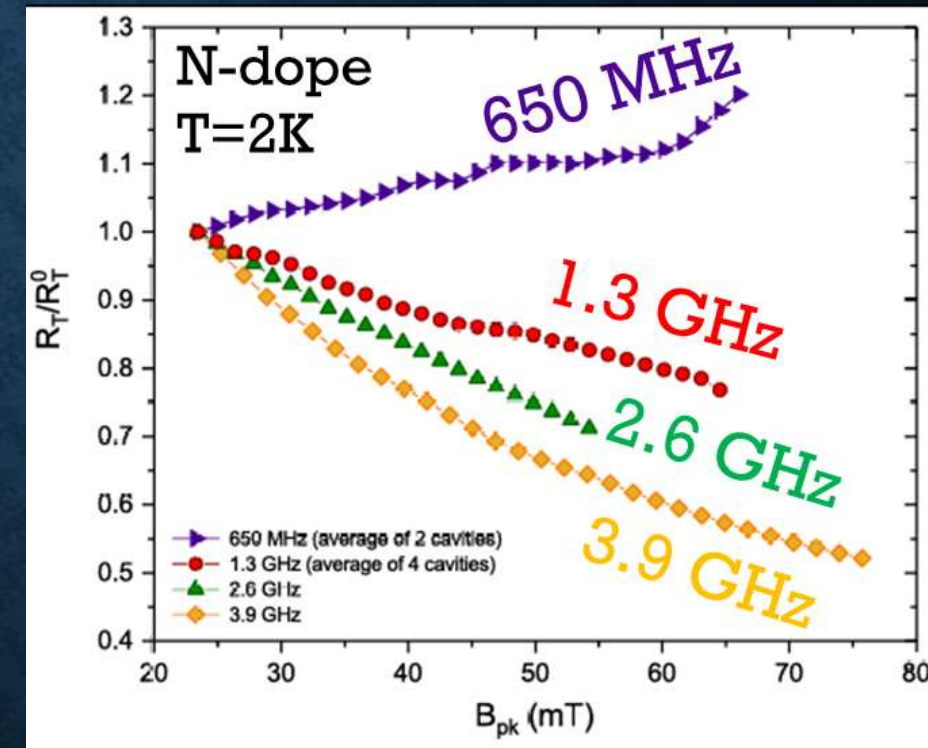
Note: this corresponds to a few (mT)

ac amplitude dependent dissipative conductivity



Experiments:
Moderately dirty ($\ell \sim \xi_0$)

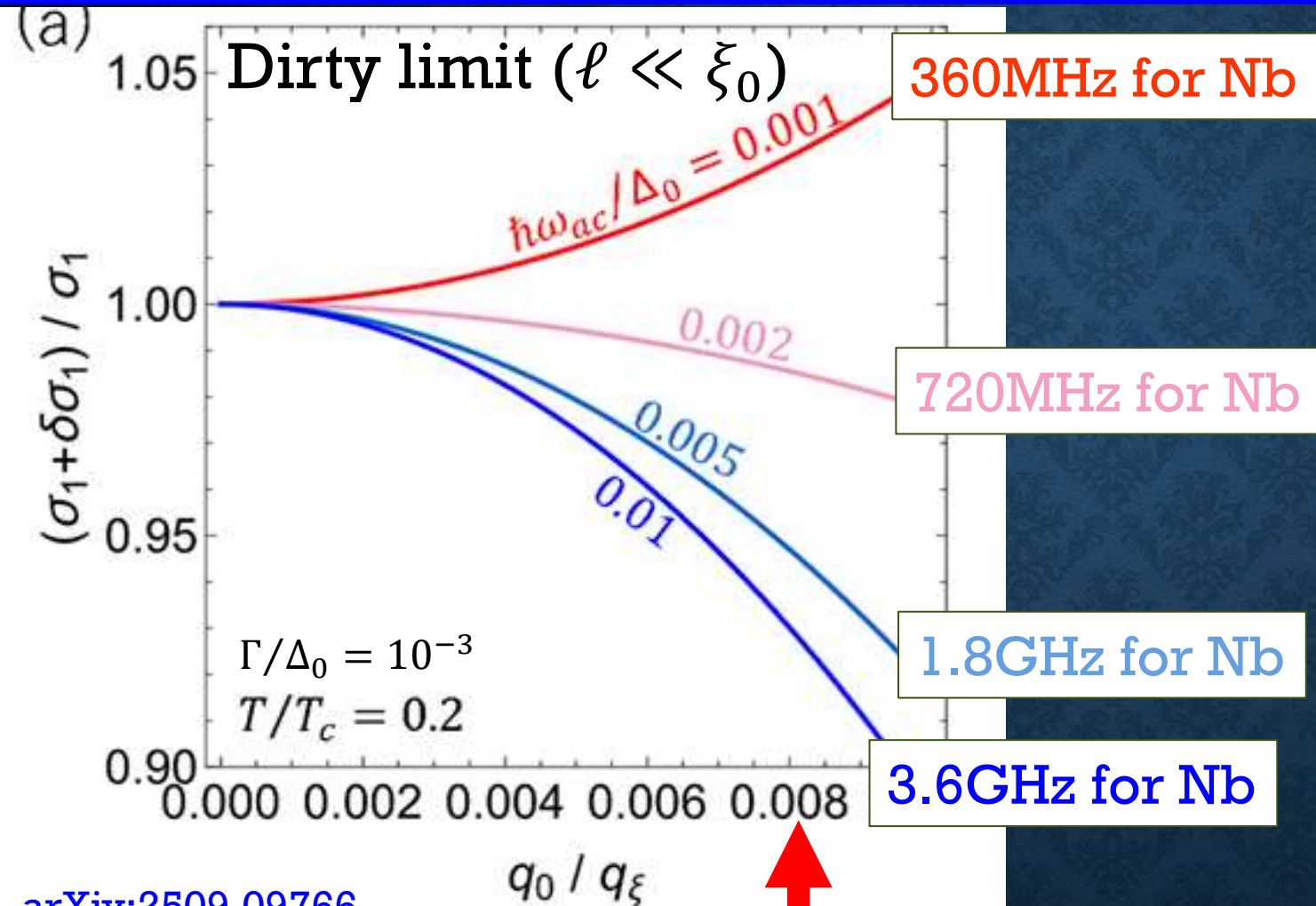
- M. Martinello et al., PRL **121**, 224801 (2018)
- P. Dhakal et al., PRAB **27**, 062001 (2024)



M. Martinello et al., Phys. Rev. Lett. **121**, 224801 (2018)
DOI: <https://doi.org/10.1103/PhysRevLett.121.224801>

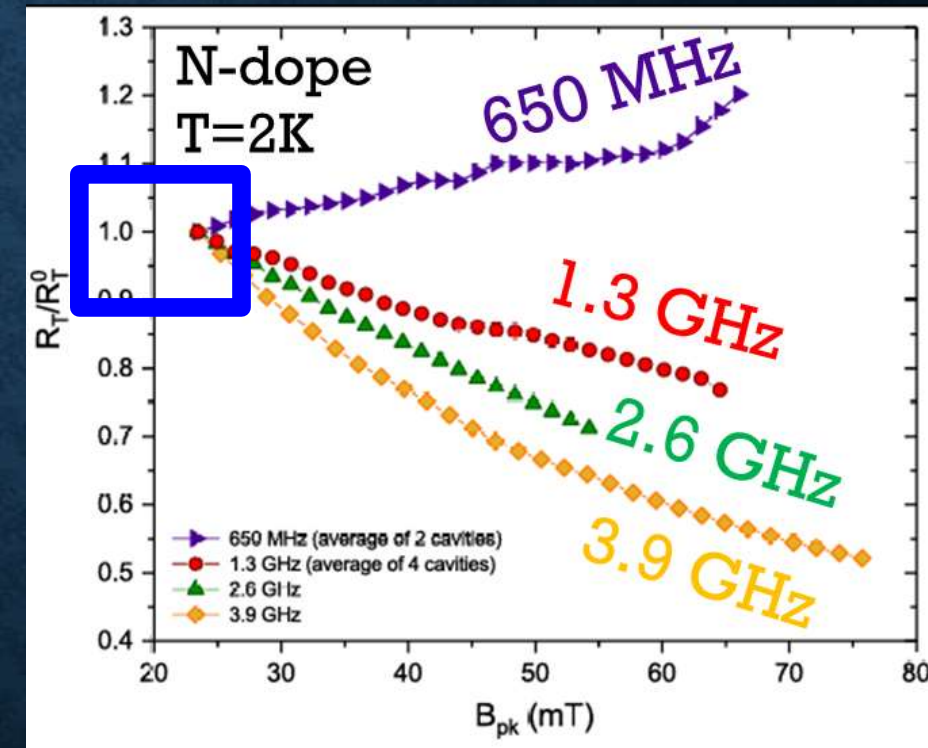
Note: this corresponds to a few (mT)

ac amplitude dependent dissipative conductivity



Experiments: Moderately dirty ($\ell \sim \xi_0$)

- M. Martinello et al., PRL **121**, 224801 (2018)
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Note: this corresponds to a few (mT)

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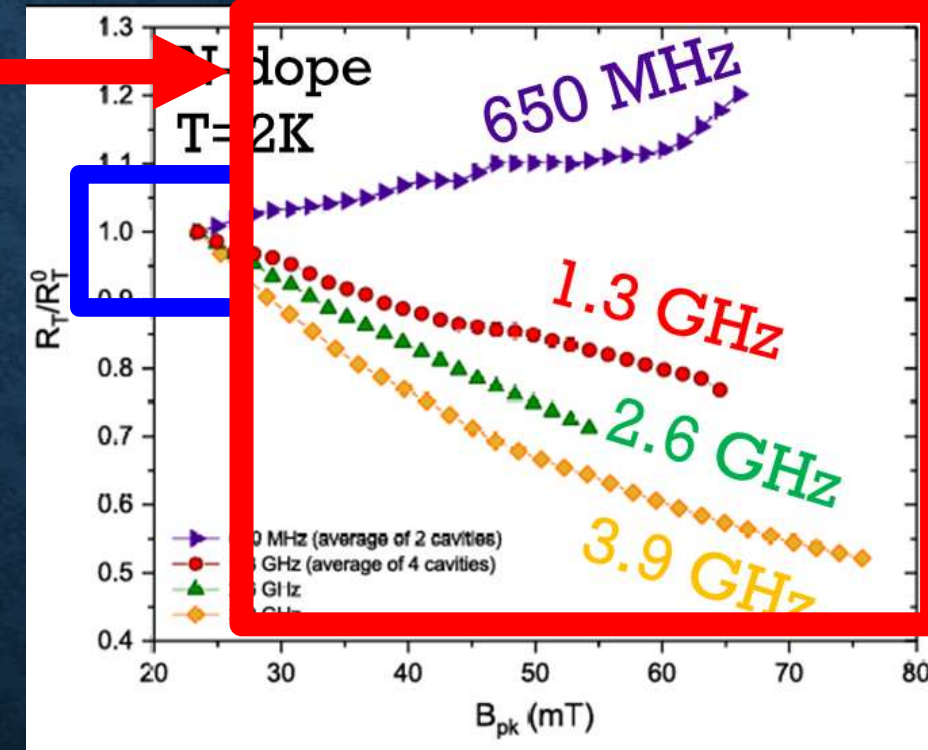
Experiments:

Moderately dirty ($\ell \sim \xi_0$)

- M. Martinello et al., PRL **121**, 224801 (2018)
- P. Dhakal et al., PRAB **27**, 062001 (2024)

To deal with **the red-framed region**, the theory must be extended nonperturbatively to capture the quasiparticle spectrum modified by strong AC currents (**DOS smearing**).

Gurevich, PRL **113**, 087001 (2014)
Kubo-Gurevich, PRB **100**, 064522 (2019)

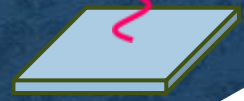
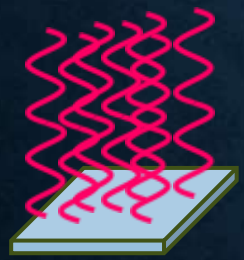


M. Martinello et al., Phys. Rev. Lett. **121**, 224801 (2018)
DOI: <https://doi.org/10.1103/PhysRevLett.121.224801>

Nonequilibrium
Dynamics
(e.g, strength of microwave)

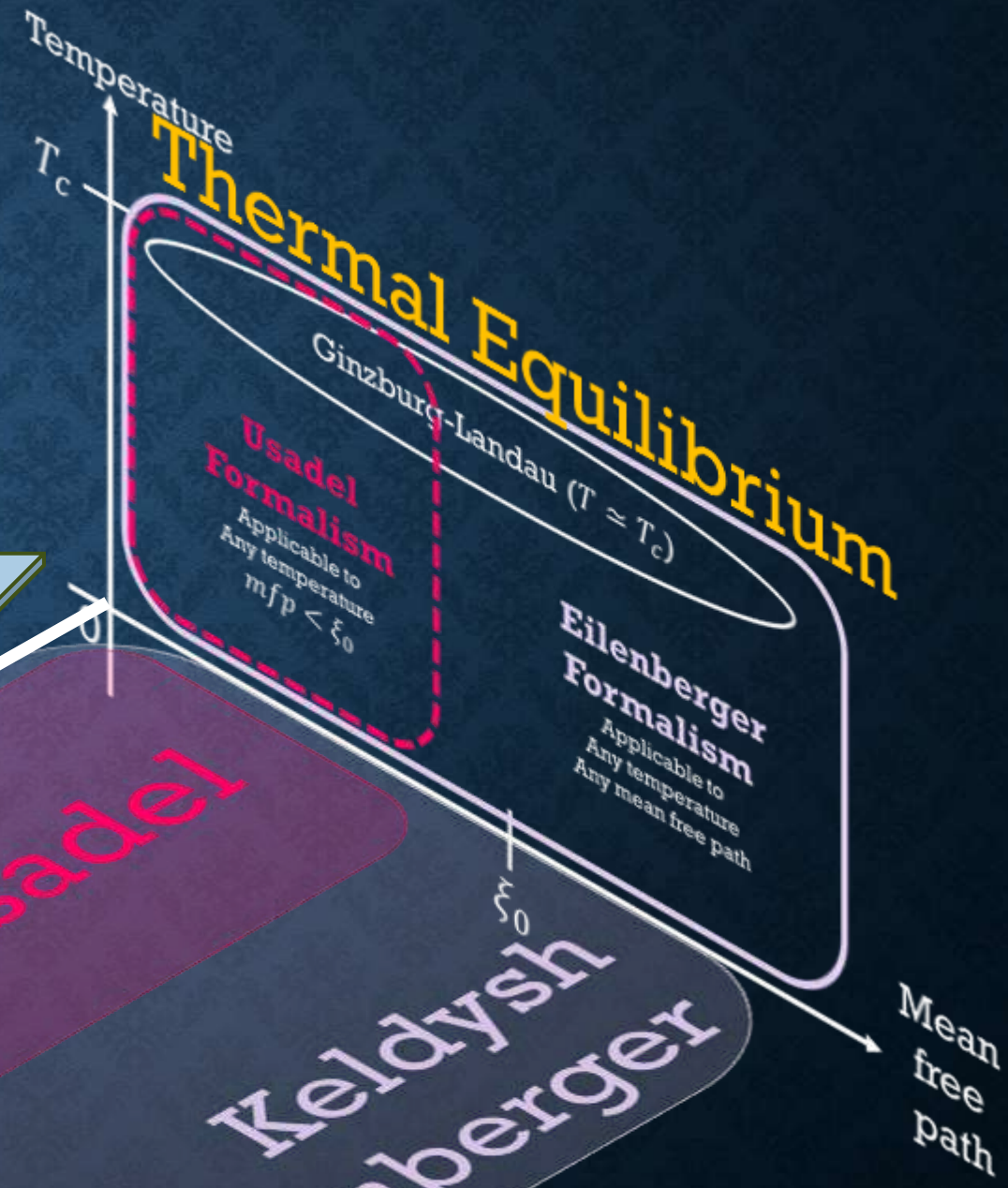
Kubo, arXiv:2509.09766
**Leading order
nonlinear
correction**

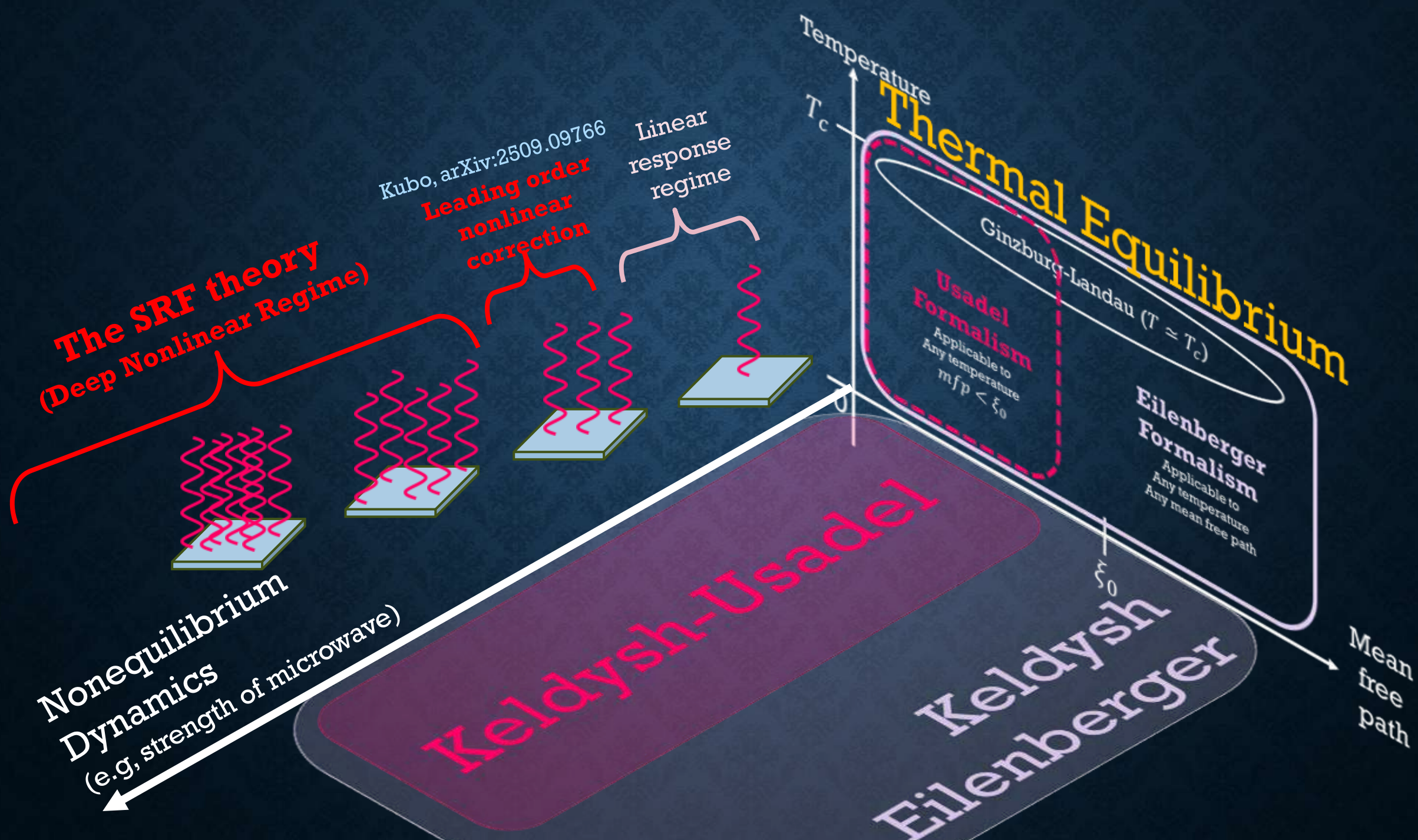
Linear
response
regime



Keldysh-Usadel

Keldysh
Eilenberger





T. Kubo, arXiv:2509.09766 (2025)

Nonperturbative treatment
(DOS smearing et al.)

Gurevich, PRL 113, 087001 (2014)
Kubo-Gurevich, PRB 100, 064522 (2019)

The SRF theory
(Deep Nonlinear Regime)

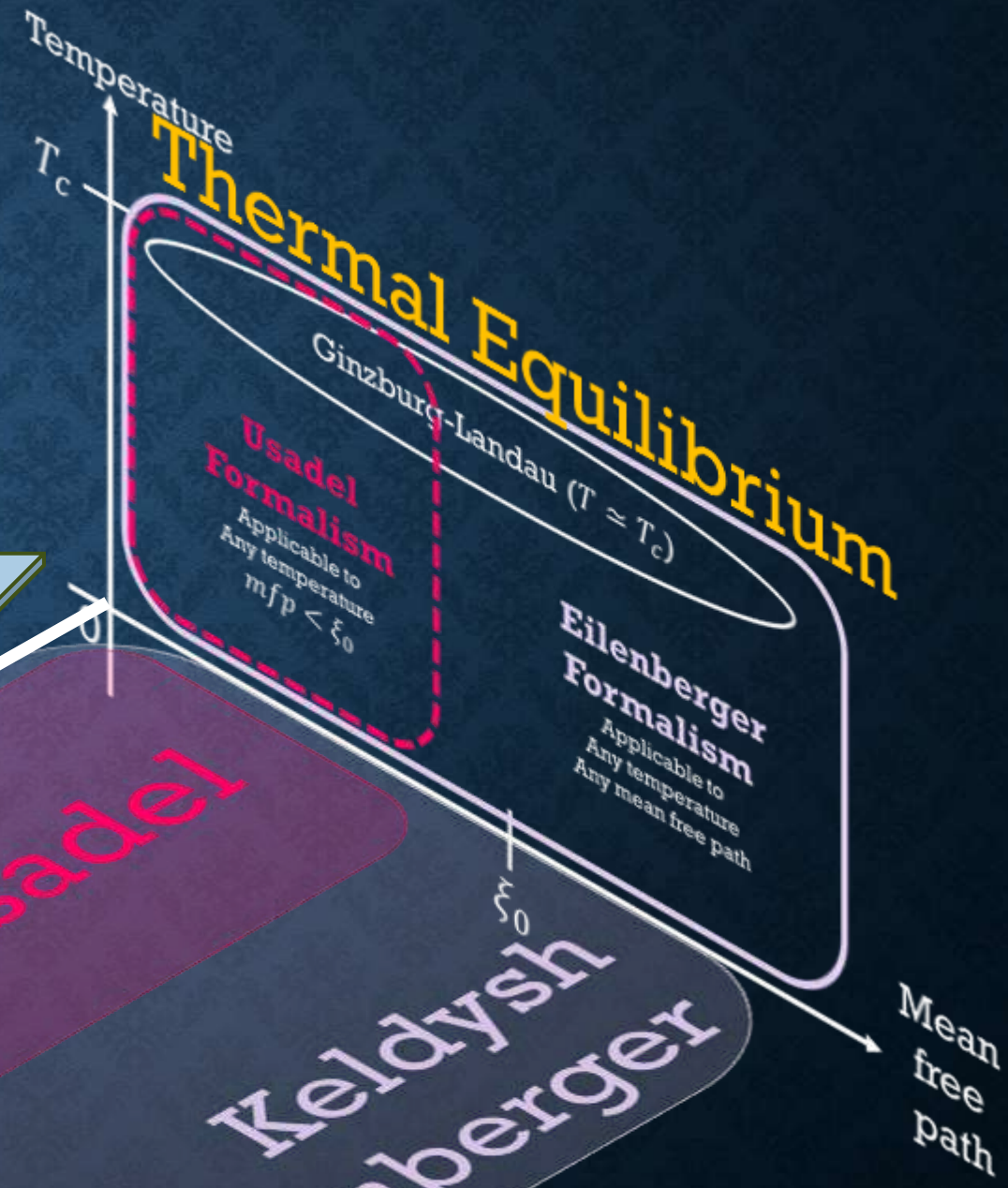
Kubo, arXiv:2509.09766
**Leading order
nonlinear
correction**

Linear
response
regime



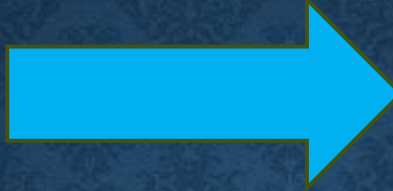
Nonequilibrium
Dynamics
(e.g, strength of microwave)

Keldysh-Usadel
**Keldysh
Eilenberger**

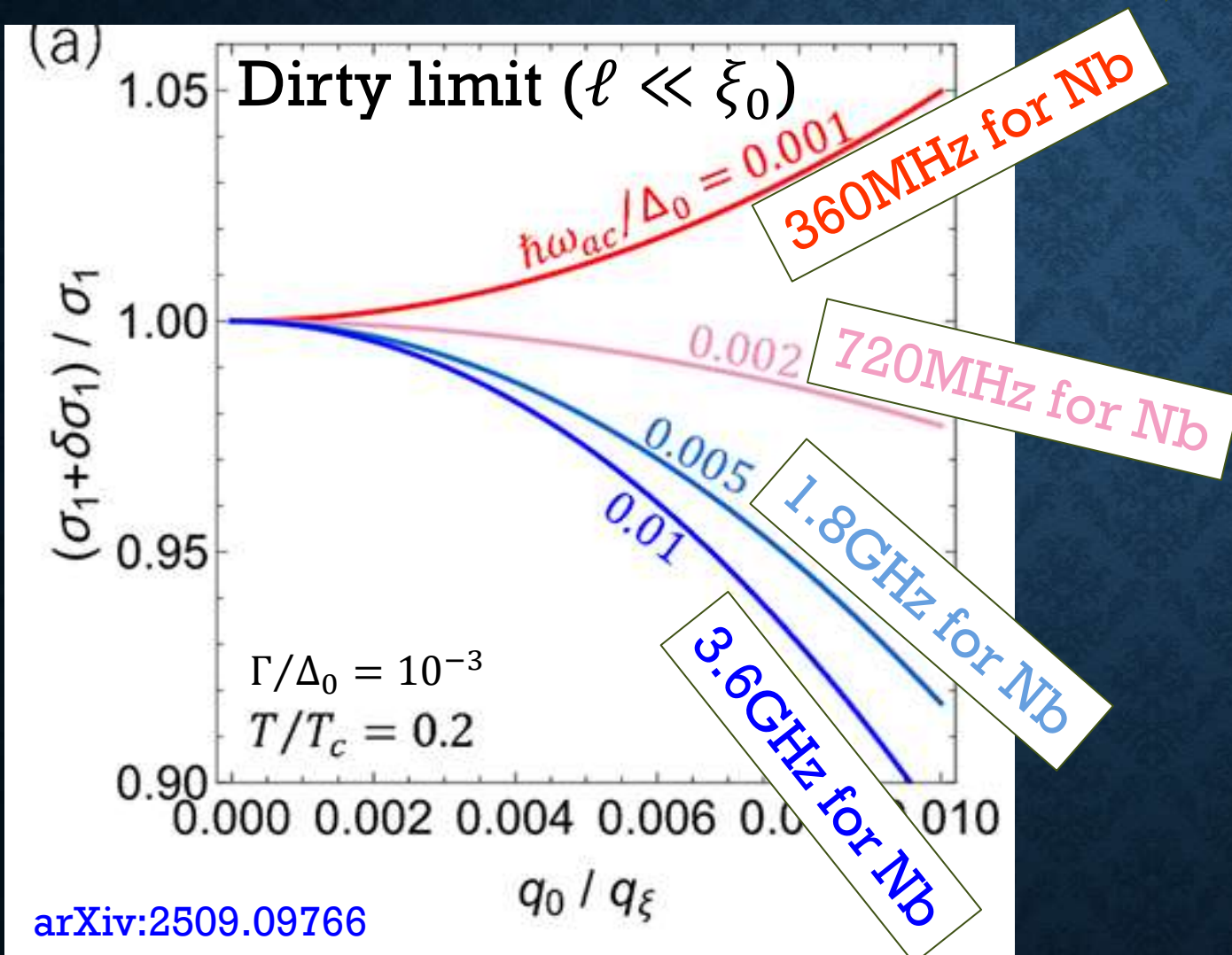


Theoretical prediction: effects of the inelastic scattering rate

$$\Gamma/\Delta_0 = 0.001$$

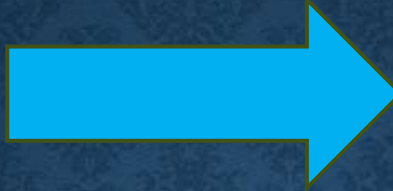


$$\Gamma/\Delta_0 = 0.01$$

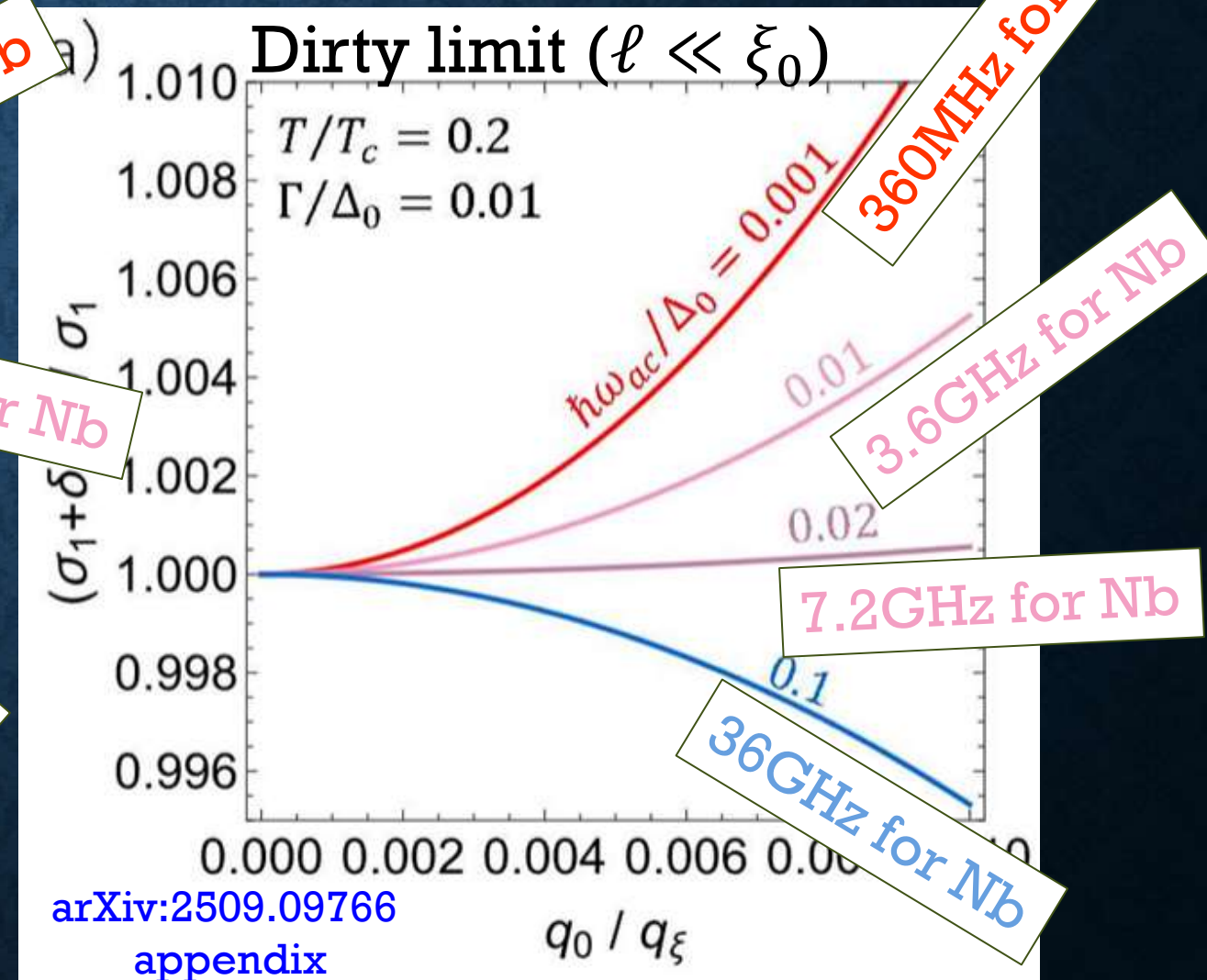
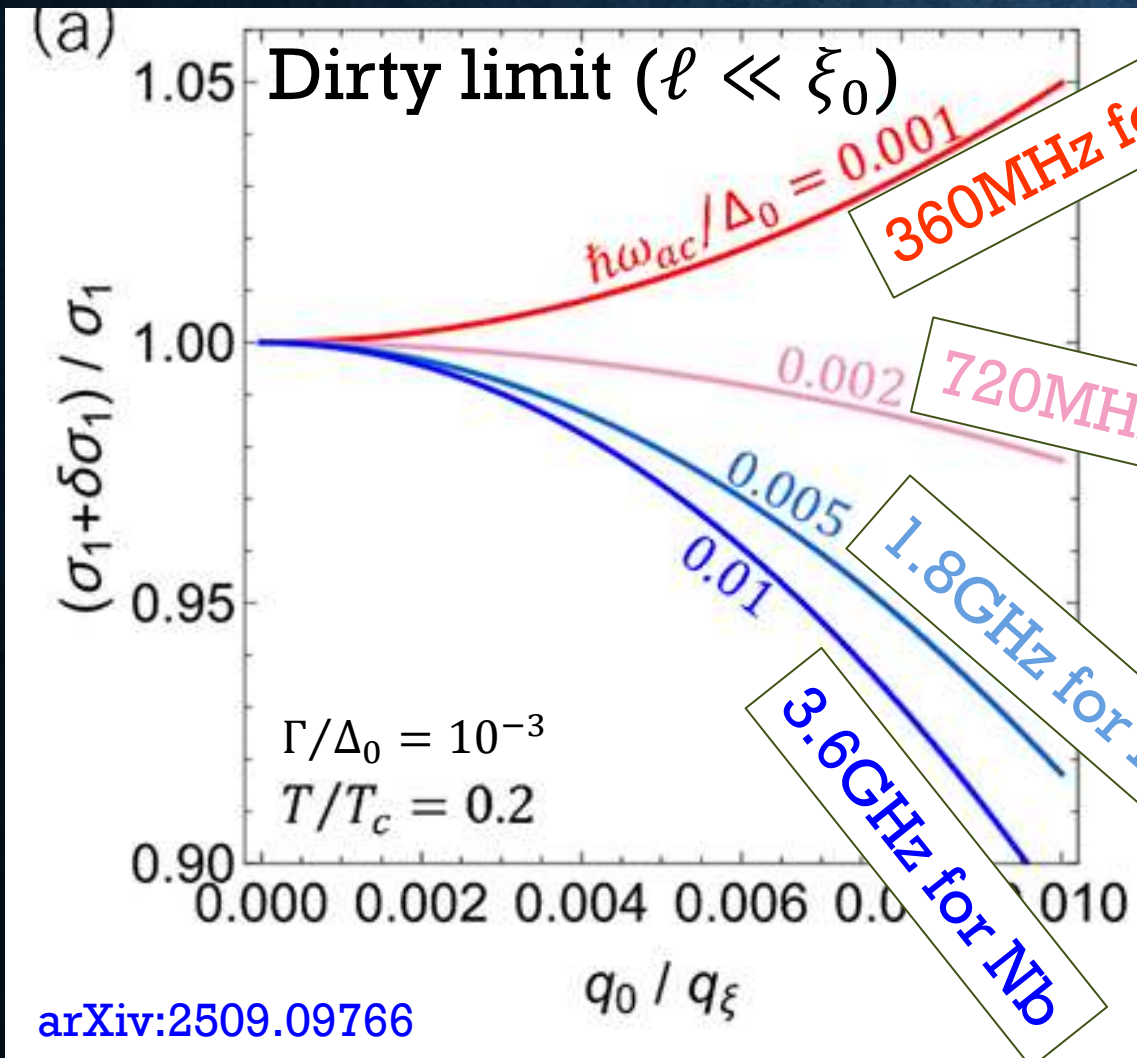


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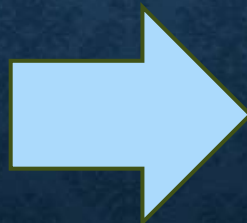
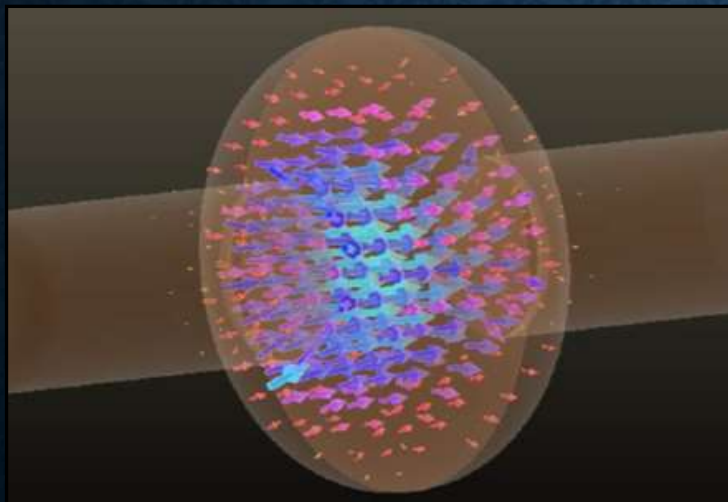


$$\Gamma/\Delta_0 = 0.01$$

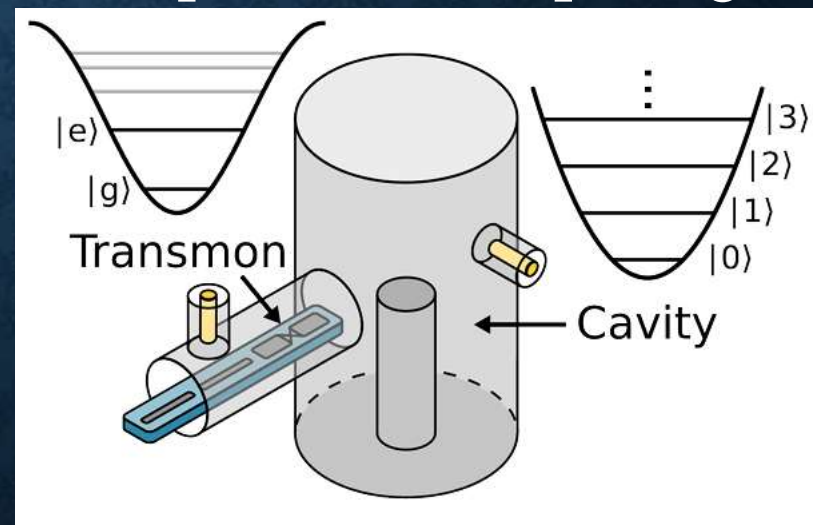


Quality factor Q_0 at lower temperatures ($T \ll 2\text{K}$)

Accelerator cavity



Quantum memory cavity for quantum computing



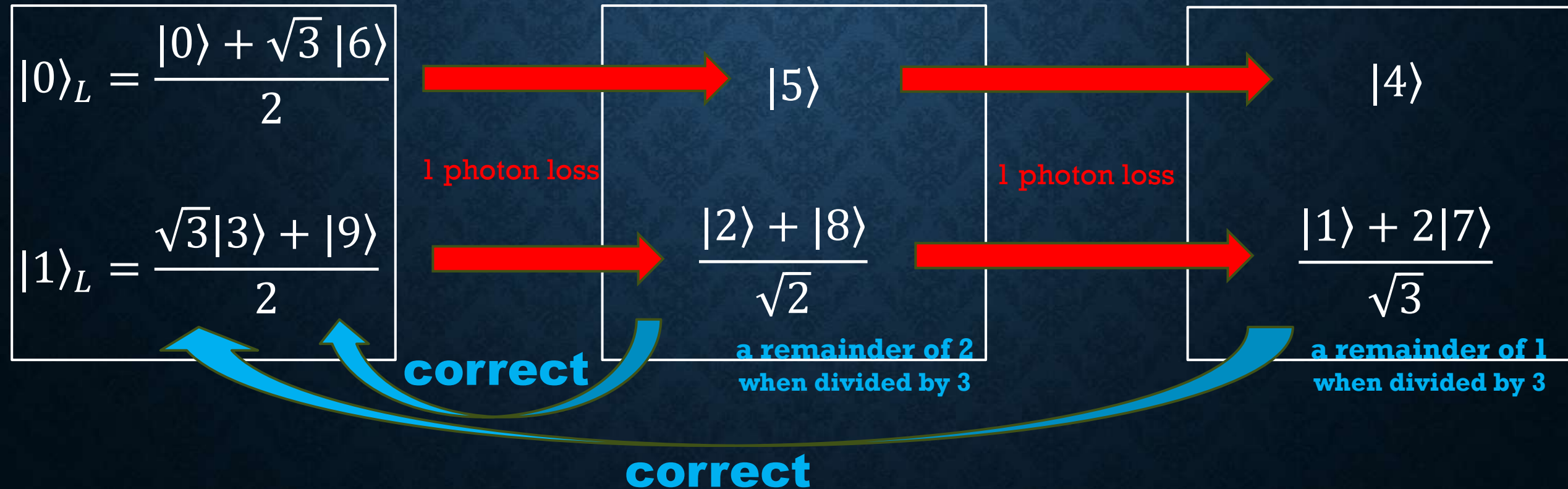
Brock et al., Nature **641**, 612 (2025)

<https://doi.org/10.1038/s41586-025-08899-y>

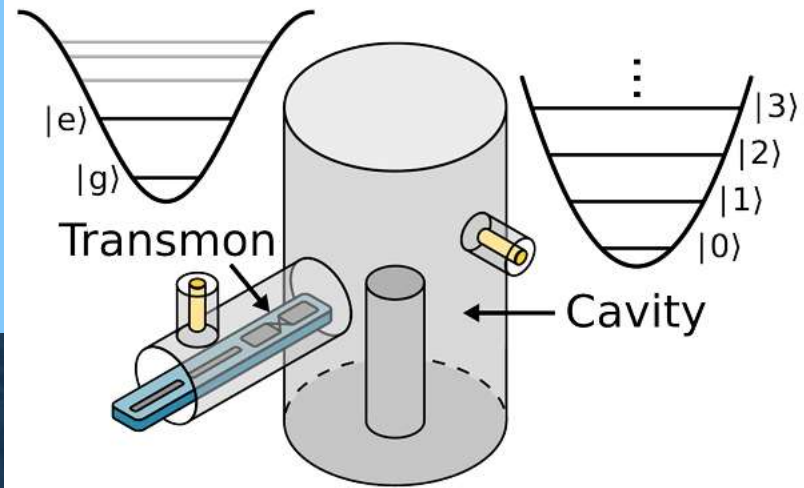
High-Q superconducting cavities allow us to store quantum information redundantly, enabling bosonic error correction.

Known to reduce the required number of qubits by one to two orders of magnitude.

e.g. binomial code



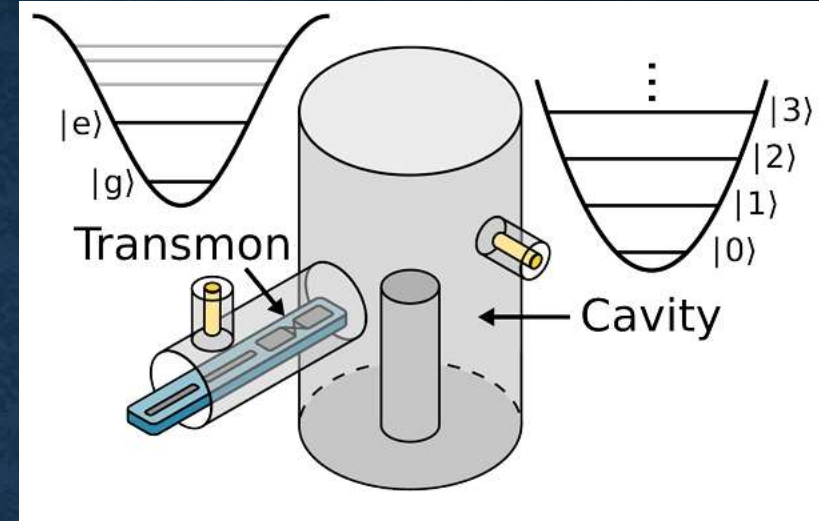
Brock et al., Nature **641**, 612 (2025)
<https://doi.org/10.1038/s41586-025-08899-y>



3D coaxial cavity progress

- Reagor et al., Phys. Rev. B **94**, 014506 (2016) Yale University
Aluminum,
 $Q = 7 \times 10^7$ (single photon level)

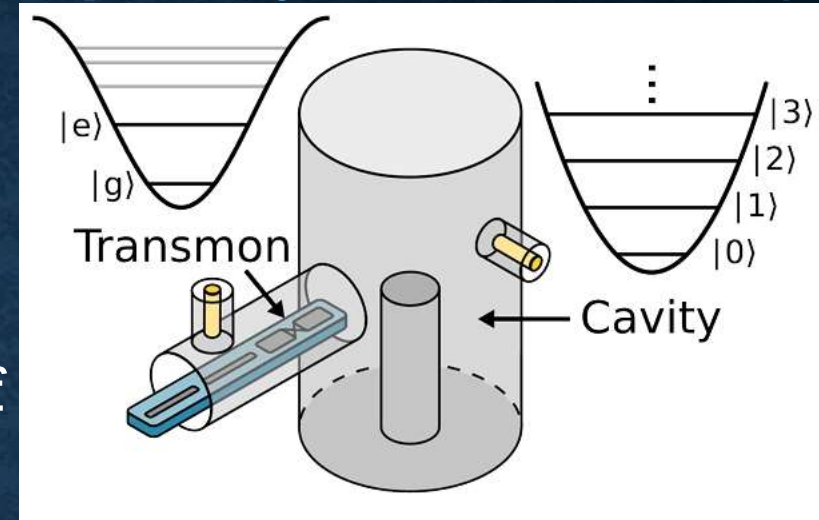
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Brock et al., Nature **641**, 612 (2025)
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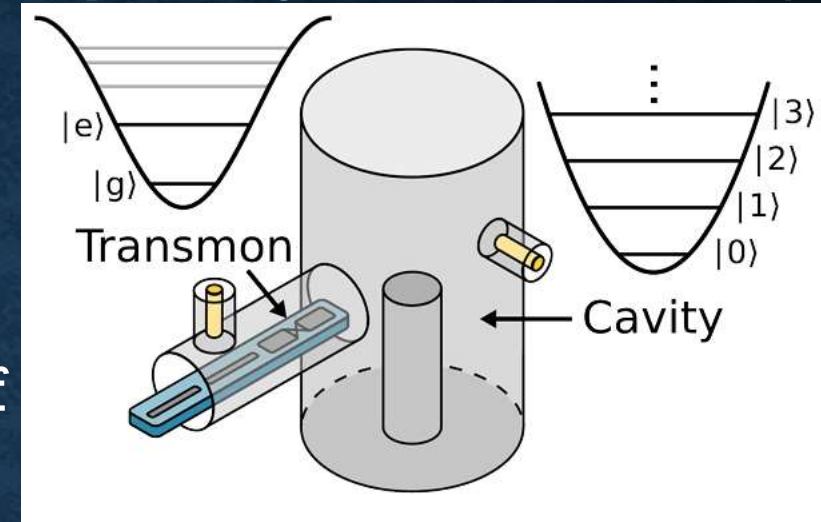
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Niobium (1:1:2 water buffered chemical polishing below 10°C),
 $Q = 1.4 \times 10^9$ (single photon level)



The world record (as of 2024)



Sometimes, I do experiments (BCP etc)

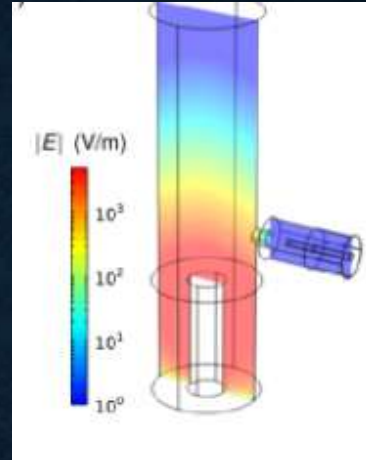


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T. Takenaka, T. Kubo et al.,
arXiv:2510.01819.



Nb coaxial cavity of **KEK-NTT** collaboration

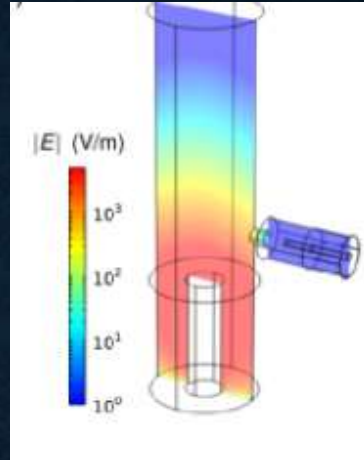


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T. Takenaka, T. Kubo et al.,
arXiv:2510.01819.



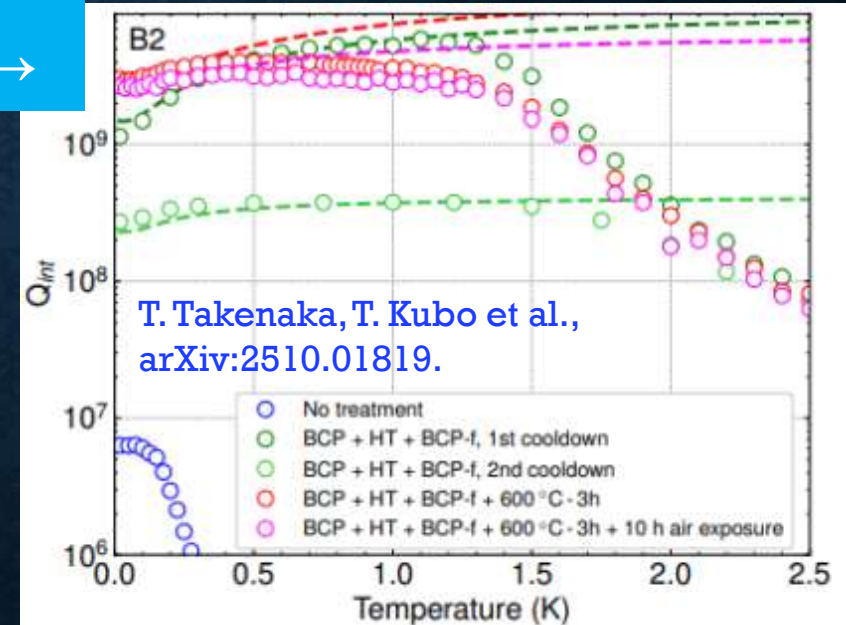
Nb coaxial cavity of **KEK-NTT** collaboration



**We broke
the world record!!!→**

Corresponding to
photon lifetime of
~0.1 sec.

20mK & Single photon level

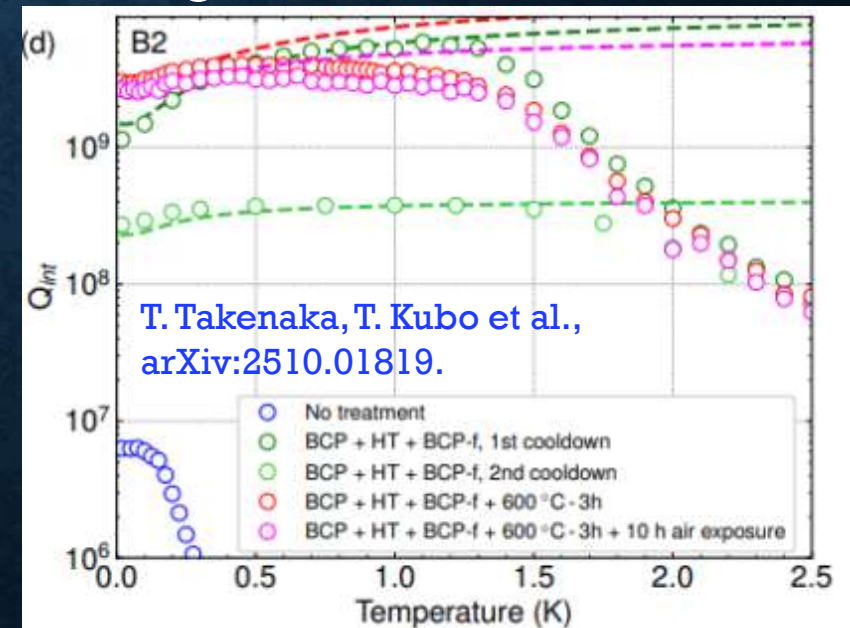
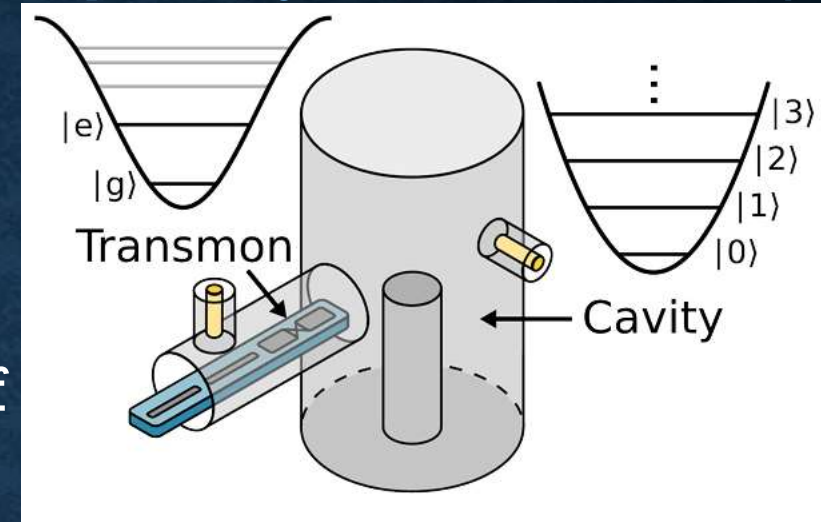


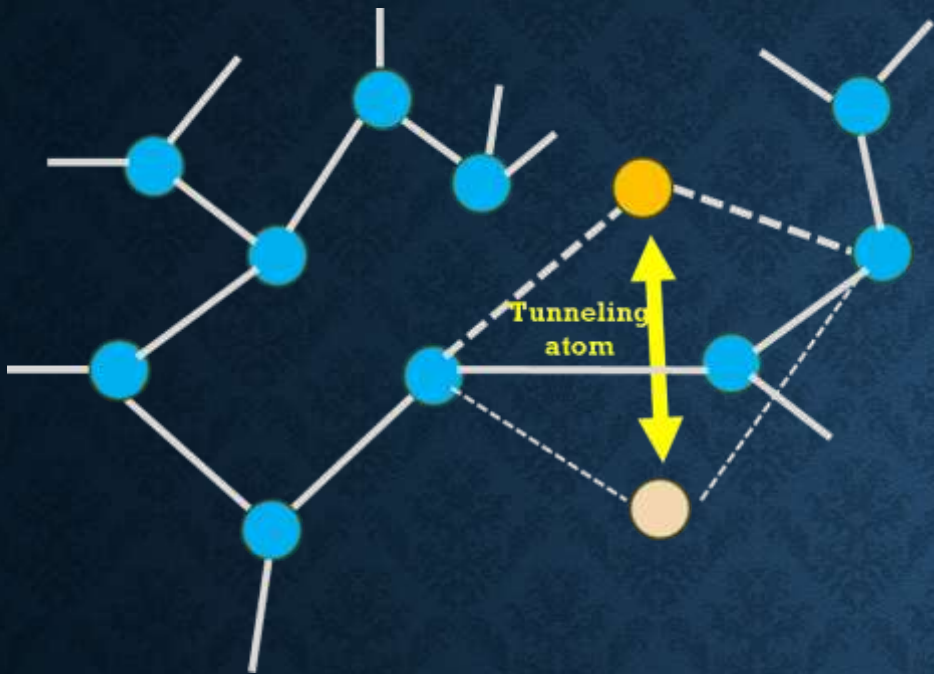
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- T. Takenaka, T. Kubo et al., arXiv:2510.01819 (2025) KEK&NTT
**Niobium (1:1:1 BCP at 25°C + 900°C anneal
+ 1:1:1 BCP flush at 25°C + 600°C anneal)**
 $Q=3 \times 10^9$ (single photon level)

The world record

Brock et al., Nature **641**, 612 (2025)
<https://doi.org/10.1038/s41586-025-08899-y>

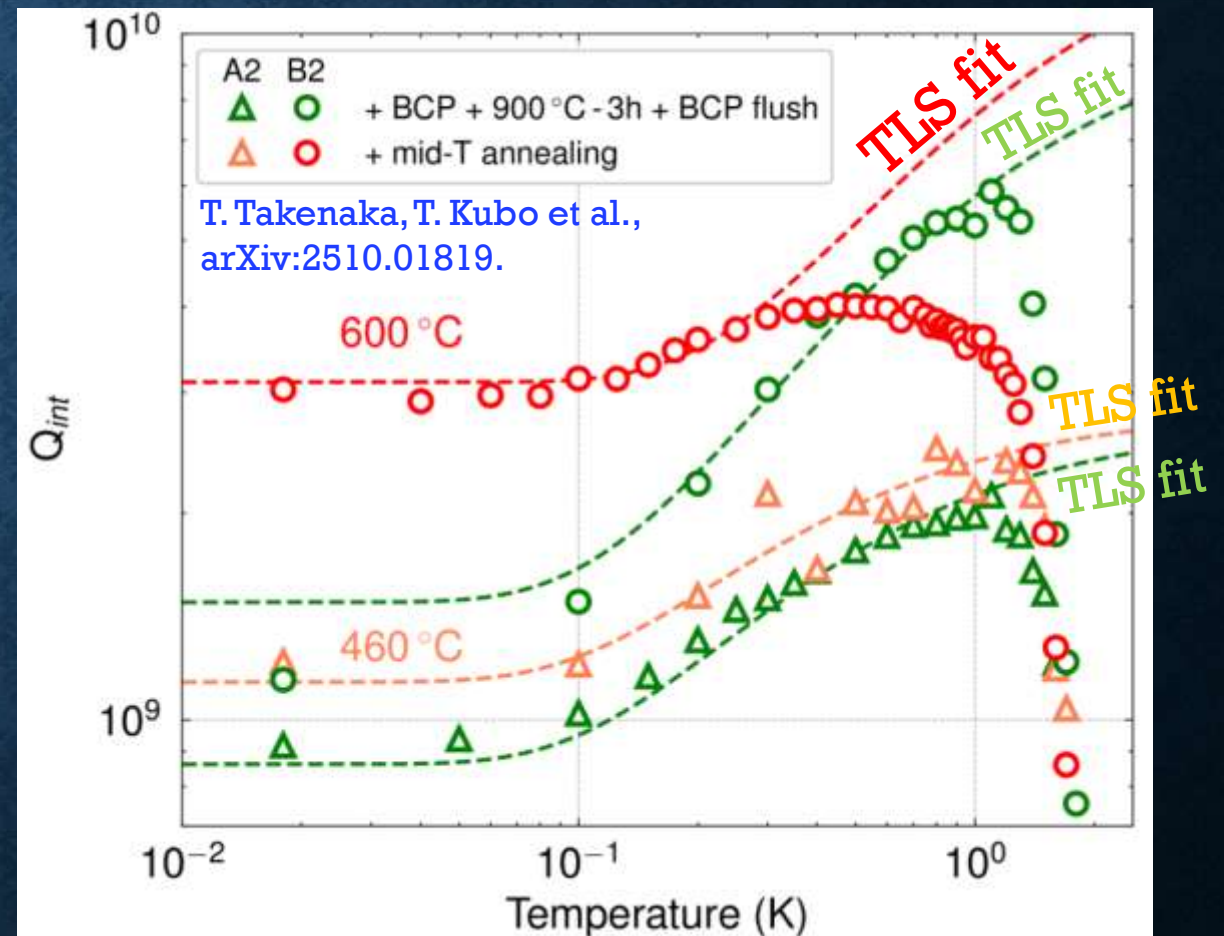




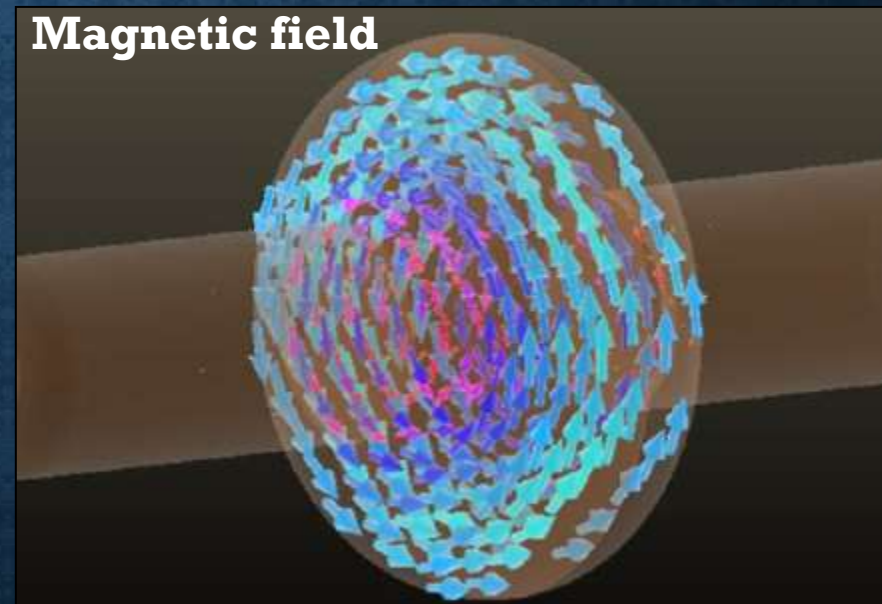
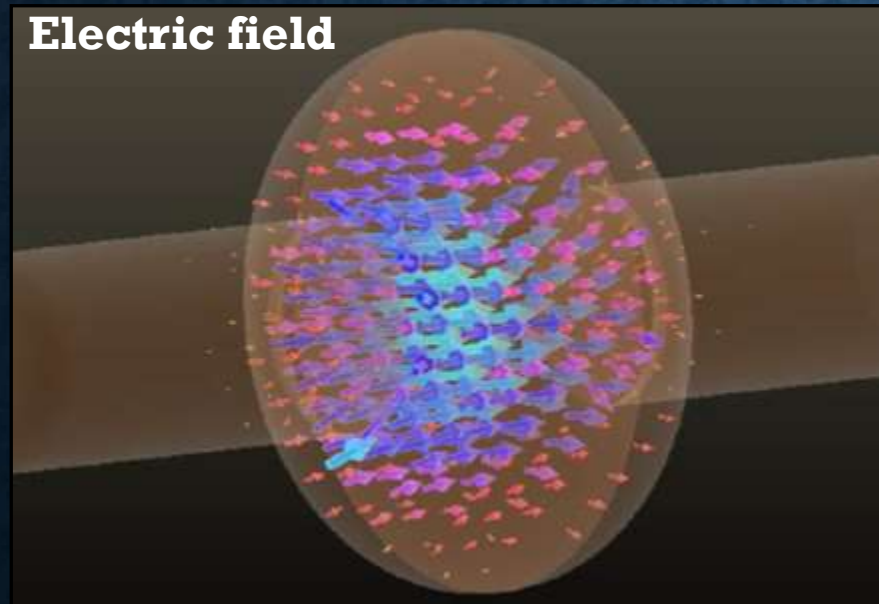
Even in this record-high-Q cavity, TLS defects still limit the quality factor.

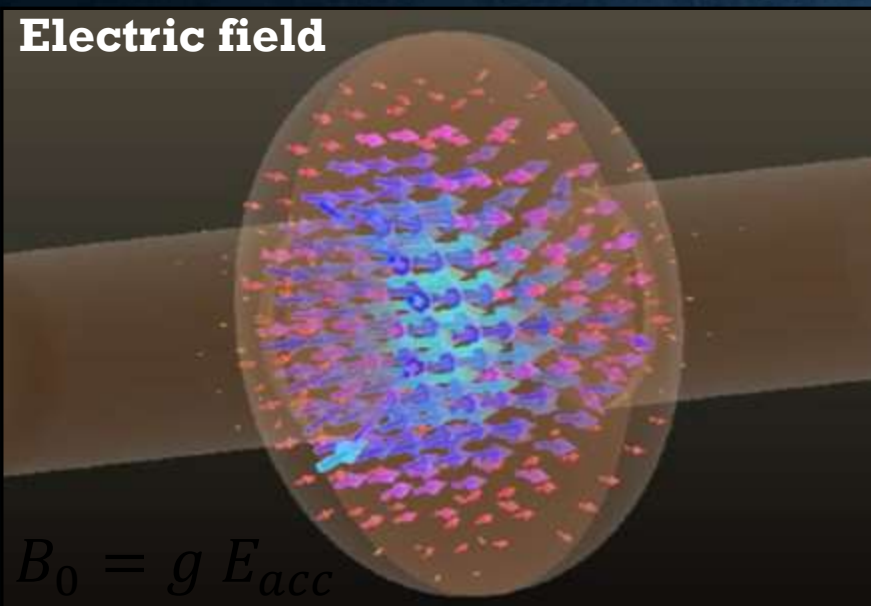
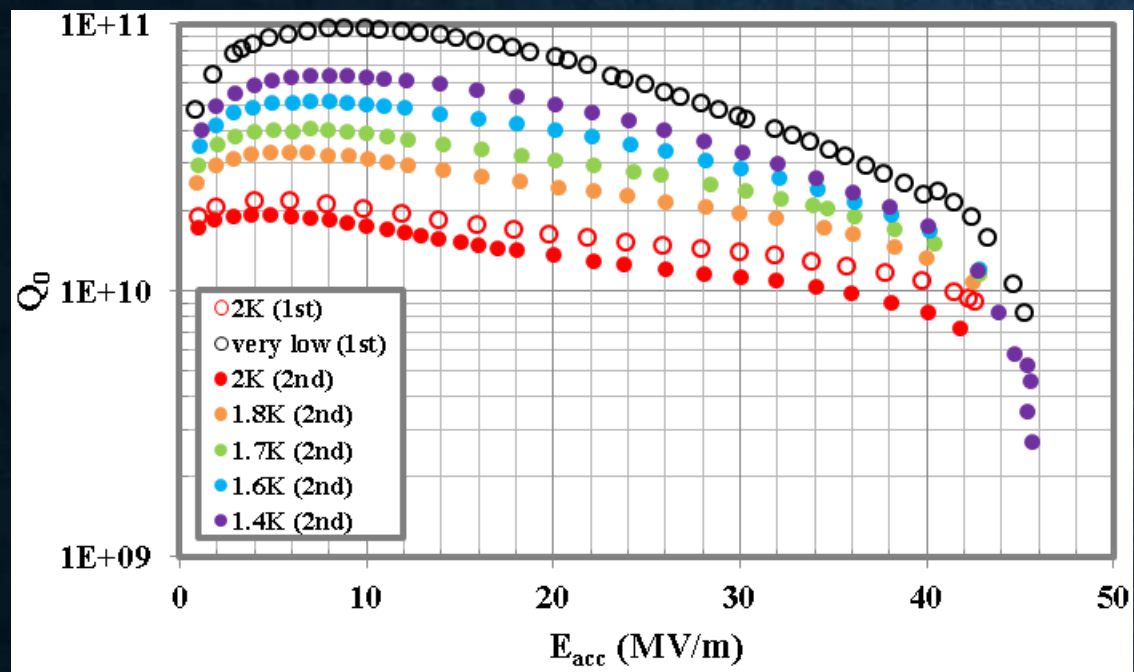
Understanding TLS physics and developing engineering methods to mitigate them are necessary to realize ultra-high-Q cavities for accelerators and quantum computing.

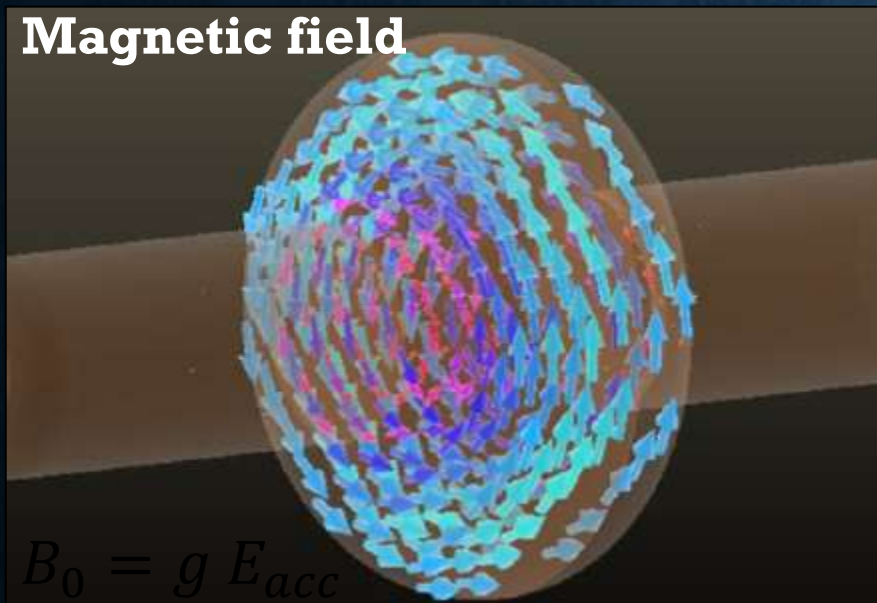
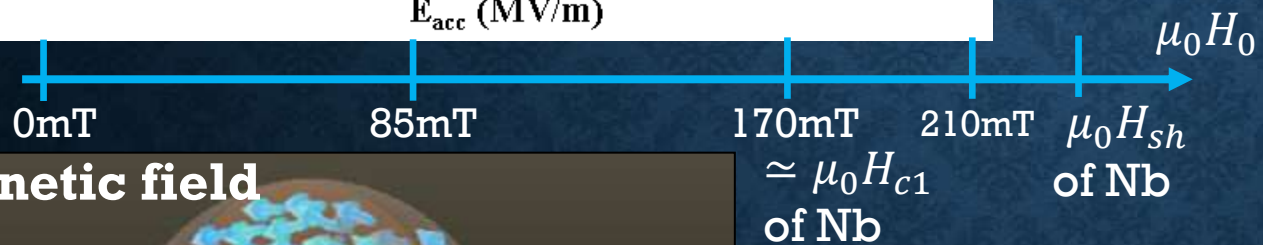
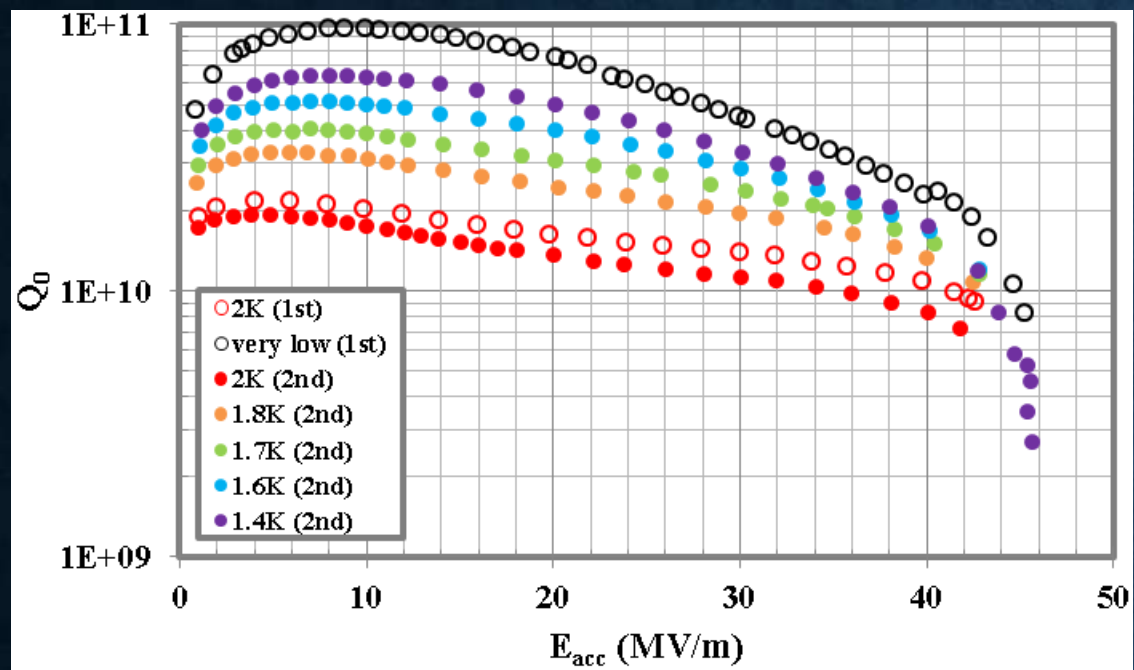
Maybe, my next project.

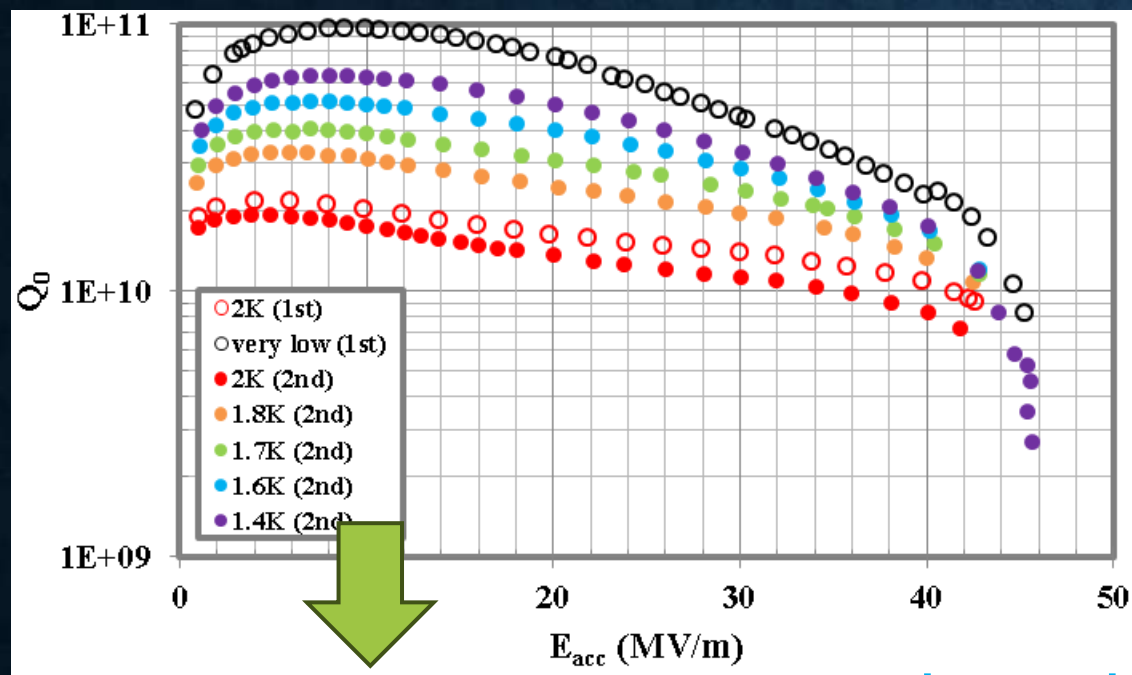


Gradient E_{acc} (microwave amplitude)

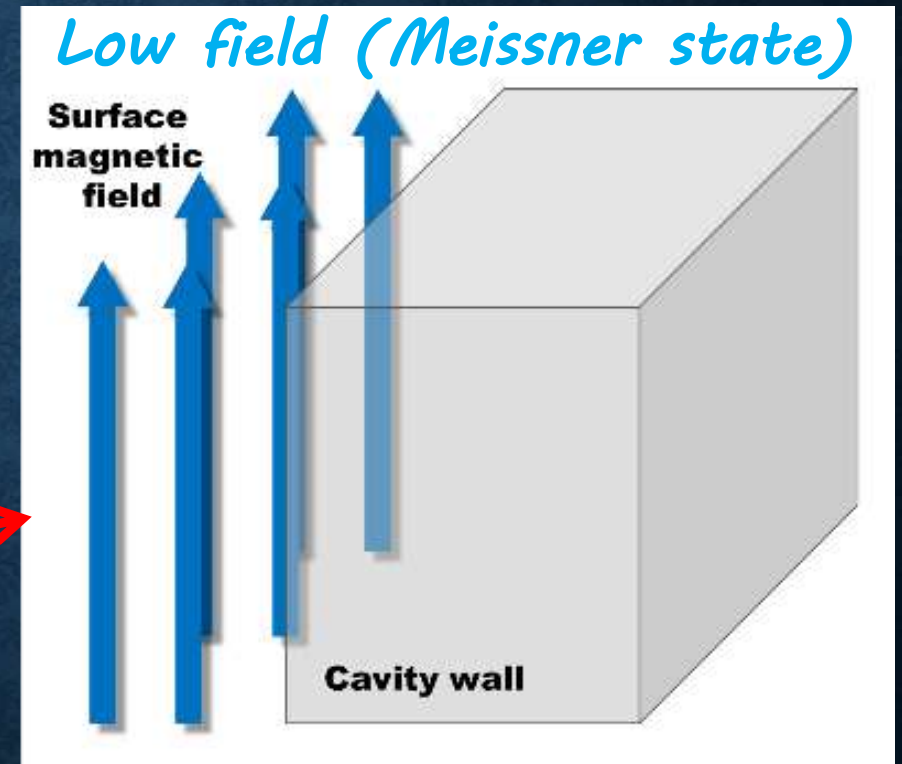
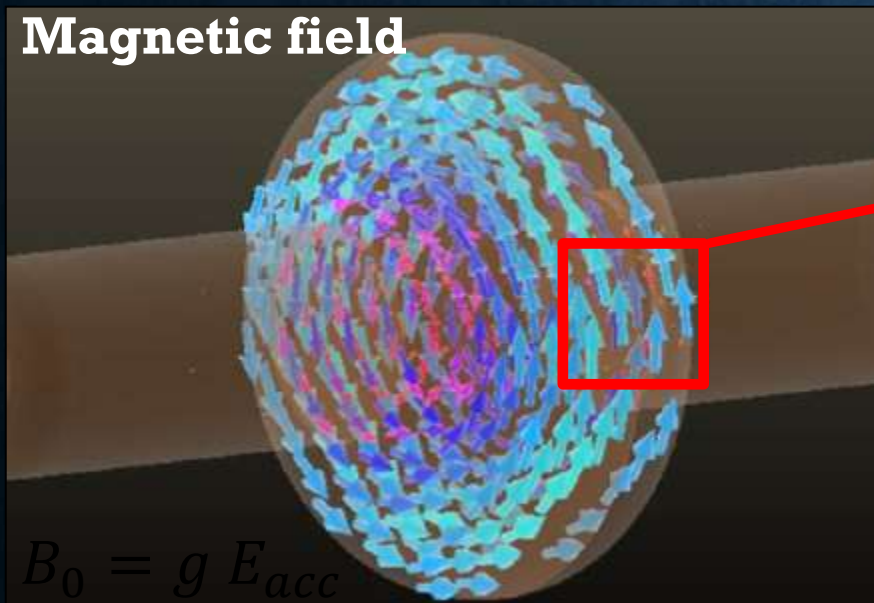


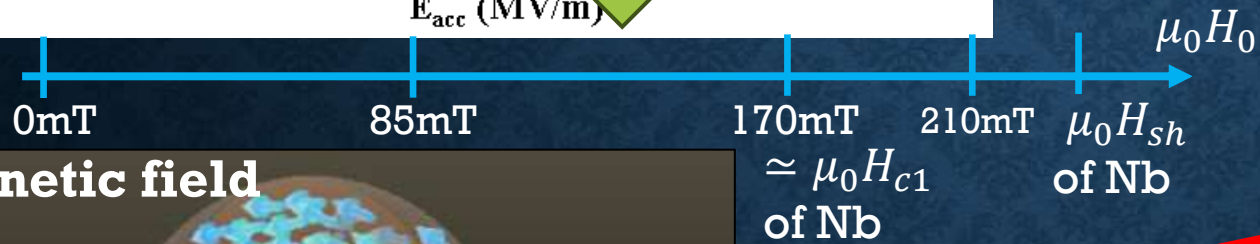
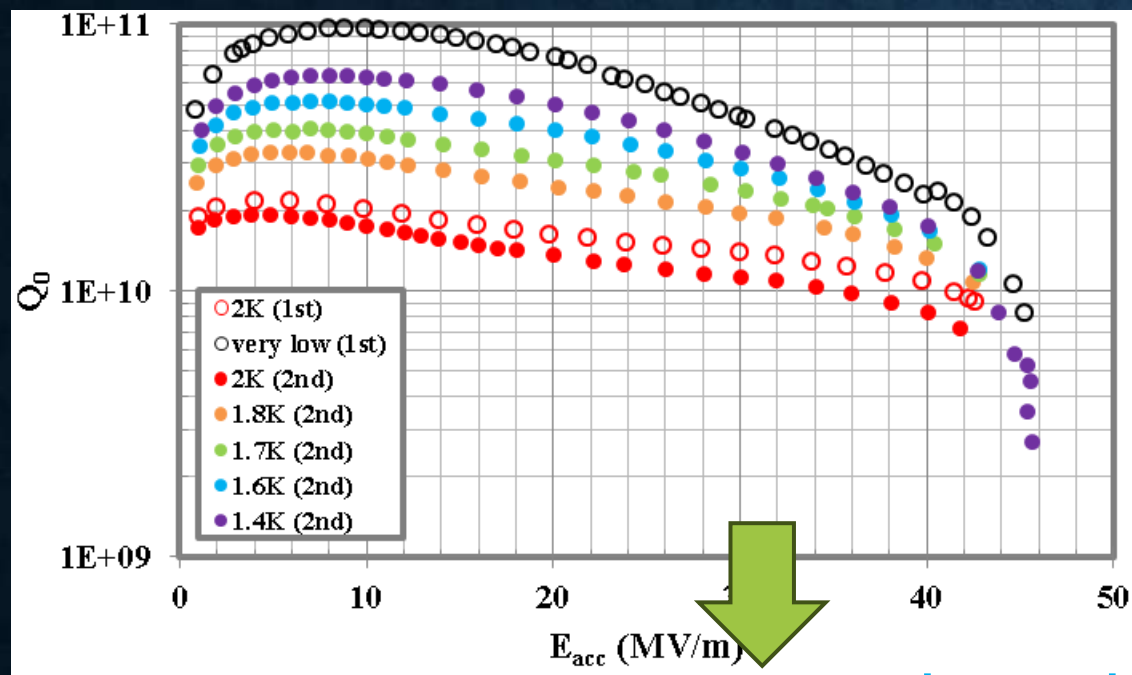




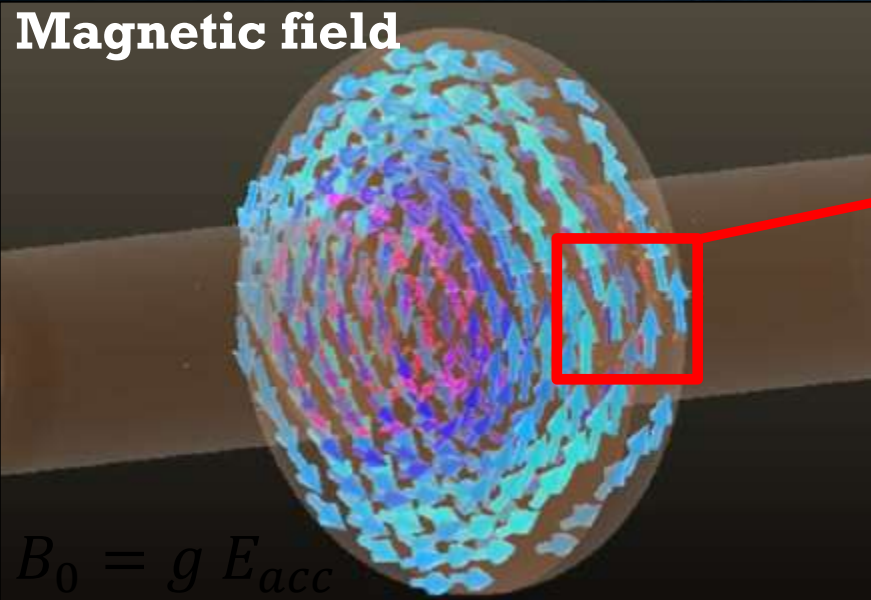


($\square$ $\square$) good

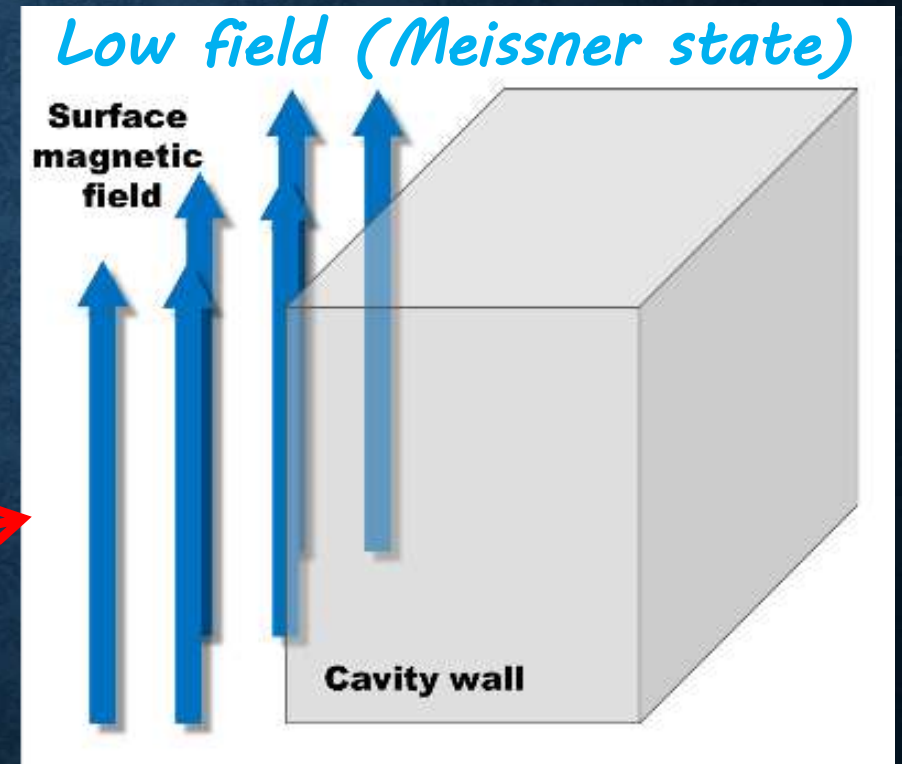


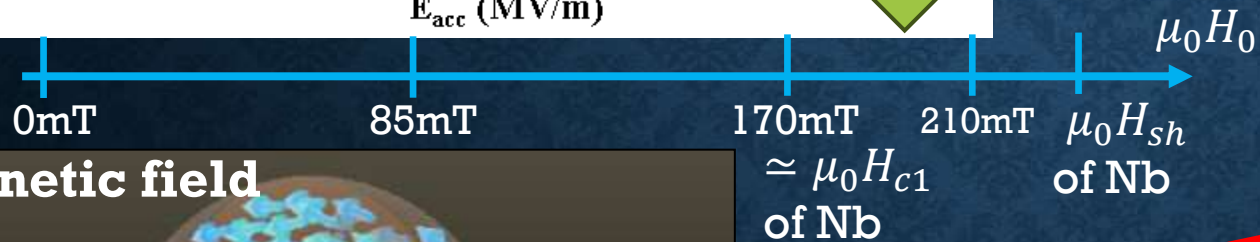
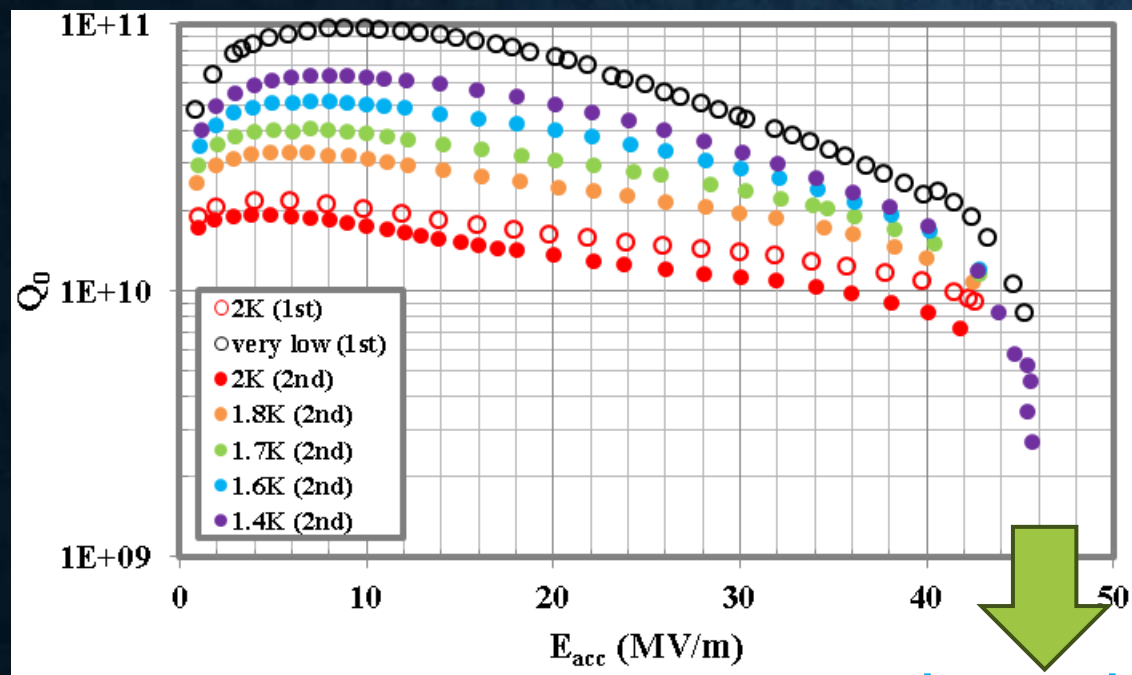


Magnetic field

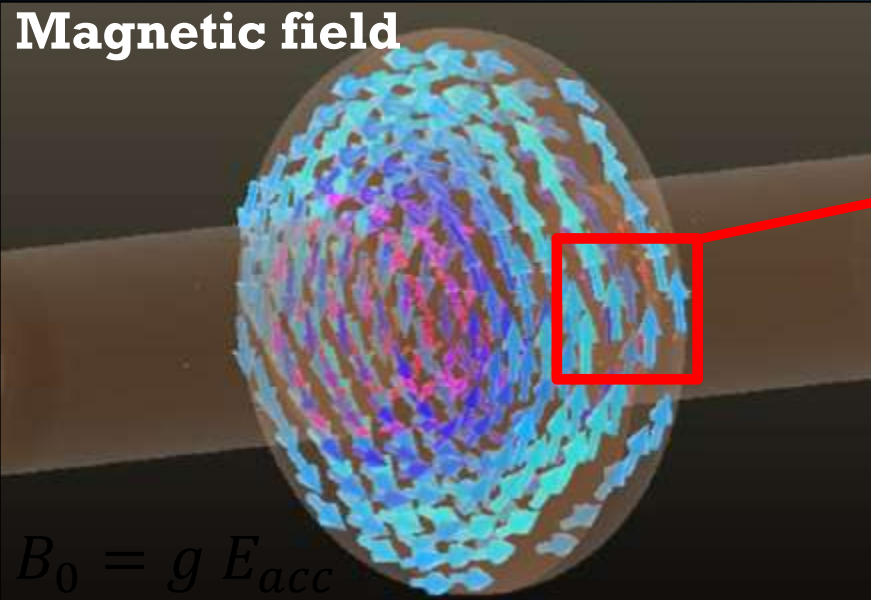


(Γ $\square$ $\square$) good

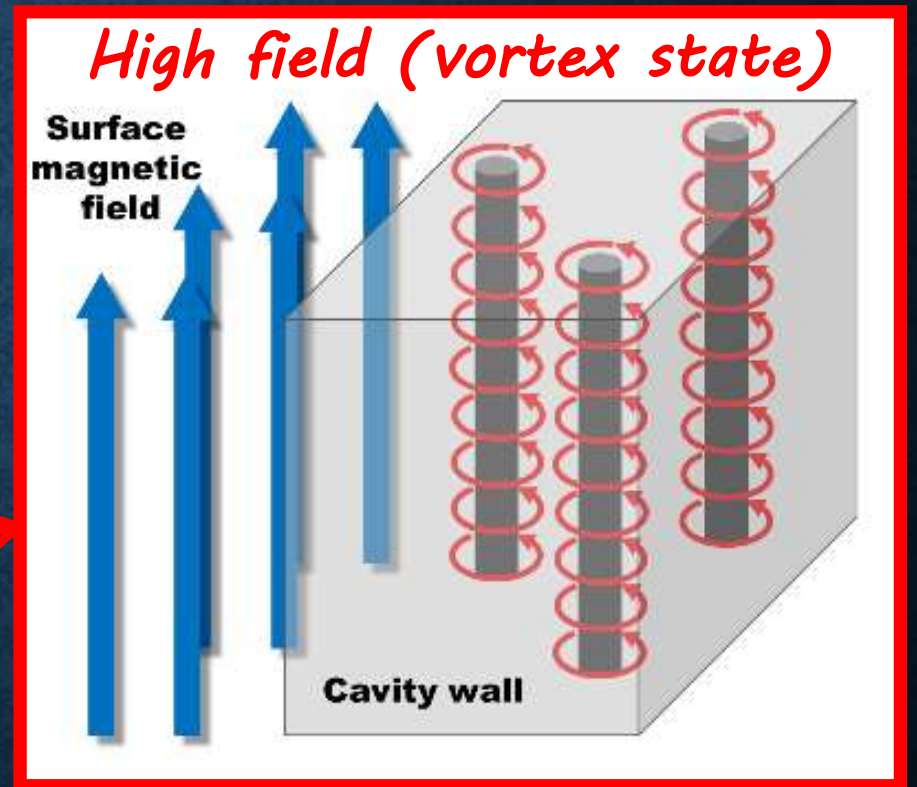


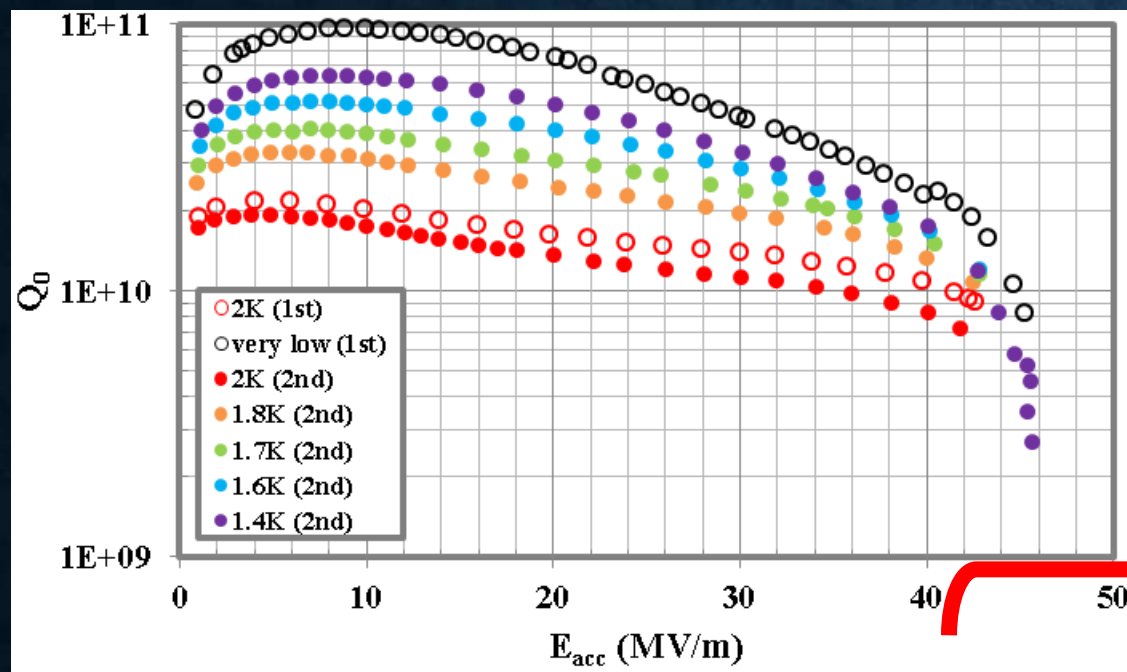


Magnetic field



∖ (^o^)/ I'm doomed

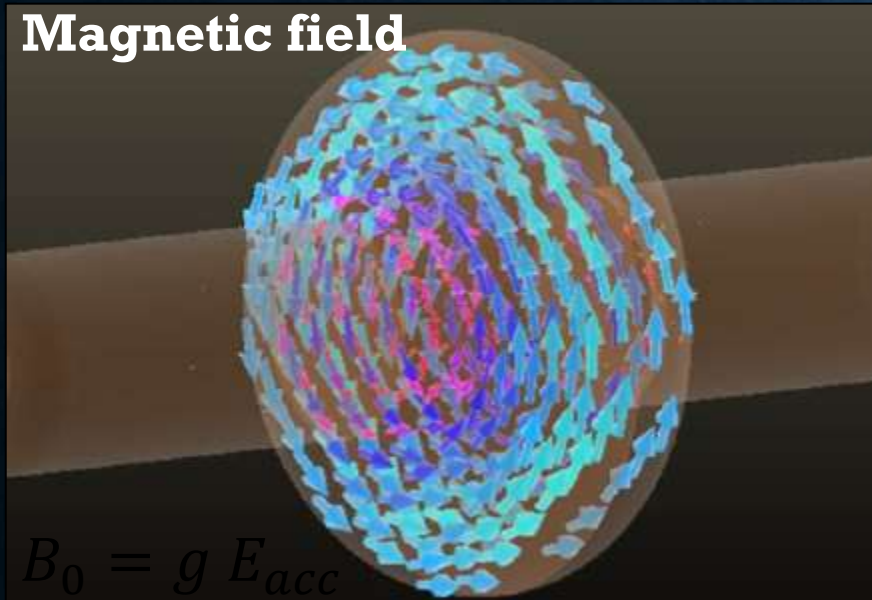




The ultimate limit of cavities made from bulk material is considered as

$$B_{c1} \leq B_0^{max} < B_{sh}$$

Magnetic field



0mT 85mT 170mT 210mT $\mu_0 H_0$
 $\simeq \mu_0 H_{c1}$ of Nb $\mu_0 H_{sh}$ of Nb

B_{c1} : the lower critical field, above which the Meissner state becomes energetically unfavorable.

B_{sh} : the superheating field, which represents the stability limit of the Meissner state. **Ultimate limit for the ideal material.**

Maximum field achieved so far

Already close to the theoretical limit?

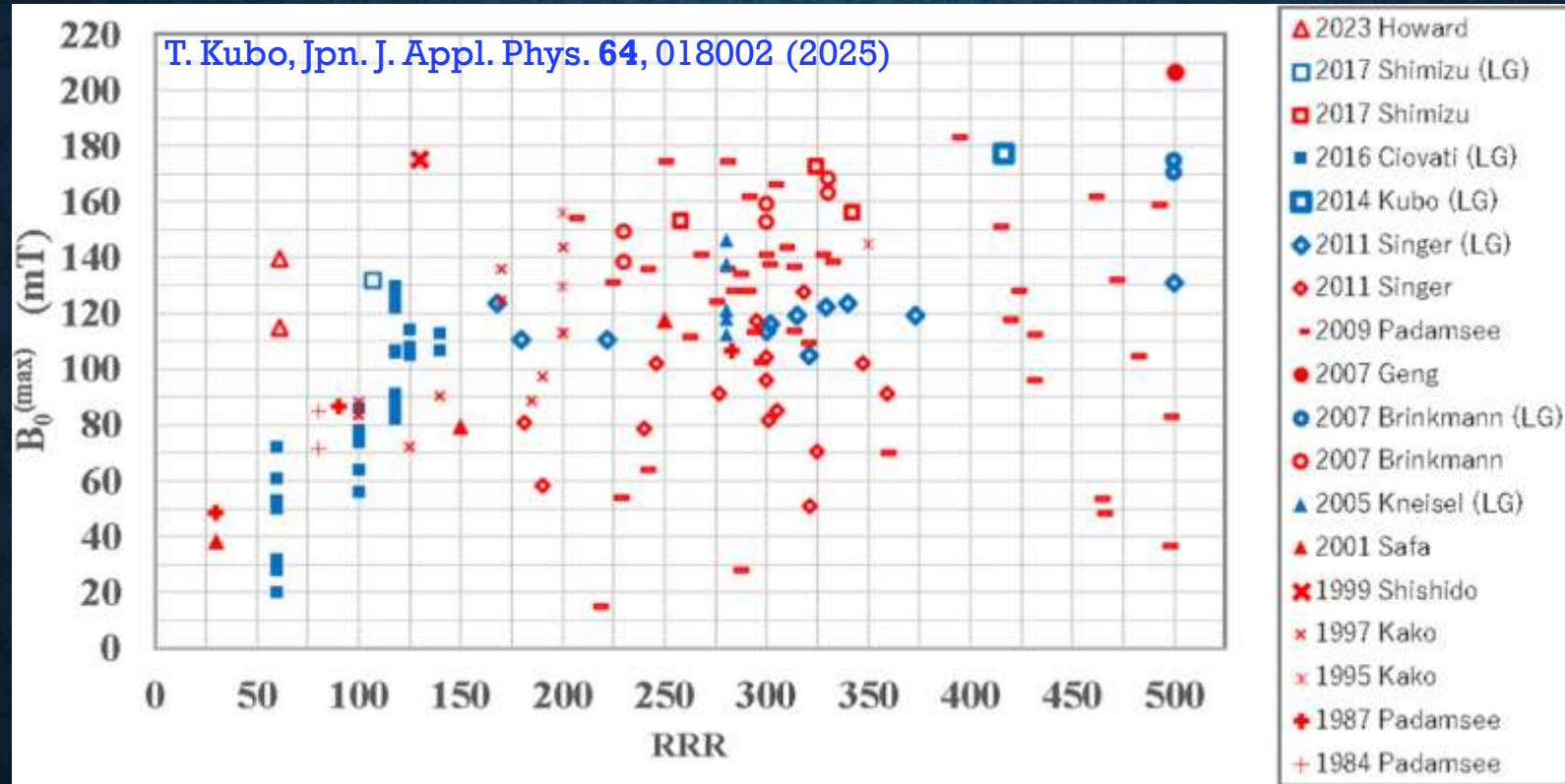


Fig. 1. Maximum surface magnetic field $B_0^{(max)}$ for 1.3 and 1.5 GHz cavities as a function of RRR. The 149 data points are sourced from Refs. 8–22, where RRR values were explicitly reported. These cavities were processed using chemical polishing and/or electropolishing, with or without low-temperature baking. Some high-performing cavities, such as those in Refs. 29, 30 achieving 190 mT and 210 mT, are not included due to the absence of reported RRR data. Nitrogen-doped cavities are excluded as doping significantly alters the surface RRR, a topic discussed in the final paragraph of this Brief Note. Similarly, cavities subjected to heat treatments above 1000 °C, which can modify the RRR value, are not included. Tests affected by multipacting-induced quenches (e.g. Test numbers 1, 2, 3, 7, and 13 of cavity F1F2 in Ref. 20) are omitted, as are cavities whose performance was limited by field emission (e.g. cavity HF-5 in Ref. 21). In cases where field emission is not explicitly mentioned or may have been overlooked, data potentially affected by it might still be included in the figure. However, such instances do not significantly impact the overall trend, particularly the primary focus of this study: the maximum fields achieved across different RRR values.

Summary of the superheating field

Intrinsic $\frac{\lambda_0}{\xi_0}$	Purity (mfp ℓ)	Theory (applicable to SRF temperature $T \ll T_c$)
-------------------------------------	-------------------------	--

Summary of the superheating field

Intrinsic $\frac{\lambda_0}{\xi_0}$	Purity (mfp ℓ)	Theory (applicable to SRF temperature $T \ll T_c$)
$\frac{\lambda_0}{\xi_0} \gg 1$ (e.g., Nb ₃ Sn, NbN, NbTiN etc.)		

Summary of the superheating field

Intrinsic $\frac{\lambda_0}{\xi_0}$	Purity (mfp ℓ)	Theory (applicable to SRF temperature $T \ll T_c$)
$\frac{\lambda_0}{\xi_0} \gg 1$ (e.g., Nb ₃ Sn, NbN, NbTiN etc.)	$\ell \gg \xi_0$ (clean)	

Summary of the superheating field

Intrinsic $\frac{\lambda_0}{\xi_0}$	Purity (mfp ℓ)	Theory (applicable to SRF temperature $T \ll T_c$)
$\frac{\lambda_0}{\xi_0} \gg 1$ (e.g., Nb ₃ Sn, NbN, NbTiN etc.)	$\ell \gg \xi_0$ (clean)	$H_{sh}(0) = 0.84H_c(0)$ $H_{sh}(T)$ for $0 \leq T \leq T_c$ Analytical result for T=0. V. P. Galaiko, Sov. Phys. JETP 23, 475 (1966) • G. Catelani and J. P. Sethna, Phys. Rev. B 78, 224509 (2008) • F. Pei-Jen Lin and A. Gurevich, Phys. Rev. B 85, 054513 (2012)

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	$\ell \ll \xi_0$ (dirty)	$H_{sh}(T)$ for $0 \leq T \leq T_c$ $H_{sh}(0) = 0.795H_c(0)$ F. Pei-Jen Lin and A. Gurevich, Phys. Rev. B 85, 054513 (2012) T. Kubo, Phys. Rev. Research 2, 033203 (2020) Analytical result for $T=0$. T. Kubo, Phys. Rev. Research 2, 033203 (2020) T. Kubo, Supercond.Sci. Technol. 34, 045006 (2021)

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$\frac{\lambda_0}{\xi_0} \lesssim 1$ (e.g., Nb)		Analytical result for T=0. T. Kubo, Phys. Rev. Research 2, 033203 (2020) T. Kubo, Supercond.Sci. Technol. 34, 045006 (2021)

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		N/A We don't know H_{sh} of clean Nb

Maximum field achieved so far

Already close to the theoretical limit?

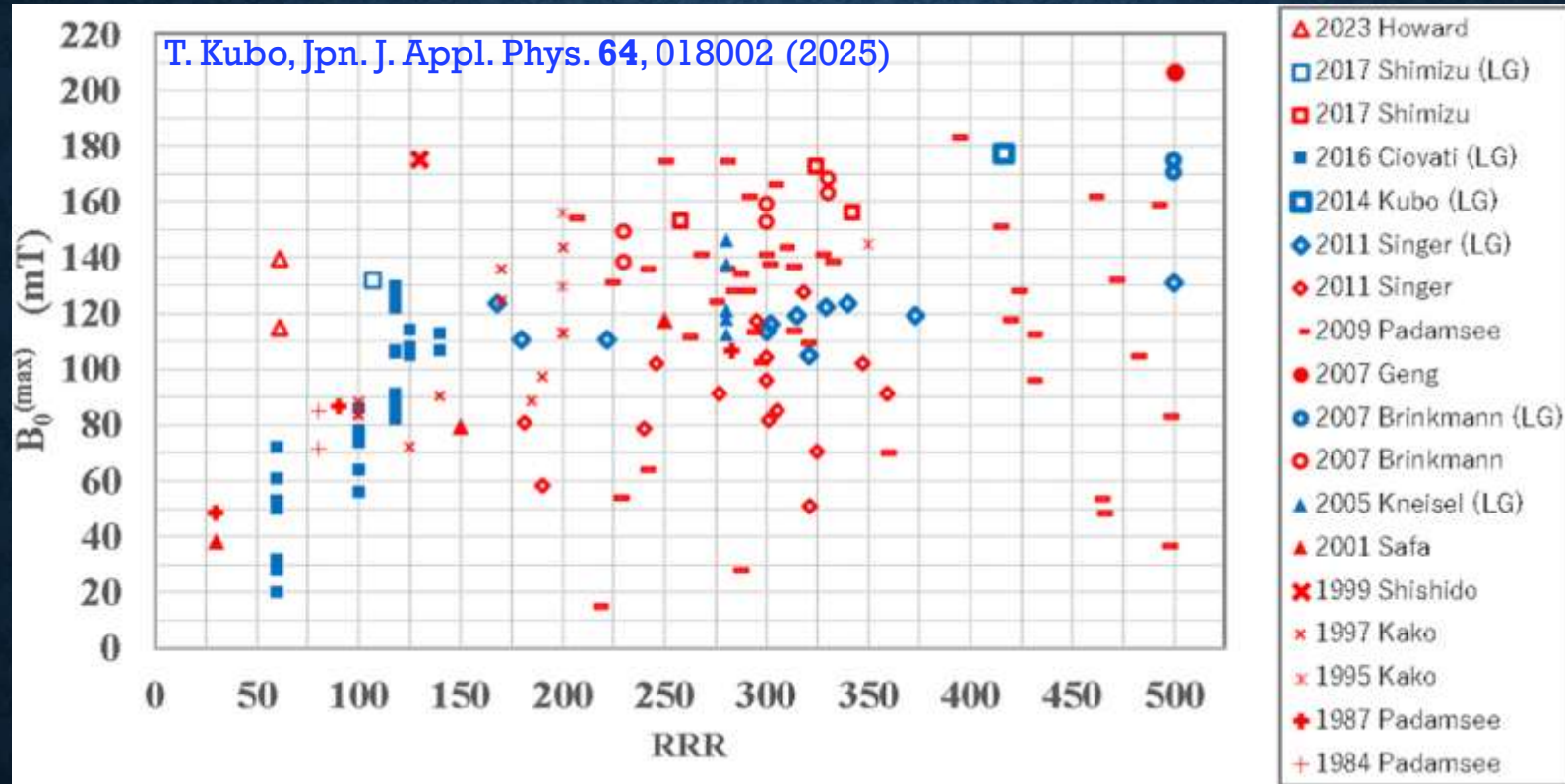
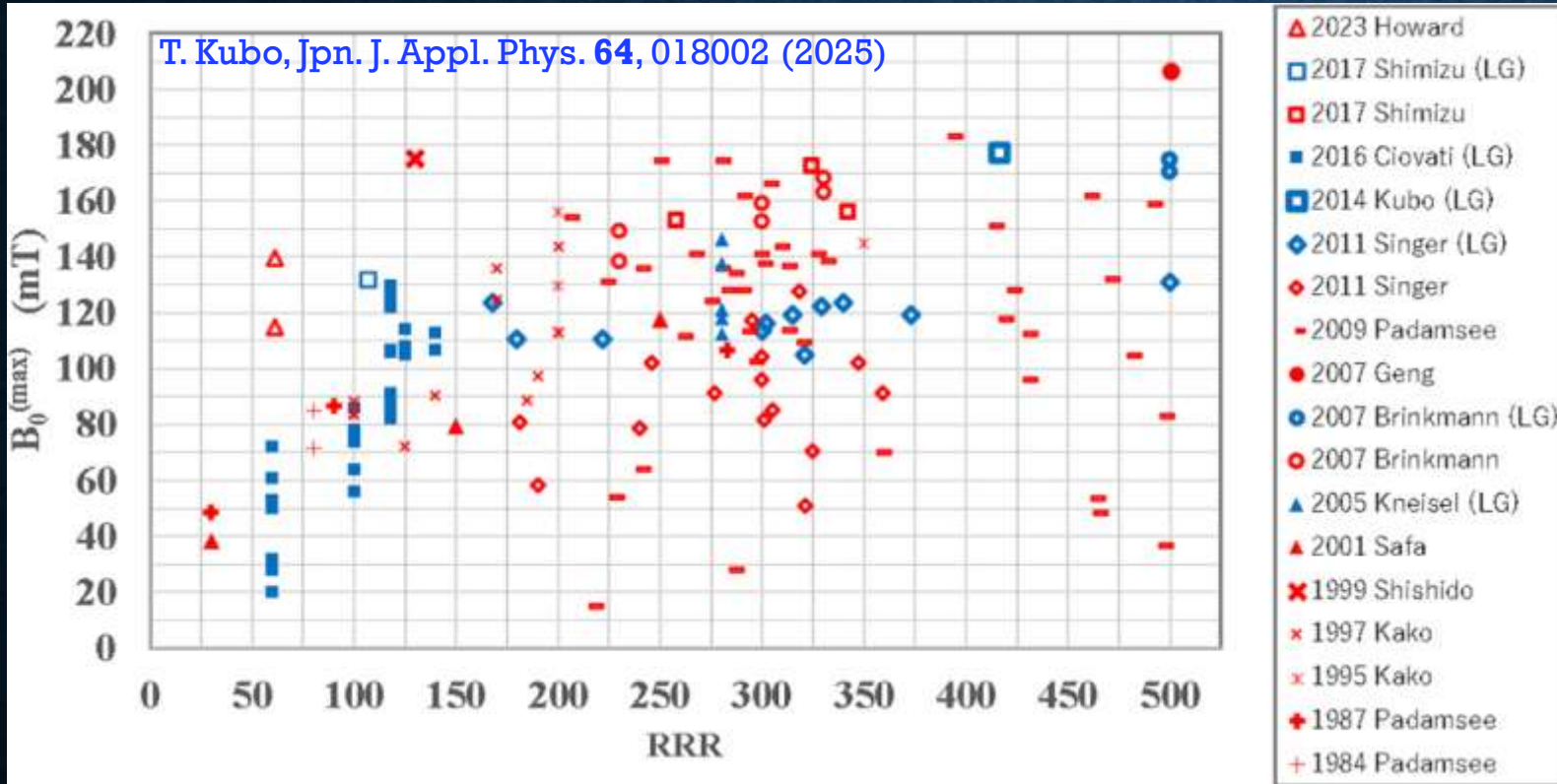


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Maximum field achieved so far

Already close to the theoretical limit?

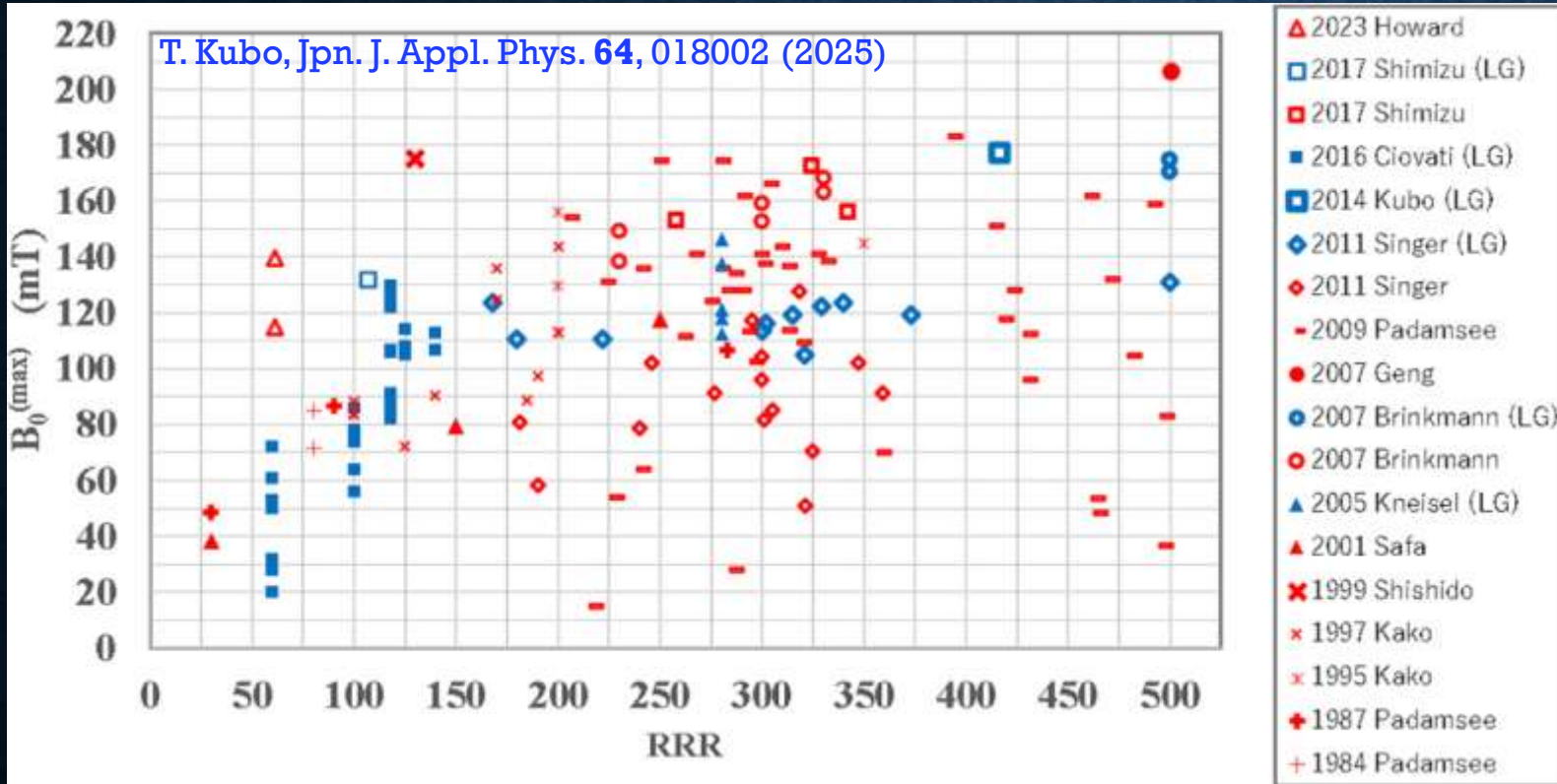


The answer: we currently have no reliable estimate, because there is **no theory of H_{sh} applicable to clean Nb at $T \ll T_c$** .

Fig. 1. Maximum surface magnetic field $B_0^{(max)}$ for 1.3 and 1.5 GHz cavities as a function of RRR. The 149 data points are sourced from Refs. 8–22, where RRR values were explicitly reported. These cavities were processed using chemical polishing and/or electropolishing, with or without low-temperature baking. Some high-performing cavities, such as those in Refs. 29, 30 achieving 190 mT and 210 mT, are not included due to the absence of reported RRR data. Nitrogen-doped cavities are excluded as doping significantly alters the surface RRR, a topic discussed in the final paragraph of this Brief Note. Similarly, cavities subjected to heat treatments above 1000 °C, which can modify the RRR value, are not included. Tests affected by multipacting-induced quenches (e.g. Test numbers 1, 2, 3, 7, and 13 of cavity F1F2 in Ref. 20) are omitted, as are cavities whose performance was limited by field emission (e.g. cavity HF-5 in Ref. 21) In cases where field emission is not explicitly mentioned or may have been overlooked, data potentially affected by it might still be included in the figure. However, such instances do not significantly impact the overall trend, particularly the primary focus of this study: the maximum fields achieved across different RRR values.

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Already close to the theoretical limit?



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Ginzburg–Landau (GL) theory, valid near $T \approx T_c$, gives

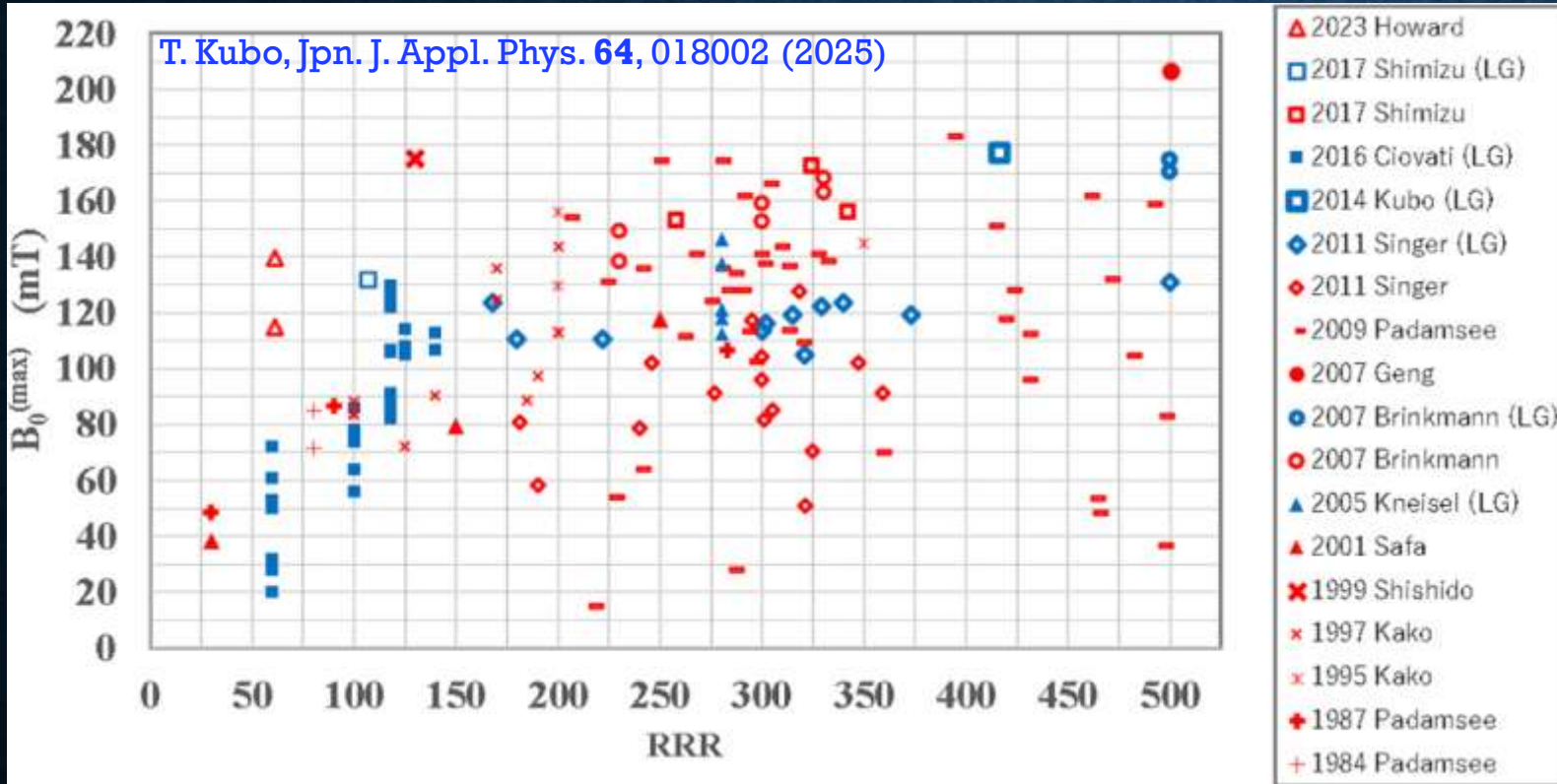
- A. J. Dolgert et al., Phys. Rev. B **53**, 5650 (1996)
- M. K. Transtrum et al., Phys. Rev. B **83**, 094505 (2011)

$$B_{sh}(T) \Big|_{T \approx T_c} = 1.26 B_c(T) \Big|_{T \approx T_c} \left(\kappa = \frac{\lambda}{\xi} = 1 \right)$$

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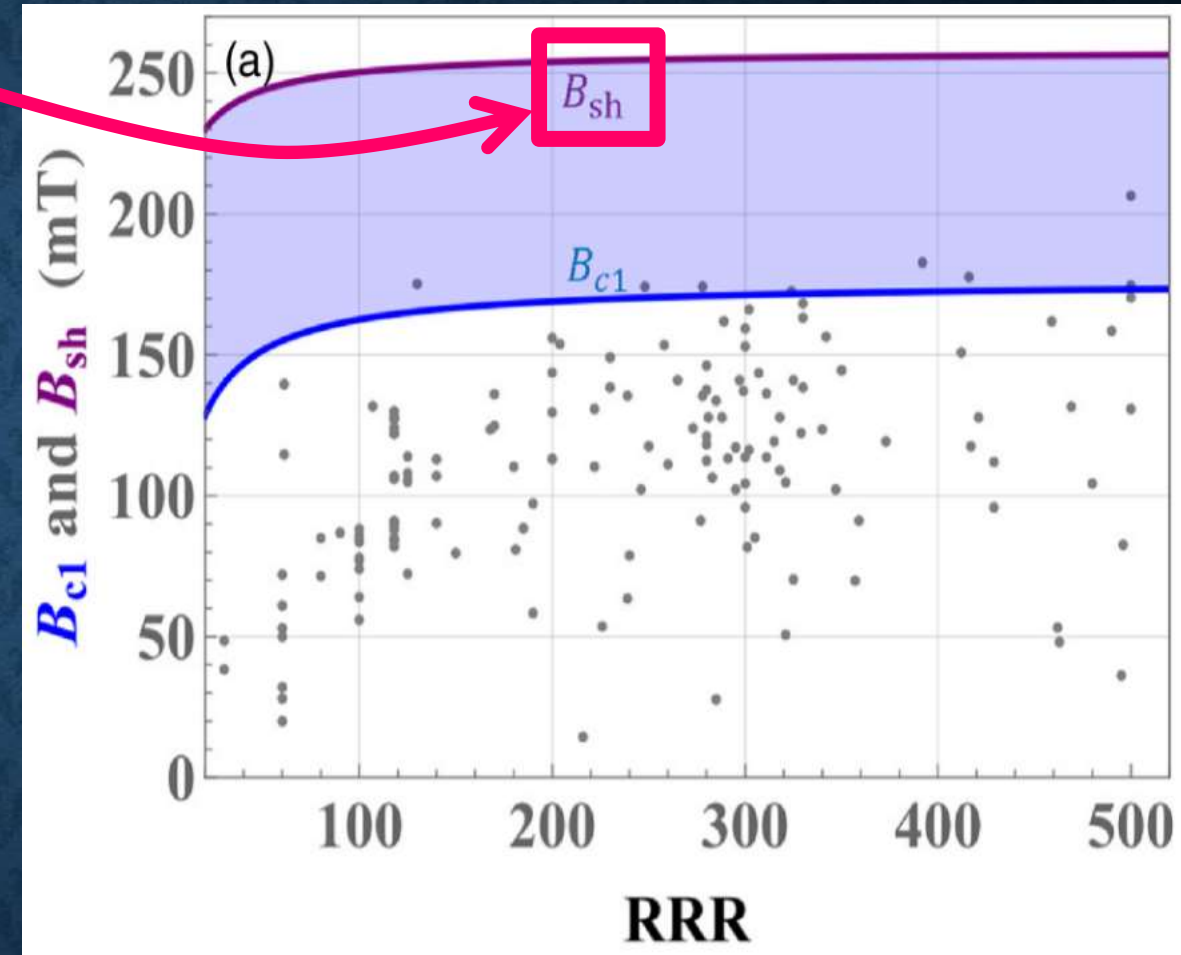
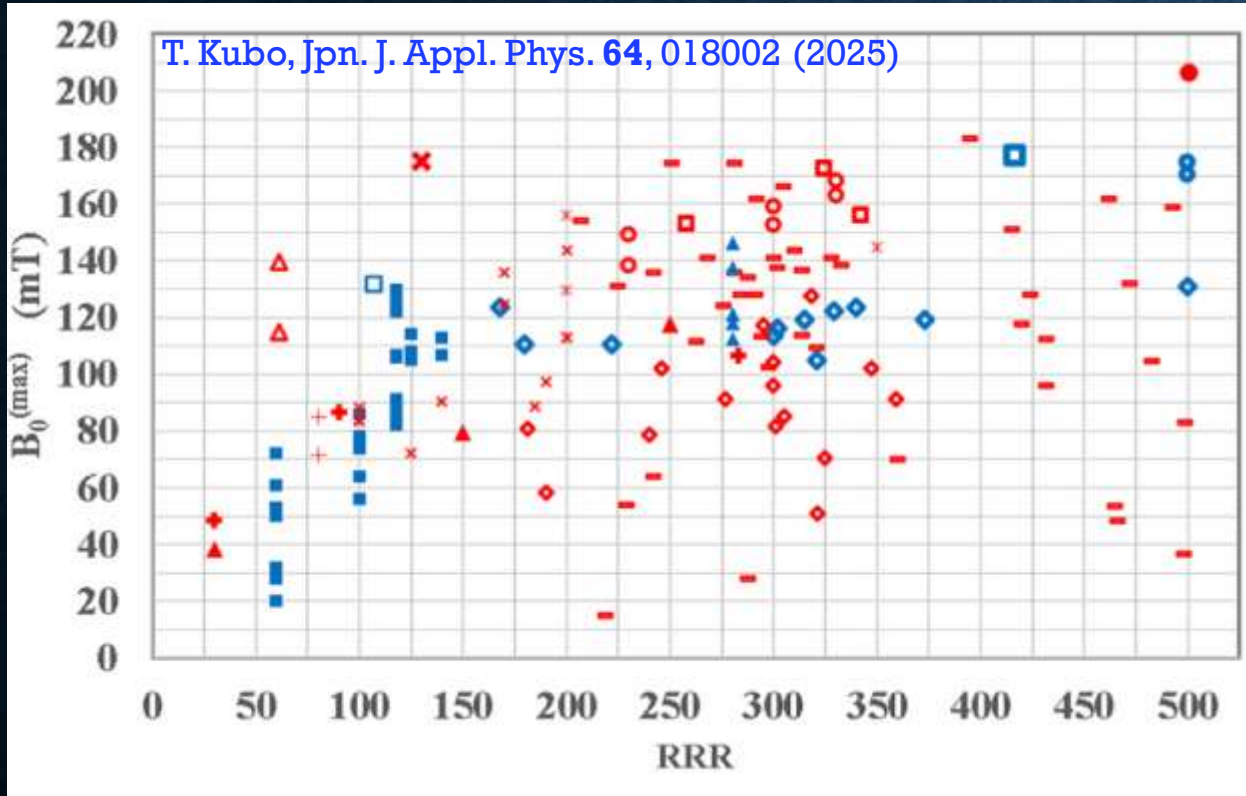
$$B_{sh}(T) \Big|_{T \approx T_c} = 1.26 B_c(T) \Big|_{T \approx T_c} \left(\kappa = \frac{\lambda}{\xi} = 1 \right)$$

If we naively extrapolate this to low temperatures — not theoretically justified — we obtain the rough estimate

$$B_{sh}(0) = 1.26 B_c(0) \text{ ?} \\ = 250 \text{ mT ?}$$

Fig. 1. Maximum surface magnetic field $B_0^{\max}$ for 1.3 and 1.5 GHz cavities as a function of RRR. The 149 data points are sourced from Refs. 8–22, where RRR values were explicitly reported. These cavities were processed using chemical polishing and/or electropolishing, with or without low-temperature baking. Some high-performing cavities, such as those in Refs. 29, 30 achieving 190 mT and 210 mT, are not included due to the absence of reported RRR data. Nitrogen-doped cavities are excluded as doping significantly alters the surface RRR, a topic discussed in the final paragraph of this Brief Note. Similarly, cavities subjected to heat treatments above 1000 °C, which can modify the RRR value, are not included. Tests affected by multipacting-induced quenches (e.g. Test numbers 1, 2, 3, 7, and 13 of cavity F1F2 in Ref. 20) are omitted, as are cavities whose performance was limited by field emission (e.g. cavity HF-5 in Ref. 21) In cases where field emission is not explicitly mentioned or may have been overlooked, data potentially affected by it might still be included in the figure. However, such instances do not significantly impact the overall trend, particularly the primary focus of this study: the maximum fields achieved across different RRR values.

If we believe this rough estimate of B_{sh} based on GL, we get



Note: To achieve B_{sh} , it is necessary to eliminate all surface defects. **However, this alone is not sufficient** to reach B_{sh} ; it is also essential to ensure that the thermal runaway field B_{run} exceeds B_{sh} . This condition is not always satisfied, particularly at 2 K. It is better to test the cavity at lower temperatures (e.g., 1.8K or 1.6K) to ensure that the condition $B_{run} > B_{sh}$ is satisfied.

Driving force of
Nonequilibrium
Dynamics
(e.g, strength of microwave)

NOTE
The above superheating-field
theories calculate the **dc**
(equilibrium) superheating field.

Temperature
 T_c
Thermal Equilibrium
Ginzburg-Landau ($T \approx T_c$)

**Usadel
Formalism**
Applicable to
Any temperature
 $mfp < \xi_0$

**Eilenberger
Formalism**
Applicable to
Any temperature
Any mean free path

Keldysh-Usadel

**Keldysh
Eilenberger**

Mean
free
path

Driving force of
Nonequilibrium
Dynamics
(e.g, strength of microwave)

Is the ac (nonequilibrium)
superheating field
equivalent to the dc one?

NOTE
The above superheating-field
theories calculate the dc
(equilibrium) superheating field.

Temperature
 T_c
Thermal Equilibrium
Ginzburg-Landau ($T \approx T_c$)

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Keldysh-Usadel

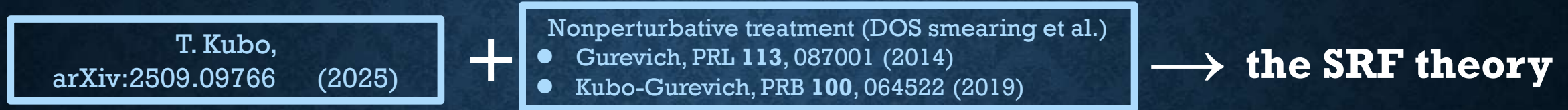
**Keldysh
Eilenberger**

Mean
free
path

Summary of my perspective

($\simeq$ what I am interested in)

- We are only one step away from establishing the SRF theory.



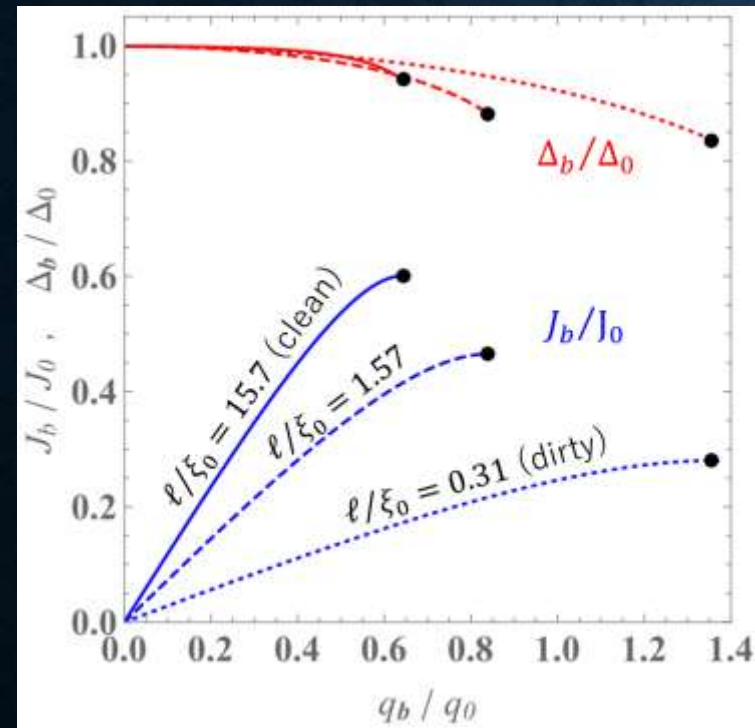
- Understanding TLS is essential to realizing ultra-high- Q cavities.
- The dc superheating field is largely understood, whereas that of clean Nb has not yet been fully understood. In addition, the ac superheating field remains largely unexplored.

Backup

dc + ac



If the AC field is absent, this reduces to a problem of equilibrium superconductivity.
That case was already well understood decades ago.



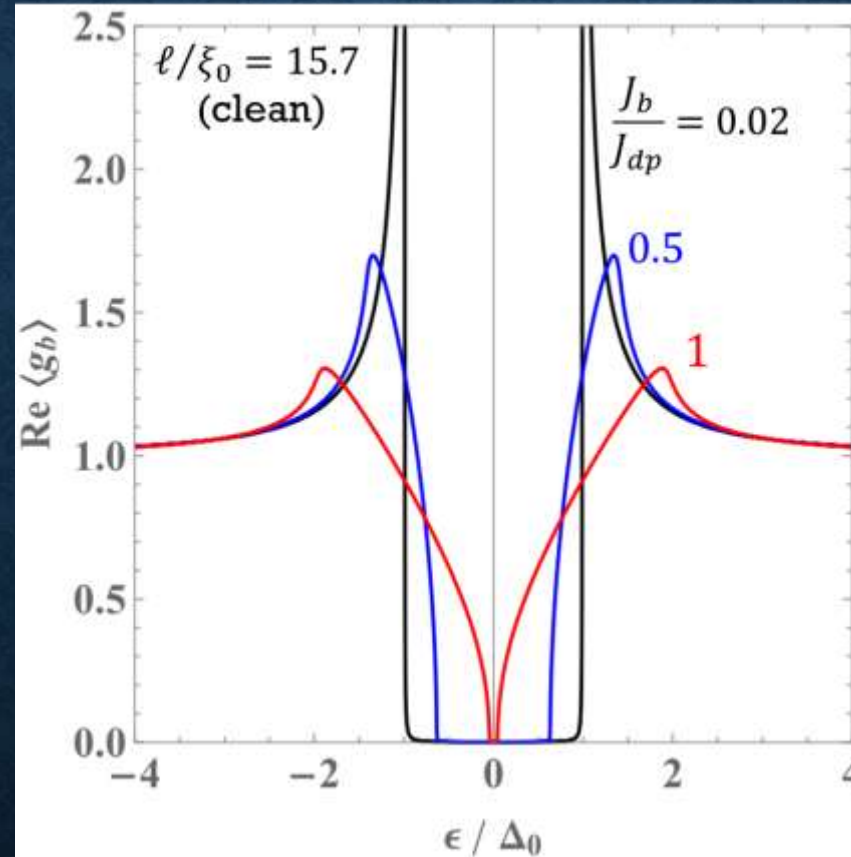
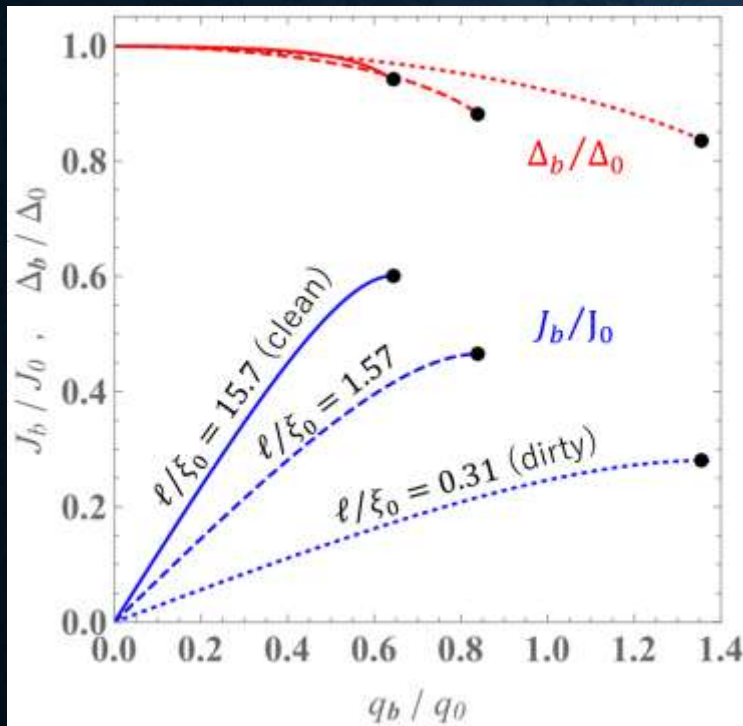
T. Kubo, Phys. Rev. Applied 22, 044042 (2024)

DOI: <https://doi.org/10.1103/PhysRevApplied.22.044042>

See also, T. Kubo, Phys. Rev. Research 2, 033203 (2020); F. Pei-Jen Lin and A. Gurevich, Phys. Rev. B 85, 054513 (2012)



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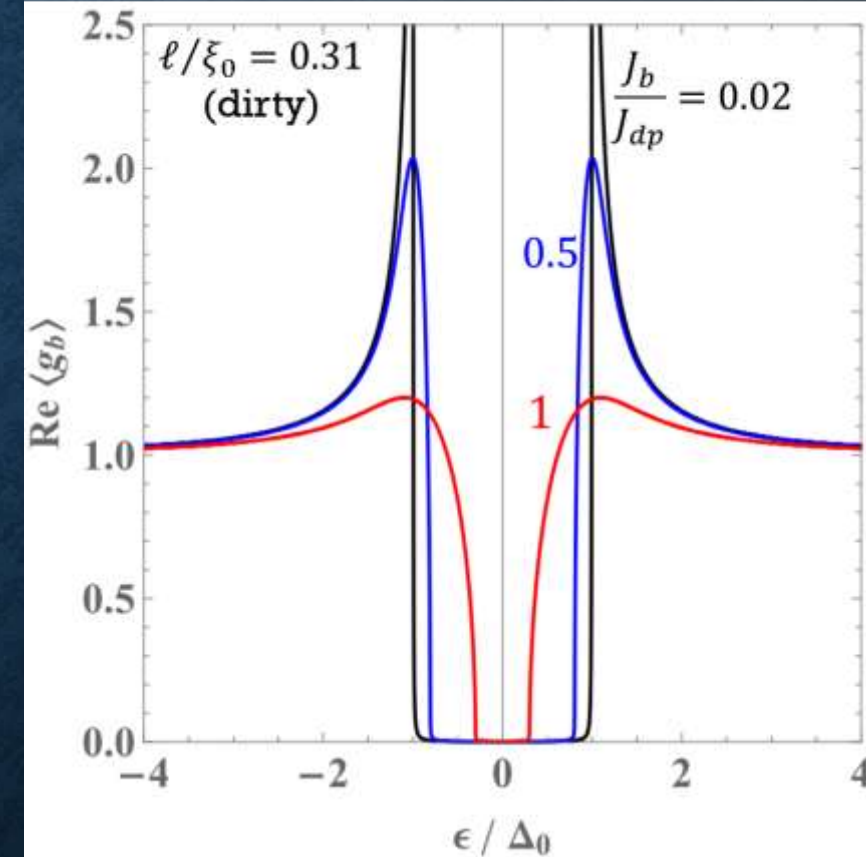
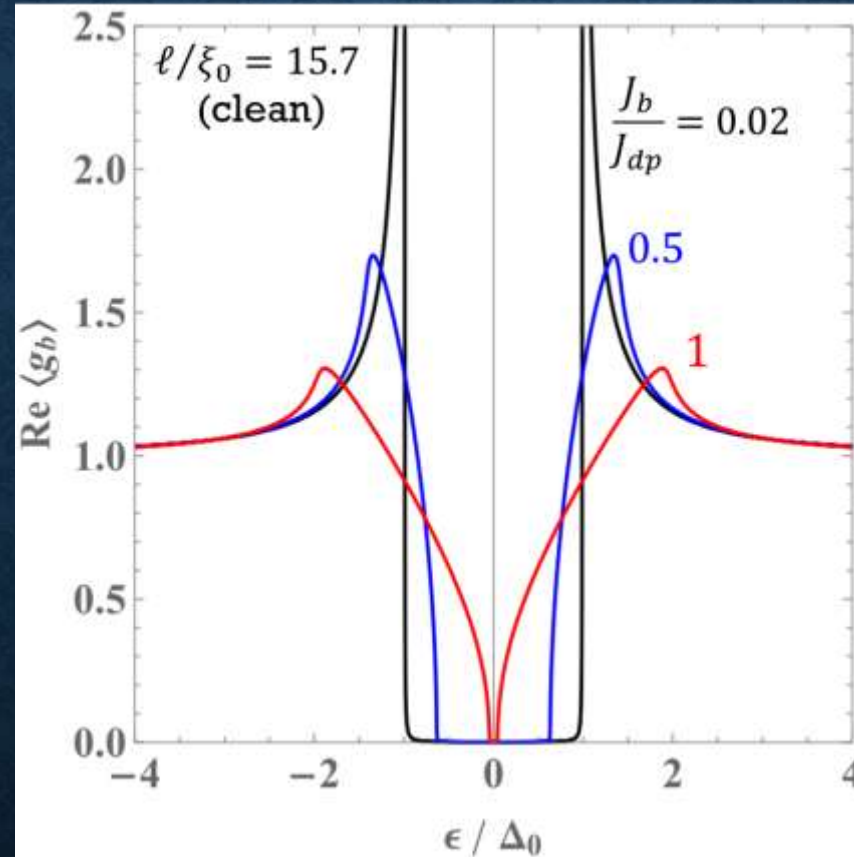
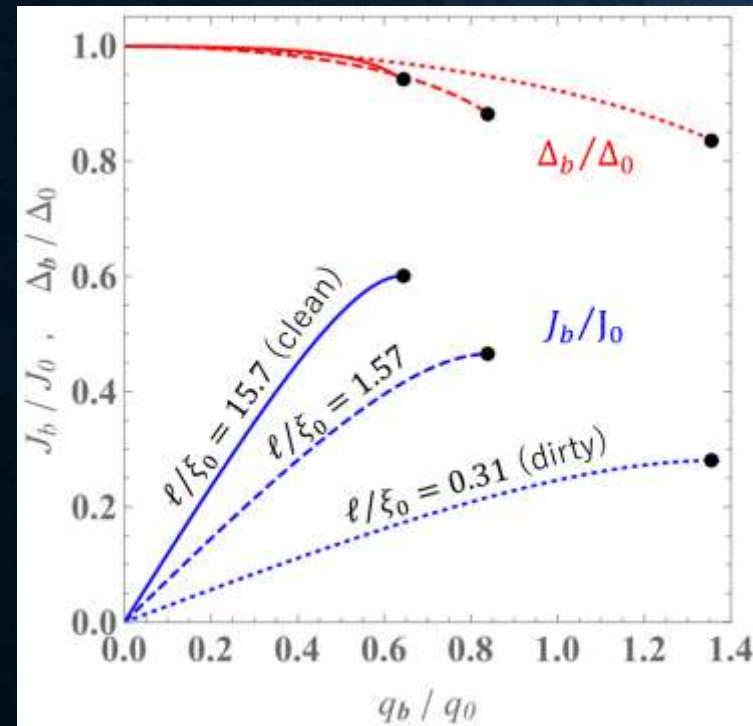
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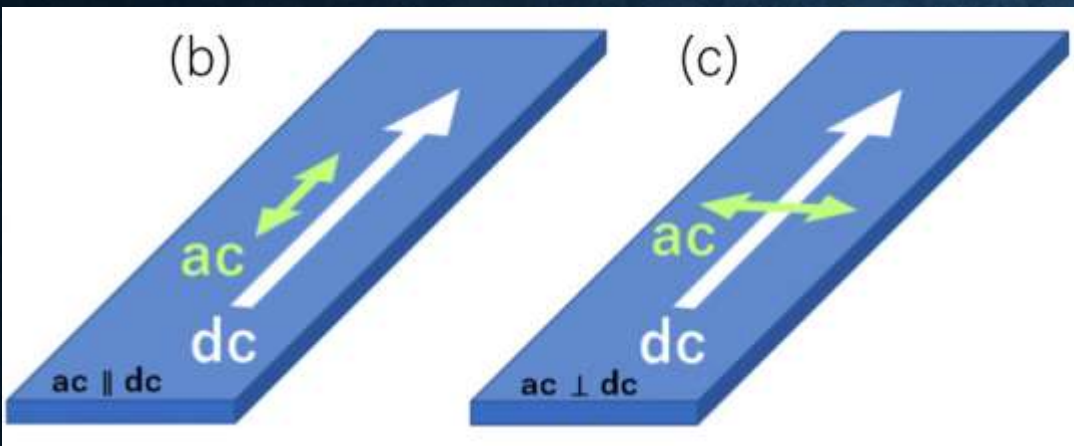
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However, once an AC field is present, the problem becomes much more complicated. In that case, we need to use the theory of **nonequilibrium** superconductivity.



A. Moor et al., Phys. Rev. Lett. **118**, 047001 (2017)

T. Jujo, J. Phys. Soc. Jpn. **91**, 074711 (2022)

T. Kubo, Phys. Rev. Applied **22**, 044042 (2024)

T. Kubo, Phys. Rev. Applied **23**, 054091 (2025)

- **The Keldysh-Eilenberger theory** is a microscopic theory of **nonequilibrium** superconductivity. It is applicable at any temperature ($0 \leq T \leq T_c$) and for arbitrary mean free path. In this sense, it serves as the “*theory of everything for conventional superconductivity*”.

T. Kubo, Phys. Rev. Applied **22**, 044042 (2024)

- **The Keldysh–Usadel theory** represents the dirty-limit reduction of the Keldysh–Eilenberger theory of nonequilibrium superconductivity, applicable at any T ($0 \leq T \leq T_c$).

T. Kubo, Phys. Rev. Applied **23**, 054091 (2025)

Keldysh–Usadel Equation: DC Current with AC Perturbation

T. Kubo, Phys. Rev. Applied **23**, 054091 (2025)

$$\begin{aligned}
 & -i(s/2) [\hat{\tau}_3 \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) - \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \delta \hat{g}^R(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) - \delta \hat{g}^R(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^A(\epsilon, \omega) - \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^A(\epsilon, \omega) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) - \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) \hat{\tau}_3] \\
 & - i(\delta W/2) [\hat{\tau}_3 \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) - \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \hat{g}_b^R(\epsilon_-) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) - \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) - \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) - \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \hat{g}_b^A(\epsilon_+) \hat{\tau}_3] \\
 & = \epsilon_+ \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) - \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \epsilon_- \\
 & + [\hat{\Delta}_b, \delta \hat{g}^K(\epsilon, \omega)] + \delta \hat{\Delta}(\omega) \hat{g}_b^K(\epsilon_-) - \hat{g}_b^K(\epsilon_+) \delta \hat{\Delta}(\omega). \quad (15)
 \end{aligned}$$

$$\begin{aligned}
 & -i(s/2) \{ \hat{\tau}_3 \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^r(\epsilon, \omega) - \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) - \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \hat{\tau}_3 \} \\
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 & - \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \hat{\tau}_3 + \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \\
 & - \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \} = \epsilon_+ \hat{\tau}_3 \delta \hat{g}^r(\epsilon, \omega) - \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \epsilon_- \\
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 \end{aligned}$$

$$\delta \Delta(\omega) = -\frac{\mathcal{G}}{8} \int d\epsilon \text{Tr} [(-i\tau_2) \delta \hat{g}^K(\epsilon, \omega)]. \quad (17)$$

Higgs
↓

We solve these equations to obtain the AC-induced **nonequilibrium corrections** ($\delta \hat{g}^{R,A,K}$ and $\delta \Delta$).

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T. Kubo, Phys. Rev. Applied 23, 054091 (2025)

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 & + \hat{t}_3 \delta \hat{g}^R(\epsilon, \omega) \hat{t}_3 \hat{g}_b^K(\epsilon_-) - \delta \hat{g}^R(\epsilon, \omega) \hat{t}_3 \hat{g}_b^K(\epsilon_-) \hat{t}_3 \\
 & + \hat{t}_3 \hat{g}_b^K(\epsilon_+) \hat{t}_3 \delta \hat{g}^A(\epsilon, \omega) - \hat{g}_b^K(\epsilon_+) \hat{t}_3 \delta \hat{g}^A(\epsilon, \omega) \hat{t}_3 \\
 & + \hat{t}_3 \delta \hat{g}^K(\epsilon, \omega) \hat{t}_3 \hat{g}_b^A(\epsilon_-) - \delta \hat{g}^K(\epsilon, \omega) \hat{t}_3 \hat{g}_b^A(\epsilon_-) \hat{t}_3] \\
 & - i(\delta W/2) [\hat{t}_3 \hat{g}_b^R(\epsilon_+) \hat{t}_3 \hat{g}_b^K(\epsilon_-) - \hat{g}_b^R(\epsilon_+) \hat{t}_3 \hat{g}_b^K(\epsilon_-) \hat{t}_3 \\
 & + \hat{t}_3 \hat{g}_b^R(\epsilon_-) \hat{t}_3 \hat{g}_b^K(\epsilon_-) - \hat{g}_b^R(\epsilon_+) \hat{t}_3 \hat{g}_b^K(\epsilon_+) \hat{t}_3 \\
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 & = \epsilon_+ \hat{t}_3 \delta \hat{g}^K(\epsilon, \omega) - \delta \hat{g}^K(\epsilon, \omega) \hat{t}_3 \epsilon_- \\
 & + [\hat{\Delta}_b, \delta \hat{g}^K(\epsilon, \omega)] + \delta \hat{\Delta}(\omega) \hat{g}_b^K(\epsilon_-) - \hat{g}_b^K(\epsilon_+) \delta \hat{\Delta}(\omega). \quad (15)
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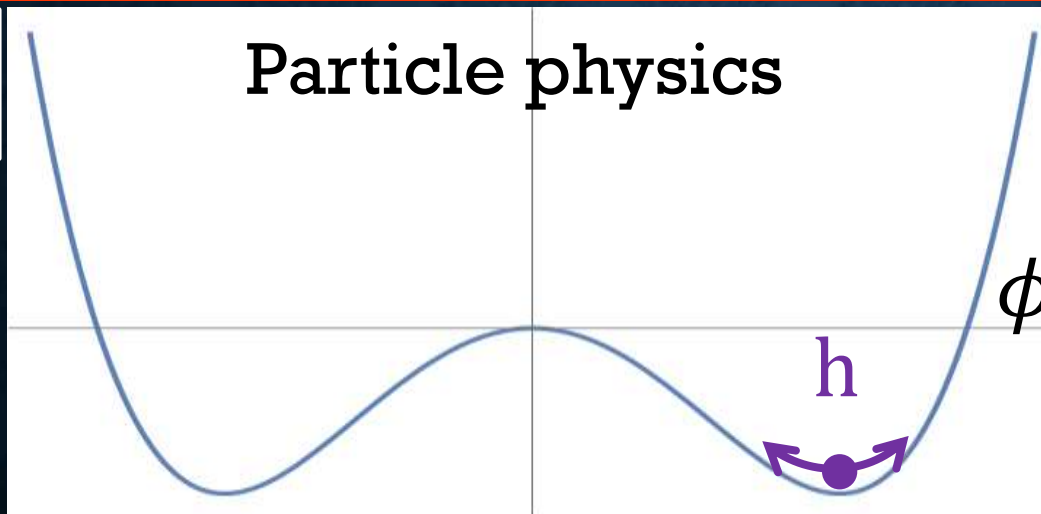
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 & - i(\delta W/2) \{ \hat{t}_3 \hat{g}_b^r(\epsilon_+) \hat{t}_3 \hat{g}_b^r(\epsilon_-) \\
 & - \hat{g}_b^r(\epsilon_+) \hat{t}_3 \hat{g}_b^r(\epsilon_-) \hat{t}_3 + \hat{t}_3 \hat{g}_b^r(\epsilon_-) \hat{t}_3 \hat{g}_b^r(\epsilon_-) \\
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 & + [\hat{\Delta}_b, \delta \hat{g}^r(\epsilon, \omega)] + \delta \hat{\Delta}(\omega) \hat{g}_b^r(\epsilon_-) - \hat{g}_b^r(\epsilon_+) \delta \hat{\Delta}(\omega), \quad (13)
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Higgs
↓

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Higgs?



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T. Kubo, Phys. Rev. Applied 23, 054091 (2025)

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 & + \hat{\tau}_3 \delta \hat{g}^R(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) - \delta \hat{g}^R(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^A(\epsilon, \omega) - \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^A(\epsilon, \omega) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) - \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) \hat{\tau}_3] \\
 & - i(\delta W/2) [\hat{\tau}_3 \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) - \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) \hat{\tau}_3 \\
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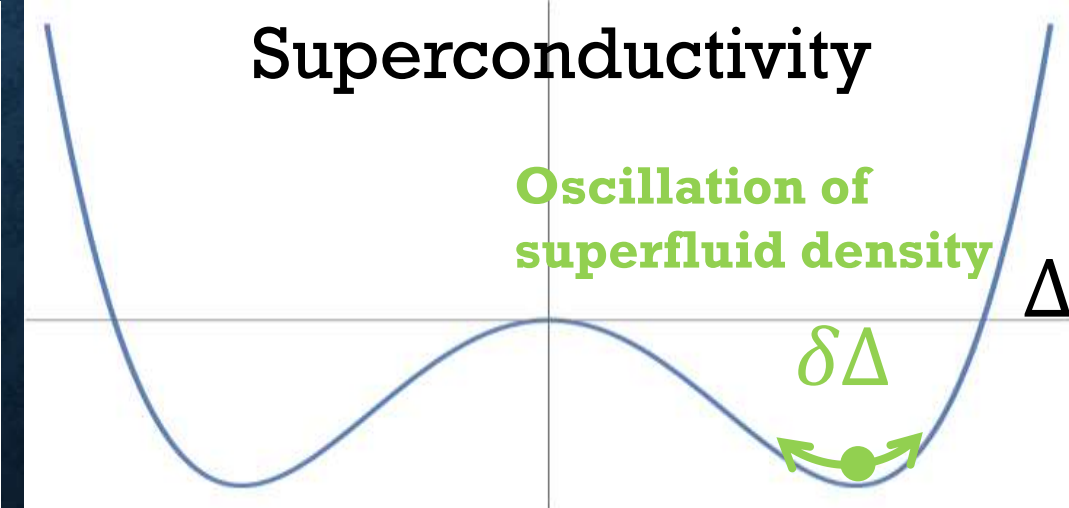
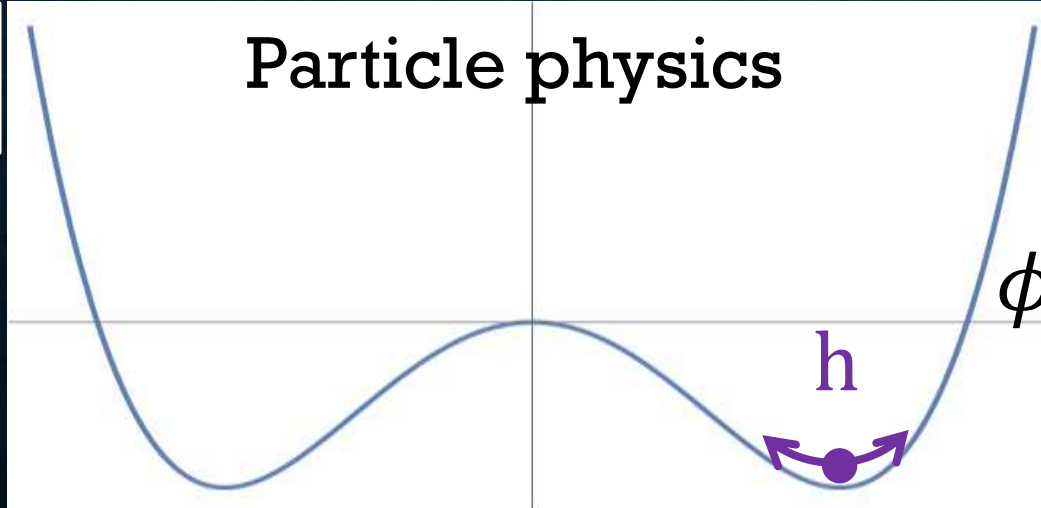
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 & + \hat{\tau}_3 \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) - \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \hat{\tau}_3 \} \\
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 & - \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \hat{\tau}_3 + \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \\
 & - \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \} = \epsilon_+ \hat{\tau}_3 \delta \hat{g}^r(\epsilon, \omega) - \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \epsilon_- \\
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T. Kubo, Phys. Rev. Applied **23**, 054091 (2025)

$$\begin{aligned}
 & -i(s/2) [\hat{\tau}_3 \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) - \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \delta \hat{g}^R(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) - \delta \hat{g}^R(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^A(\epsilon, \omega) - \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^A(\epsilon, \omega) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) - \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) \hat{\tau}_3] \\
 & - i(\delta W/2) [\hat{\tau}_3 \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) - \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \hat{g}_b^R(\epsilon_-) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) - \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) - \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) - \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \hat{g}_b^A(\epsilon_+) \hat{\tau}_3] \\
 & = \epsilon_+ \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) - \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \epsilon_- \\
 & + [\hat{\Delta}_b, \delta \hat{g}^K(\epsilon, \omega)] + \delta \hat{\Delta}(\omega) \hat{g}_b^K(\epsilon_-) - \hat{g}_b^K(\epsilon_+) \delta \hat{\Delta}(\omega). \quad (15)
 \end{aligned}$$

$$\begin{aligned}
 & -i(s/2) \{ \hat{\tau}_3 \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^r(\epsilon, \omega) - \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) - \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \hat{\tau}_3 \} \\
 & - i(\delta W/2) \{ \hat{\tau}_3 \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \\
 & - \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \hat{\tau}_3 + \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \\
 & - \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \} = \epsilon_+ \hat{\tau}_3 \delta \hat{g}^r(\epsilon, \omega) - \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \epsilon_- \\
 & + [\hat{\Delta}_b, \delta \hat{g}^r(\epsilon, \omega)] + \delta \hat{\Delta}(\omega) \hat{g}_b^r(\epsilon_-) - \hat{g}_b^r(\epsilon_+) \delta \hat{\Delta}(\omega), \quad (13)
 \end{aligned}$$

$$\delta \Delta(\omega) = -\frac{\mathcal{G}}{8} \int d\epsilon \text{Tr} [(-i\tau_2) \delta \hat{g}^K(\epsilon, \omega)]. \quad (17)$$

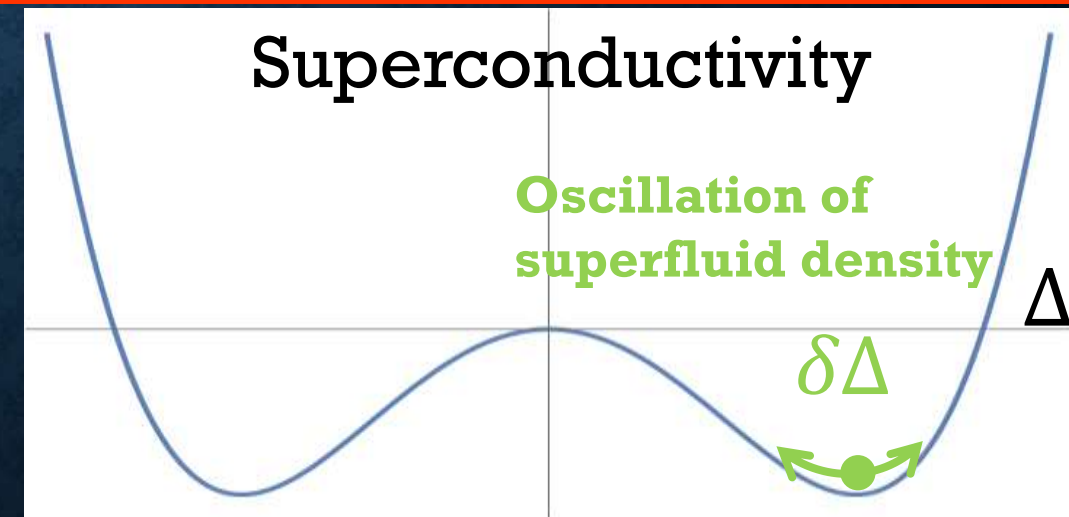
Higgs
↓

We solve these equations to obtain the AC-induced **nonequilibrium corrections** ($\delta \hat{g}^{R,A,K}$ and $\delta \Delta$).

Higgs?

See the famous review paper:
R. Shimano and N. Tsuji,
Annu. Rev. Condens. Matter Phys. **11**, 103 (2020)

The Higgs mode in superconductivity is an $\mathcal{O}(A^2)$ effect. It does not appear in standard linear-response theories such as Mattis–Bardeen, but it does emerge in nonlinear response. In our case, we apply both DC and AC fields. As a result, the Higgs mode shows up as $\delta \Delta \propto A_{dc} \cdot A$, and responds linearly to the AC field.



Keldysh–Usadel Equation: DC Current with AC Perturbation

T. Kubo, Phys. Rev. Applied **23**, 054091 (2025)

$$\begin{aligned}
 & -i(s/2) [\hat{\tau}_3 \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) - \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \delta \hat{g}^R(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) - \delta \hat{g}^R(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^A(\epsilon, \omega) - \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^A(\epsilon, \omega) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) - \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) \hat{\tau}_3] \\
 & - i(\delta W/2) [\hat{\tau}_3 \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) - \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \hat{g}_b^R(\epsilon_-) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) - \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) - \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) - \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \hat{g}_b^A(\epsilon_+) \hat{\tau}_3] \\
 & = \epsilon_+ \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) - \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \epsilon_- \\
 & + [\hat{\Delta}_b, \delta \hat{g}^K(\epsilon, \omega)] + \delta \hat{\Delta}(\omega) \hat{g}_b^K(\epsilon_-) - \hat{g}_b^K(\epsilon_+) \delta \hat{\Delta}(\omega). \quad (15)
 \end{aligned}$$

$$\begin{aligned}
 & -i(s/2) \{ \hat{\tau}_3 \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^r(\epsilon, \omega) - \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) - \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \hat{\tau}_3 \} \\
 & - i(\delta W/2) \{ \hat{\tau}_3 \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \\
 & - \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \hat{\tau}_3 + \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \\
 & - \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \} = \epsilon_+ \hat{\tau}_3 \delta \hat{g}^r(\epsilon, \omega) - \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \epsilon_- \\
 & + [\hat{\Delta}_b, \delta \hat{g}^r(\epsilon, \omega)] + \delta \hat{\Delta}(\omega) \hat{g}_b^r(\epsilon_-) - \hat{g}_b^r(\epsilon_+) \delta \hat{\Delta}(\omega), \quad (13)
 \end{aligned}$$

$$\delta \Delta(\omega) = -\frac{\mathcal{G}}{8} \int d\epsilon \text{Tr} [(-i\tau_2) \delta \hat{g}^K(\epsilon, \omega)]. \quad (17)$$

Higgs
↓

We solve these equations to obtain the AC-induced **nonequilibrium corrections** ($\delta \hat{g}^{R,A,K}$ and $\delta \Delta$).

Keldysh–Usadel Equation: DC Current with AC Perturbation

T. Kubo, Phys. Rev. Applied 23, 054091 (2025)

$$\begin{aligned}
 & -i(s/2) [\hat{\tau}_3 \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) - \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \delta \hat{g}^R(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) - \delta \hat{g}^R(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^A(\epsilon, \omega) - \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^A(\epsilon, \omega) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) - \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) \hat{\tau}_3] \\
 & - i(\delta W/2) [\hat{\tau}_3 \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) - \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) - \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) - \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) - \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \hat{g}_b^A(\epsilon_+) \hat{\tau}_3] \\
 & = \epsilon_+ \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) - \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \epsilon_- \\
 & + [\hat{\Delta}_b, \delta \hat{g}^K(\epsilon, \omega)] + \delta \hat{\Delta}(\omega) \hat{g}_b^K(\epsilon_-) - \hat{g}_b^K(\epsilon_+) \delta \hat{\Delta}(\omega). \quad (15)
 \end{aligned}$$

$$\begin{aligned}
 & -i(s/2) \{ \hat{\tau}_3 \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^r(\epsilon, \omega) - \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) - \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \hat{\tau}_3 \} \\
 & - i(\delta W/2) \{ \hat{\tau}_3 \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \\
 & - \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \hat{\tau}_3 + \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \\
 & - \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \} = \epsilon_+ \hat{\tau}_3 \delta \hat{g}^r(\epsilon, \omega) - \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \epsilon_- \\
 & + [\hat{\Delta}_b, \delta \hat{g}^r(\epsilon, \omega)] + \delta \hat{\Delta}(\omega) \hat{g}_b^r(\epsilon_-) - \hat{g}_b^r(\epsilon_+) \delta \hat{\Delta}(\omega), \quad (13)
 \end{aligned}$$

$$\delta \Delta(\omega) = -\frac{\mathcal{G}}{8} \int d\epsilon \text{Tr} [(-i\tau_2) \delta \hat{g}^K(\epsilon, \omega)]. \quad (17)$$

Higgs
↓

We solve these equations to obtain the AC-induced **nonequilibrium corrections** ($\delta \hat{g}^{R,A,K}$ and $\delta \Delta$).

To obtain the ac response,
we substitute the solutions ($\delta \hat{g}^{R,A,K}$, $\delta \Delta$) into

$$\delta \mathbf{J}(\omega) = -i \frac{\sigma_n}{e} \int d\epsilon \delta \mathbf{S}(\epsilon, \omega), \quad (18)$$

$$\begin{aligned}
 \delta \mathbf{S}(\epsilon, \omega) = & (i/16) \text{Tr} [i \mathbf{q}_b \\
 & \times \{ \hat{\tau}_3 \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) + \hat{\tau}_3 \delta \hat{g}^R(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) \\
 & + \hat{\tau}_3 \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^A(\epsilon, \omega) + \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) \} \\
 & + i \delta \mathbf{q}_\omega \{ \hat{\tau}_3 \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) + \hat{\tau}_3 \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) \}]. \quad (19)
 \end{aligned}$$

Instead of showing the rigorous algebraic calculations, we will look at schematic illustrations that represent the final results.

$$J \sim Agg + A_{dc}g\delta g$$

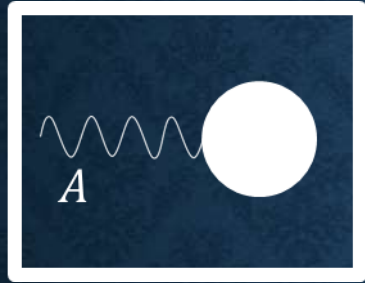


$$\delta \mathbf{J}(\omega) = -i \frac{\sigma_n}{e} \int d\epsilon \delta \mathbf{S}(\epsilon, \omega), \quad (18)$$

$$\begin{aligned} \delta \mathbf{S}(\epsilon, \omega) = & (i/16) \text{Tr} [i \mathbf{q}_b \\ & \times \{ \hat{\tau}_3 \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) + \hat{\tau}_3 \delta \hat{g}^R(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) \\ & + \hat{\tau}_3 \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^A(\epsilon, \omega) + \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) \} \\ & + i \delta \mathbf{q}_\omega \{ \hat{\tau}_3 \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) + \hat{\tau}_3 \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) \}]. \end{aligned} \quad (19)$$

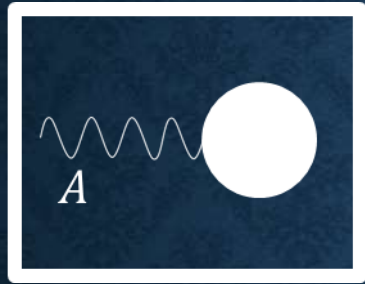
Instead of showing the rigorous algebraic calculations, we will look at schematic illustrations that represent the final results.

$$J \sim \boxed{A g g} + A_{dc} g \delta g$$



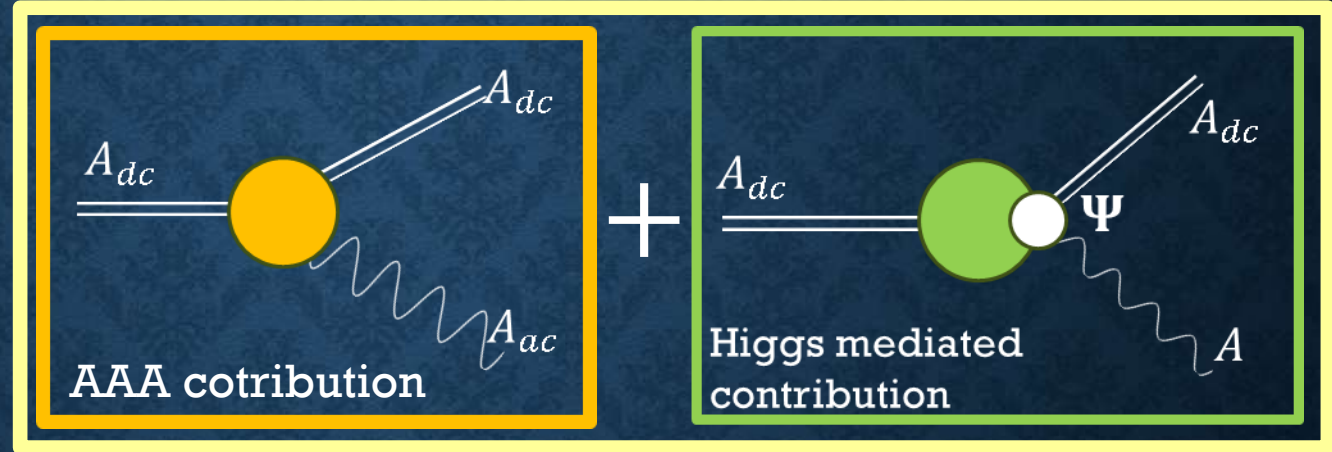
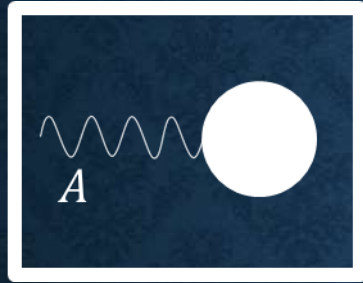
Instead of showing the rigorous algebraic calculations, we will look at schematic illustrations that represent the final results.

$$J \sim \boxed{A g g} + A_{dc} g \delta g \quad \delta g \sim \mathbf{A}_{dc} \cdot \mathbf{A} + \delta \Delta \quad \text{and} \quad \delta \Delta \sim \Psi \mathbf{A}_{dc} \cdot \mathbf{A}$$



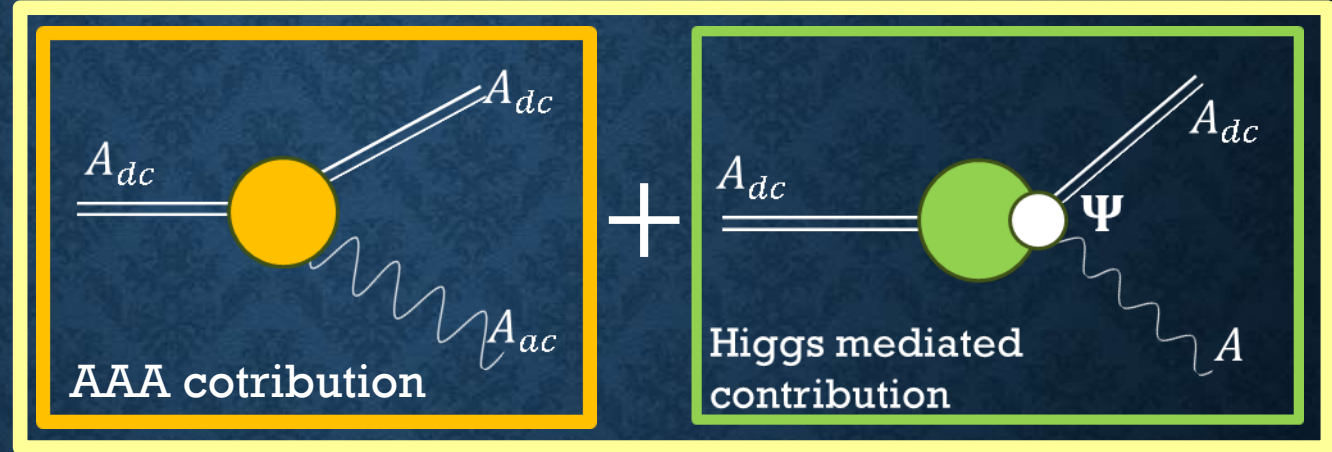
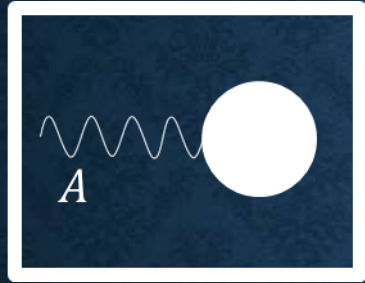
Instead of showing the rigorous algebraic calculations, we will look at schematic illustrations that represent the final results.

$$J \sim \boxed{A g g} + \boxed{A_{dc} g \delta g} \quad \delta g \sim \textcolor{brown}{A}_{dc} \cdot \textcolor{brown}{A} + \textcolor{green}{\delta\Delta} \quad \text{and} \quad \textcolor{green}{\delta\Delta} \sim \Psi \textcolor{green}{A}_{dc} \cdot \textcolor{green}{A}$$



Instead of showing the rigorous algebraic calculations, we will look at schematic illustrations that represent the final results.

$$J \sim \boxed{A g g} + \boxed{A_{dc} g \delta g} \quad \delta g \sim \mathbf{A}_{dc} \cdot \mathbf{A} + \delta\Delta \quad \text{and} \quad \delta\Delta \sim \Psi \mathbf{A}_{dc} \cdot \mathbf{A}$$



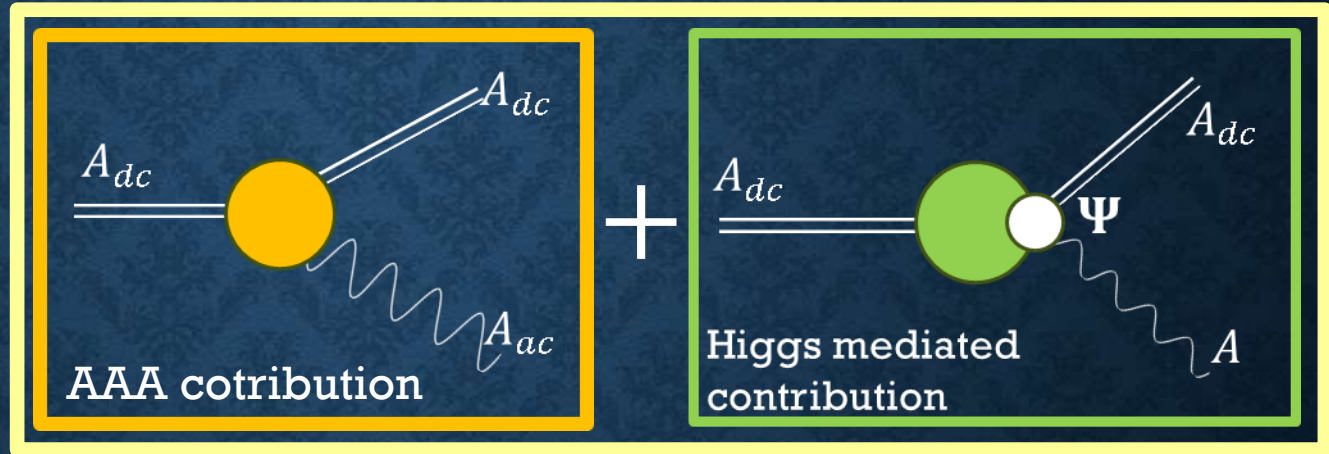
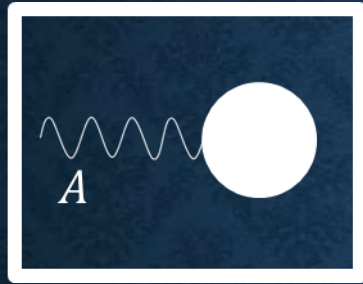
$$\propto \mathbf{A}_{dc} \cdot \mathbf{A}$$

$$\propto \mathbf{A}_{dc} \cdot \mathbf{A}$$

Nonequilibrium corrections due to the
Doppler fluctuation of flow $\propto \mathbf{A}_{dc} \cdot \mathbf{A}$

Instead of showing the rigorous algebraic calculations, we will look at schematic illustrations that represent the final results.

$$J \sim \boxed{A g g} + \boxed{A_{dc} g \delta g} \quad \delta g \sim \textcolor{brown}{A}_{dc} \cdot \textcolor{brown}{A} + \textcolor{green}{\delta\Delta} \quad \text{and} \quad \delta\Delta \sim \Psi \textcolor{green}{A}_{dc} \cdot \textcolor{green}{A}$$



Then, the complex conductivity is given by

$$\sigma = \frac{J}{E} \sim \frac{\boxed{\text{wavy } A \text{ into white circle}} + \boxed{\text{AAA cotribution}} + \boxed{\text{Higgs mediated contribution}}}{A}$$

Complex conductivity formula

T. Kubo, Phys. Rev. Applied **23**, 054091 (2025)

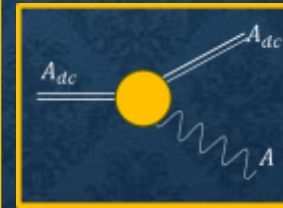
$ac \parallel dc$ case

$$\sigma = \sigma^{(0)} + \sigma^{(1)} + \sigma^{(2)}$$

$$\begin{aligned} \frac{\sigma^{(0)}}{\sigma_n} &= \int \frac{d\epsilon}{\hbar\omega} (\text{Re } G_b \text{Re } G'_b + \text{Re } F_b \text{Re } F'_b) (f_{\text{FD}} - f'_{\text{FD}}) \\ &+ i \int \frac{d\epsilon}{\hbar\omega} (\text{Re } G_b \text{Im } G'_b + \text{Re } F_b \text{Im } F'_b) (2f_{\text{FD}} - 1), \end{aligned} \quad (41)$$

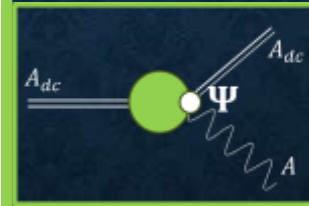


$$\begin{aligned} \frac{\sigma^{(1)}}{\sigma_n} &= \frac{8s}{\hbar\omega} \int \frac{d\epsilon}{\hbar\omega} \text{Re } F_b \text{Im } F_b \text{Re } G'_b (f_{\text{FD}} - f'_{\text{FD}}) \\ &+ i \frac{2s}{\hbar\omega} \int \frac{d\epsilon}{\hbar\omega} [2 \text{Re } F_b \text{Im } F_b \text{Im } \{G_b + G'_b\} + \{(\text{Re } F'_b)^2 \\ &- (\text{Re } F_b)^2 + (\text{Im } F_b)^2 - (\text{Im } F'_b)^2\} \text{Re } G_b] (2f_{\text{FD}} - 1), \end{aligned} \quad (42)$$



$$\propto A_{dc} \cdot A$$

$$\begin{aligned} \frac{\sigma^{(2)}}{\sigma_n} &= \frac{2s\Psi}{\hbar\omega} \int \frac{d\epsilon}{\hbar\omega} (\text{Re } F_b \text{Re } G'_b - \text{Re } G_b \text{Re } F'_b) \\ &\times (f_{\text{FD}} - f'_{\text{FD}}) + i \frac{2s\Psi}{\hbar\omega} \int \frac{d\epsilon}{\hbar\omega} \{ \text{Re } G_b \text{Im } (F_b - F'_b) \\ &+ \text{Re } F_b \text{Im } (G_b + G'_b) \} (2f_{\text{FD}} - 1). \end{aligned} \quad (43)$$



$$\propto A_{dc} \cdot A$$

AAA
contribution

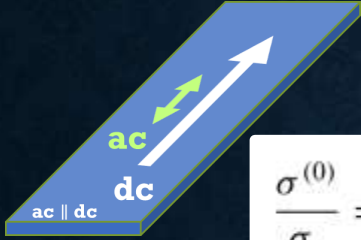
Higgs
mediated
contribution

Nonequilibrium corrections due
to the Doppler fluctuation of flow
 $\propto A_{dc} \cdot A$

Complex conductivity formula

T. Kubo, Phys. Rev. Applied **23**, 054091 (2025)

$ac \parallel dc$ case

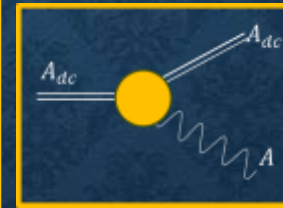


$$\sigma = \sigma^{(0)} + \sigma^{(1)} + \sigma^{(2)}$$

$$\frac{\sigma^{(0)}}{\sigma_n} = \int \frac{d\epsilon}{\hbar\omega} (\text{Re } G_b \text{Re } G'_b + \text{Re } F_b \text{Re } F'_b) (f_{\text{FD}} - f'_{\text{FD}}) + i \int \frac{d\epsilon}{\hbar\omega} (\text{Re } G_b \text{Im } G'_b + \text{Re } F_b \text{Im } F'_b) (2f_{\text{FD}} - 1), \quad (41)$$

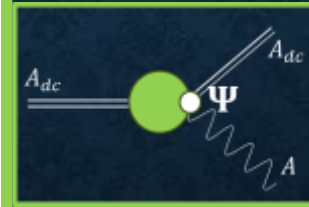


$$\frac{\sigma^{(1)}}{\sigma_n} = \frac{8s}{\hbar\omega} \int \frac{d\epsilon}{\hbar\omega} \text{Re } F_b \text{Im } F_b \text{Re } G'_b (f_{\text{FD}} - f'_{\text{FD}}) + i \frac{2s}{\hbar\omega} \int \frac{d\epsilon}{\hbar\omega} [2 \text{Re } F_b \text{Im } F_b \text{Im } \{G_b + G'_b\} + \{(\text{Re } F'_b)^2 - (\text{Re } F_b)^2 + (\text{Im } F_b)^2 - (\text{Im } F'_b)^2\} \text{Re } G_b] (2f_{\text{FD}} - 1), \quad (42)$$



$$\propto A_{dc} \cdot A$$

$$\frac{\sigma^{(2)}}{\sigma_n} = \frac{2s\Psi}{\hbar\omega} \int \frac{d\epsilon}{\hbar\omega} (\text{Re } F_b \text{Re } G'_b - \text{Re } G_b \text{Re } F'_b) \times (f_{\text{FD}} - f'_{\text{FD}}) + i \frac{2s\Psi}{\hbar\omega} \int \frac{d\epsilon}{\hbar\omega} \{ \text{Re } G_b \text{Im } (F_b - F'_b) + \text{Re } F_b \text{Im } (G_b + G'_b) \} (2f_{\text{FD}} - 1). \quad (43)$$



$$\propto A_{dc} \cdot A$$

AAA contribution

Higgs mediated contribution

$ac \perp dc$ case



$$\sigma = \sigma^{(0)}$$

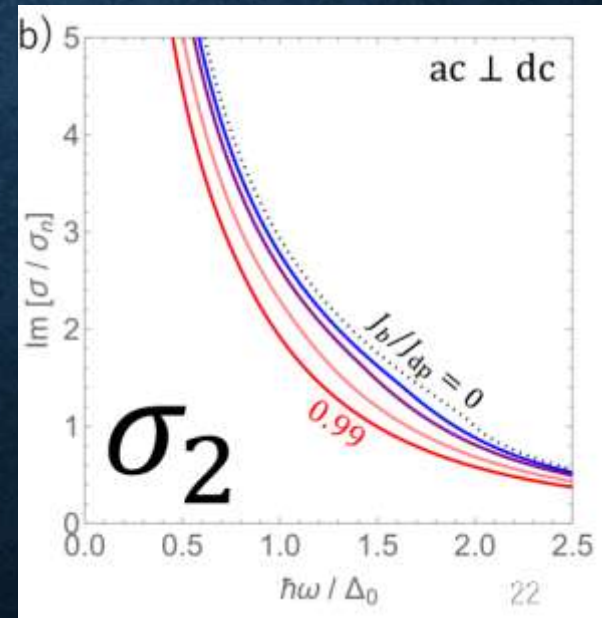
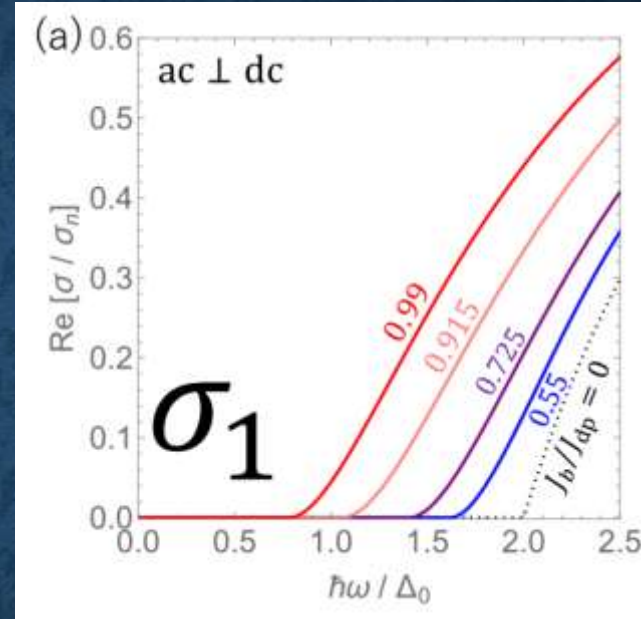
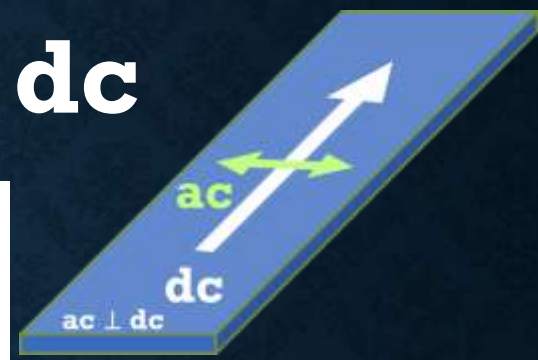


$$\frac{\sigma^{(0)}}{\sigma_n} = \int \frac{d\epsilon}{\hbar\omega} (\text{Re } G_b \text{Re } G'_b + \text{Re } F_b \text{Re } F'_b) (f_{\text{FD}} - f'_{\text{FD}}) + i \int \frac{d\epsilon}{\hbar\omega} (\text{Re } G_b \text{Im } G'_b + \text{Re } F_b \text{Im } F'_b) (2f_{\text{FD}} - 1), \quad (41)$$

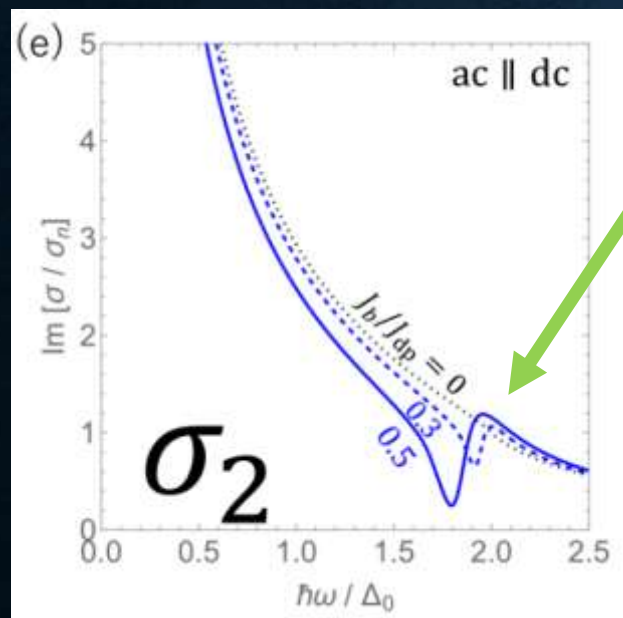
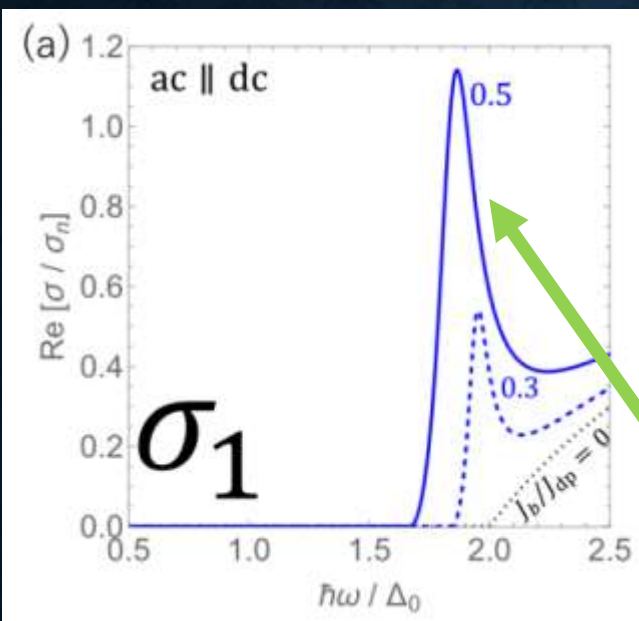
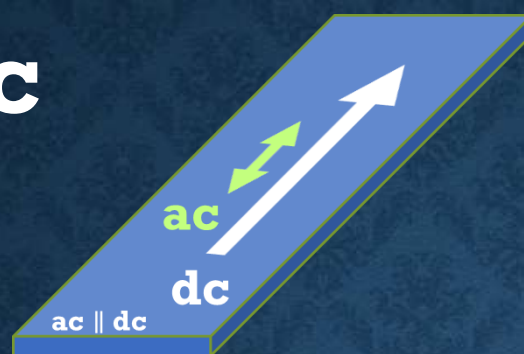
No Doppler fluctuation

Nonequilibrium corrections due to the Doppler fluctuation of flow
 $\propto A_{dc} \cdot A$

ac $\perp$ dc

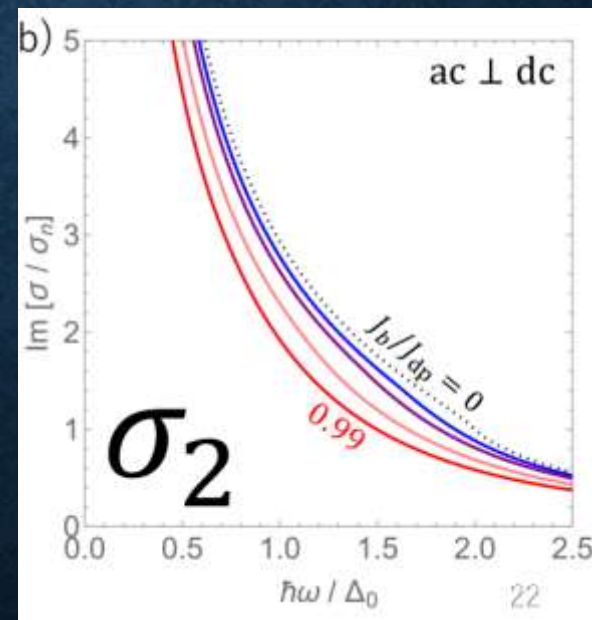
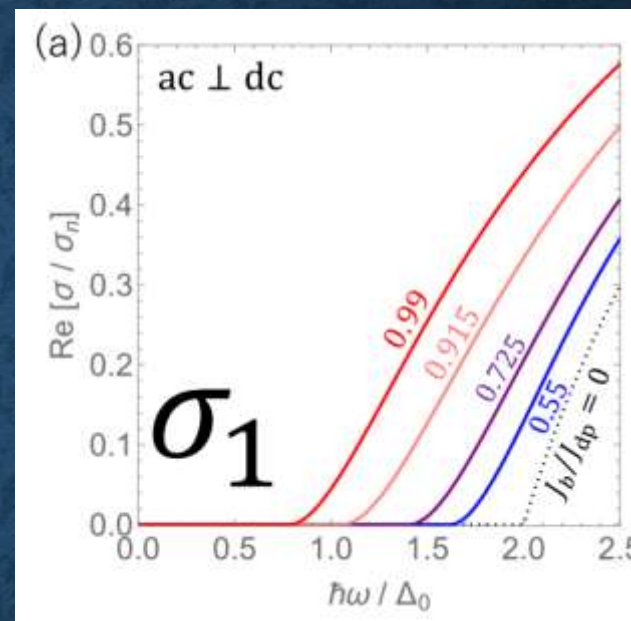


ac || dc



T. Kubo,
Phys. Rev. Applied **23**, 054091 (2025)

ac ⊥ dc



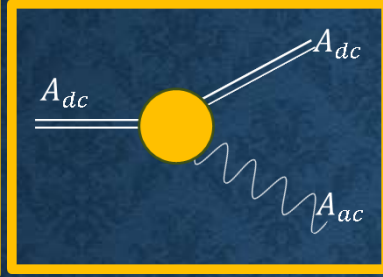
T. Kubo,
Phys. Rev. Applied **23**, 054091 (2025)

The contributions from the nonequilibrium corrections are significant even at low frequencies

$$\sigma = \sigma^{(0)} + \sigma^{(1)} + \sigma^{(2)}$$

Direct AAA photon action

$$\begin{aligned} \frac{\sigma^{(1)}}{\sigma_n} = & \frac{8s}{\hbar\omega} \int \frac{d\epsilon}{\hbar\omega} \text{Re } F_b \text{Im } F_b \text{Re } G'_b (f_{\text{FD}} - f'_{\text{FD}}) \\ & + i \frac{2s}{\hbar\omega} \int \frac{d\epsilon}{\hbar\omega} [2 \text{Re } F_b \text{Im } F_b \text{Im} \{G_b + G'_b\} + \{(\text{Re } F'_b)^2 \\ & - (\text{Re } F_b)^2 + (\text{Im } F_b)^2 - (\text{Im } F'_b)^2\} \text{Re } G_b] (2f_{\text{FD}} - 1), \end{aligned} \quad (42)$$



Significantly contributes to
 $Re[\sigma]$, $Im[\sigma]$
 even at low frequencies ($\hbar\omega \ll \Delta$)

Higgs mediated contribution

$$\begin{aligned} \frac{\sigma^{(2)}}{\sigma_n} = & \frac{2s\Psi}{\hbar\omega} \int \frac{d\epsilon}{\hbar\omega} (\text{Re } F_b \text{Re } G'_b - \text{Re } G_b \text{Re } F'_b) \\ & \times (f_{\text{FD}} - f'_{\text{FD}}) + i \frac{2s\Psi}{\hbar\omega} \int \frac{d\epsilon}{\hbar\omega} \{ \text{Re } G_b \text{Im}(F_b - F'_b) \\ & + \text{Re } F_b \text{Im}(G_b + G'_b) \} (2f_{\text{FD}} - 1). \end{aligned} \quad (43)$$



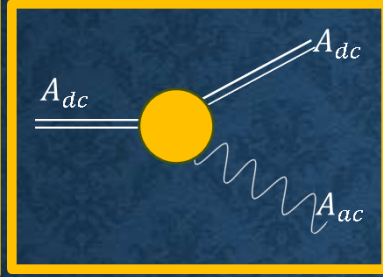
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Significantly contributes to

$$\boxed{Re[\sigma]}, \quad Im[\sigma]$$

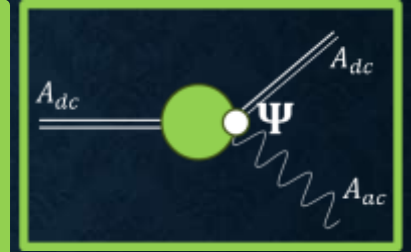
even at low frequencies ($\hbar\omega \ll \Delta$)



Surface resistance
or
quality factor

Higgs mediated contribution

$$\begin{aligned} \frac{\sigma^{(2)}}{\sigma_n} = & \frac{2s\Psi}{\hbar\omega} \int \frac{d\epsilon}{\hbar\omega} (\text{Re } F_b \text{Re } G'_b - \text{Re } G_b \text{Re } F'_b) \\ & \times (f_{\text{FD}} - f'_{\text{FD}}) + i \frac{2s\Psi}{\hbar\omega} \int \frac{d\epsilon}{\hbar\omega} \{ \text{Re } G_b \text{Im}(F_b - F'_b) \\ & + \text{Re } F_b \text{Im}(G_b + G'_b) \} (2f_{\text{FD}} - 1). \end{aligned} \quad (43)$$



Significantly contributes to

$$Im[\sigma]$$

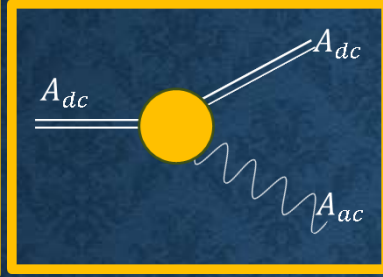
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Significantly contributes to

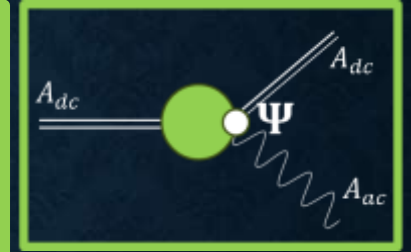
$\text{Re}[\sigma]$, $\text{Im}[\sigma]$

even at low frequencies ($\hbar\omega \ll \Delta$)

Surface resistance
or
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Higgs mediated contribution

$$\begin{aligned} \frac{\sigma^{(2)}}{\sigma_n} = & \frac{2s\Psi}{\hbar\omega} \int \frac{d\epsilon}{\hbar\omega} (\text{Re } F_b \text{Re } G'_b - \text{Re } G_b \text{Re } F'_b) \\ & \times (f_{\text{FD}} - f'_{\text{FD}}) + i \frac{2s\Psi}{\hbar\omega} \int \frac{d\epsilon}{\hbar\omega} \{ \text{Re } G_b \text{Im}(F_b - F'_b) \\ & + \text{Re } F_b \text{Im}(G_b + G'_b) \} (2f_{\text{FD}} - 1). \end{aligned} \quad (43)$$



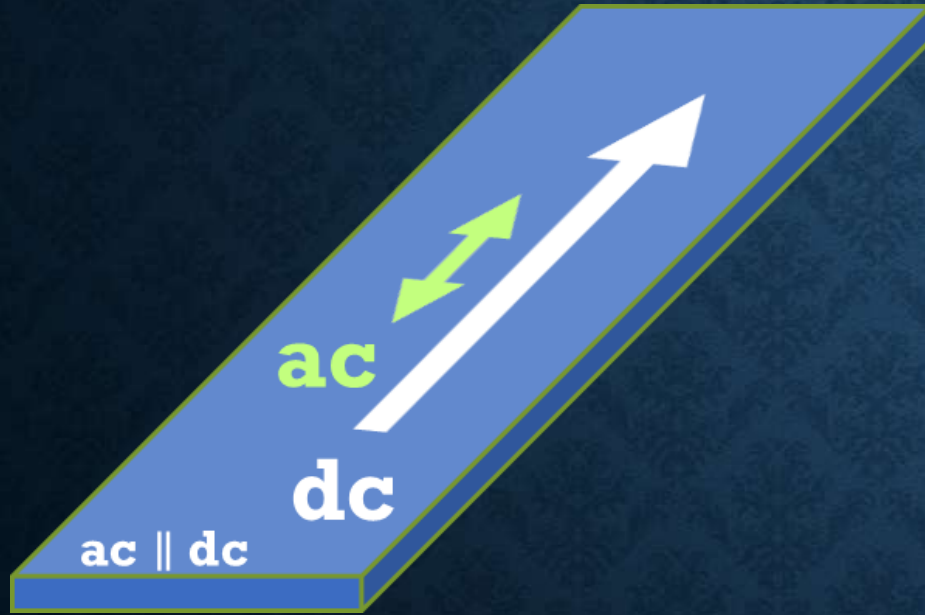
Significantly contributes to

$\text{Im}[\sigma]$

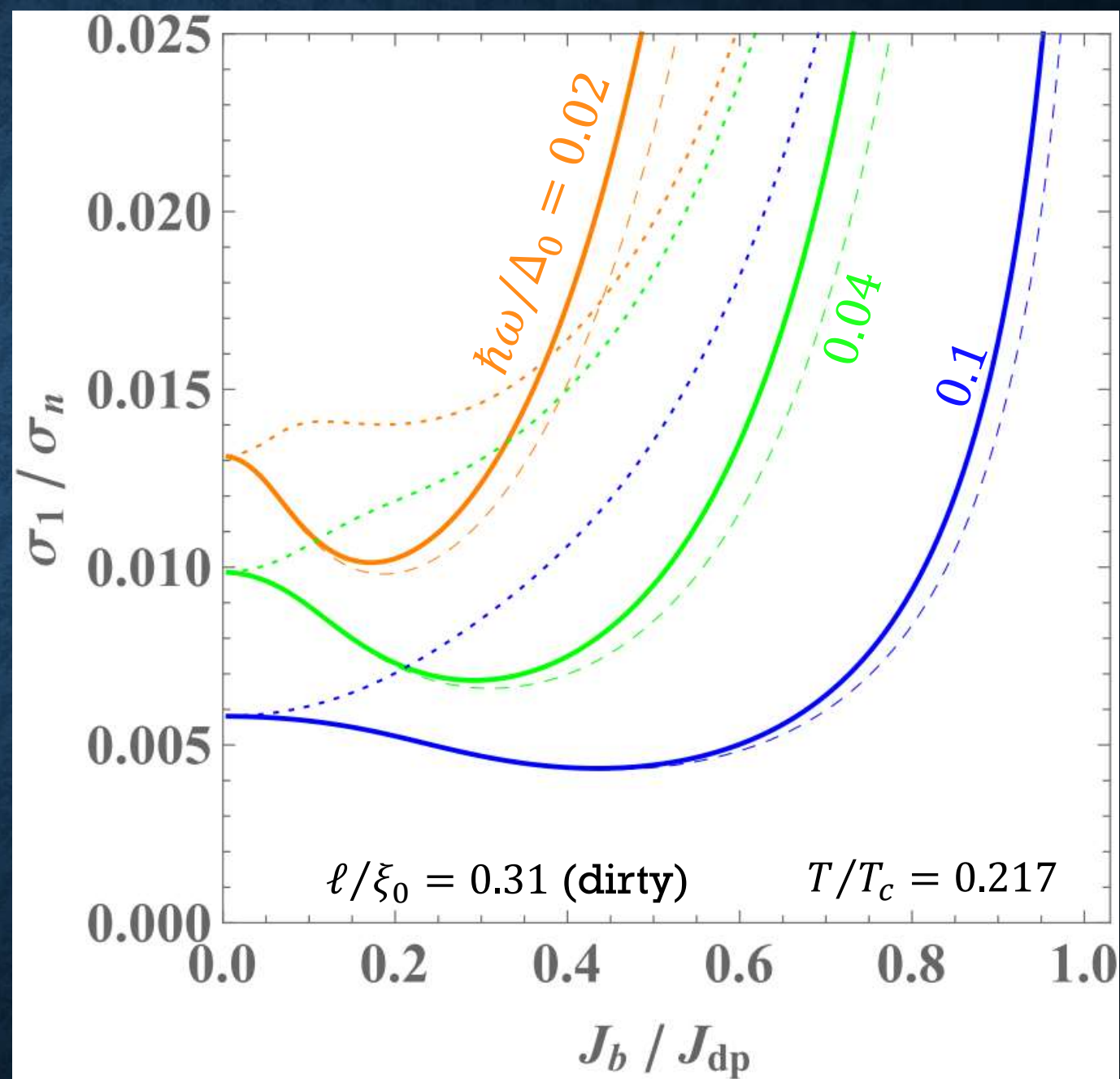
even at low frequencies ($\hbar\omega \ll \Delta$)

Kinetic inductance

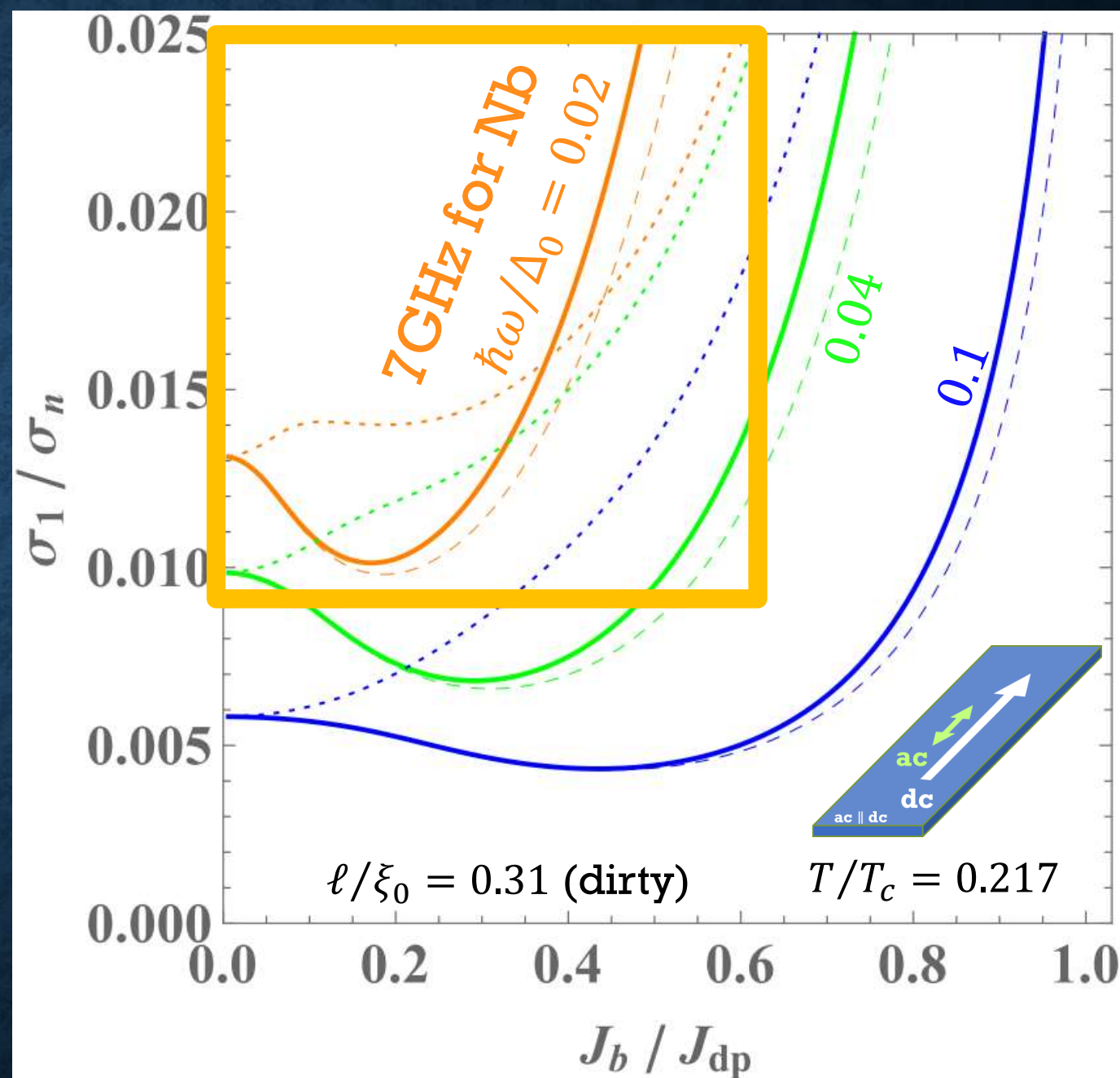
Current dependent $\sigma_1(J_b)$ curve in ac+dc system



dc: arbitrary strength
ac: perturbation



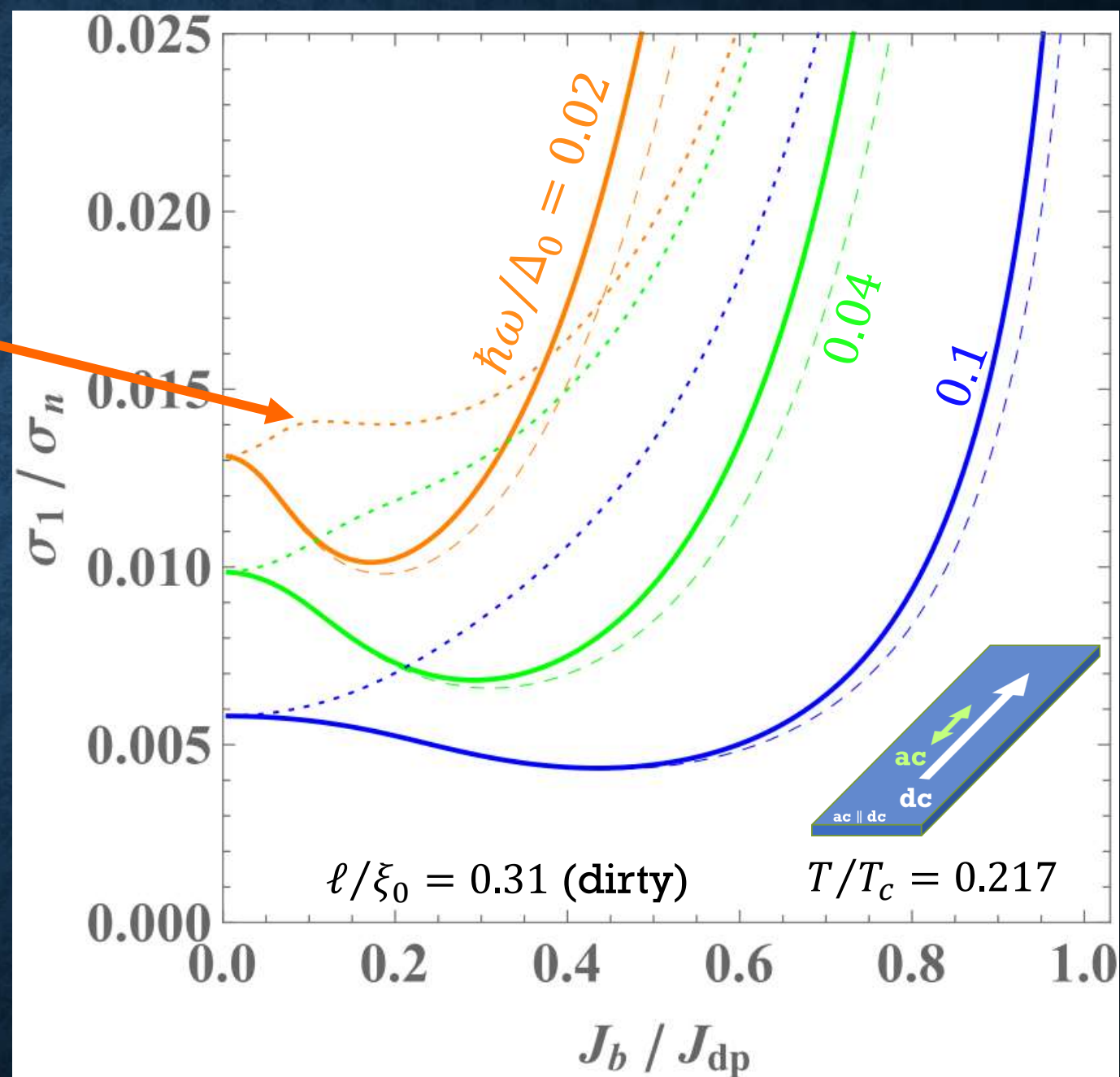
Let's see the orange curves ($\hbar\omega/\Delta_0 = 0.02$), which correspond to 7GHz for Nb.



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Dotted curve:
 $\text{Re}[\sigma^{(0)}]$



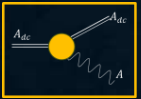
Let's see the orange curves ($\hbar\omega/\Delta_0 = 0.02$), which correspond to 7GHz for Nb.



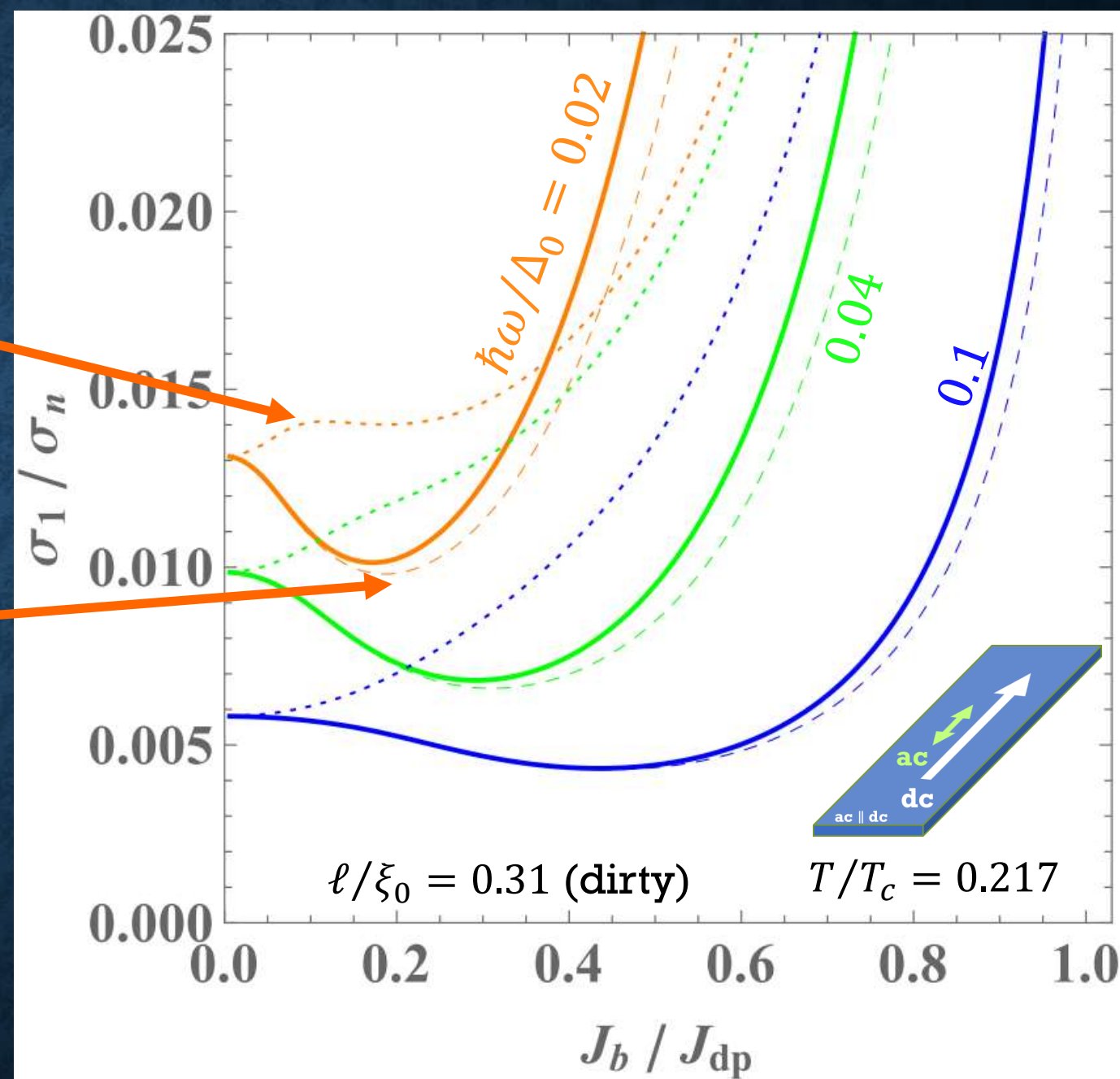
Dotted curve:
 $\text{Re}[\sigma^{(0)}]$



Dashed curve:
 $\text{Re}[\sigma^{(0)} + \sigma^{(1)}]$



Significant contribution from the direct AAA action term $\sigma^{(1)}$



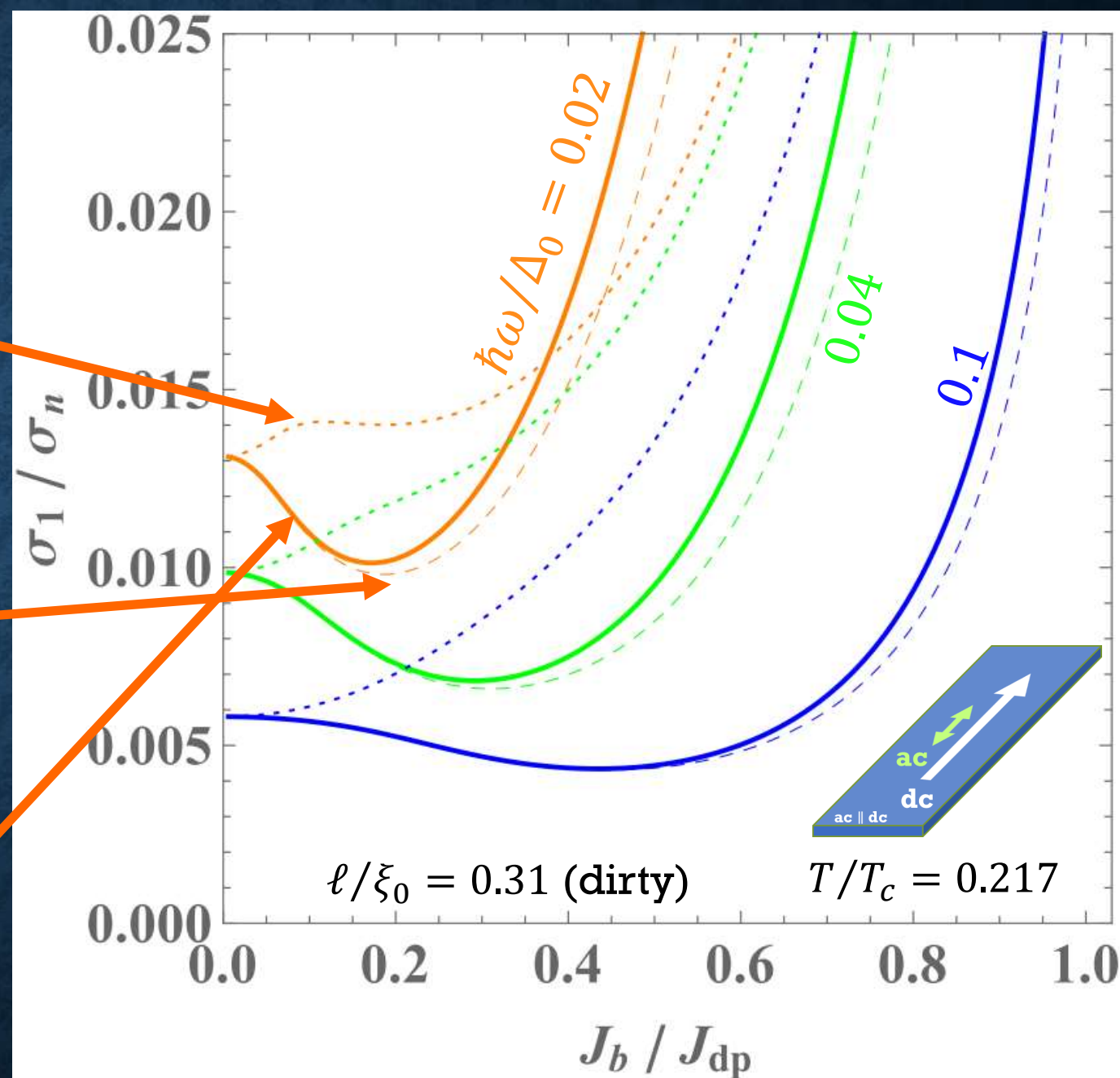
Let's see the orange curves ($\hbar\omega/\Delta_0 = 0.02$), which correspond to 7GHz for Nb.

Dotted curve:
 $\text{Re}[\sigma^{(0)}]$

Dashed curve:
 $\text{Re}[\sigma^{(0)} + \sigma^{(1)}]$

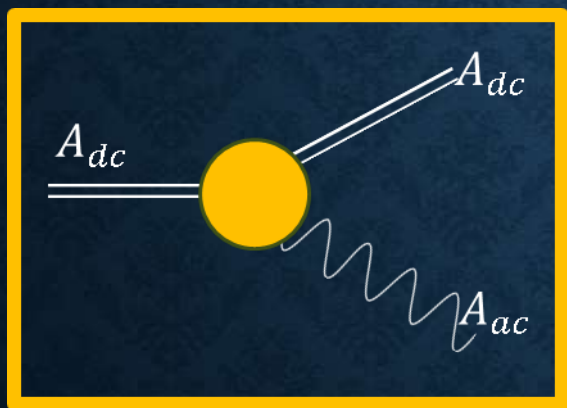
Solid curve:
 $\text{Re}[\sigma^{(0)} + \sigma^{(1)} + \sigma^{(2)}]$

Significant contribution from the direct AAA action term $\sigma^{(1)}$



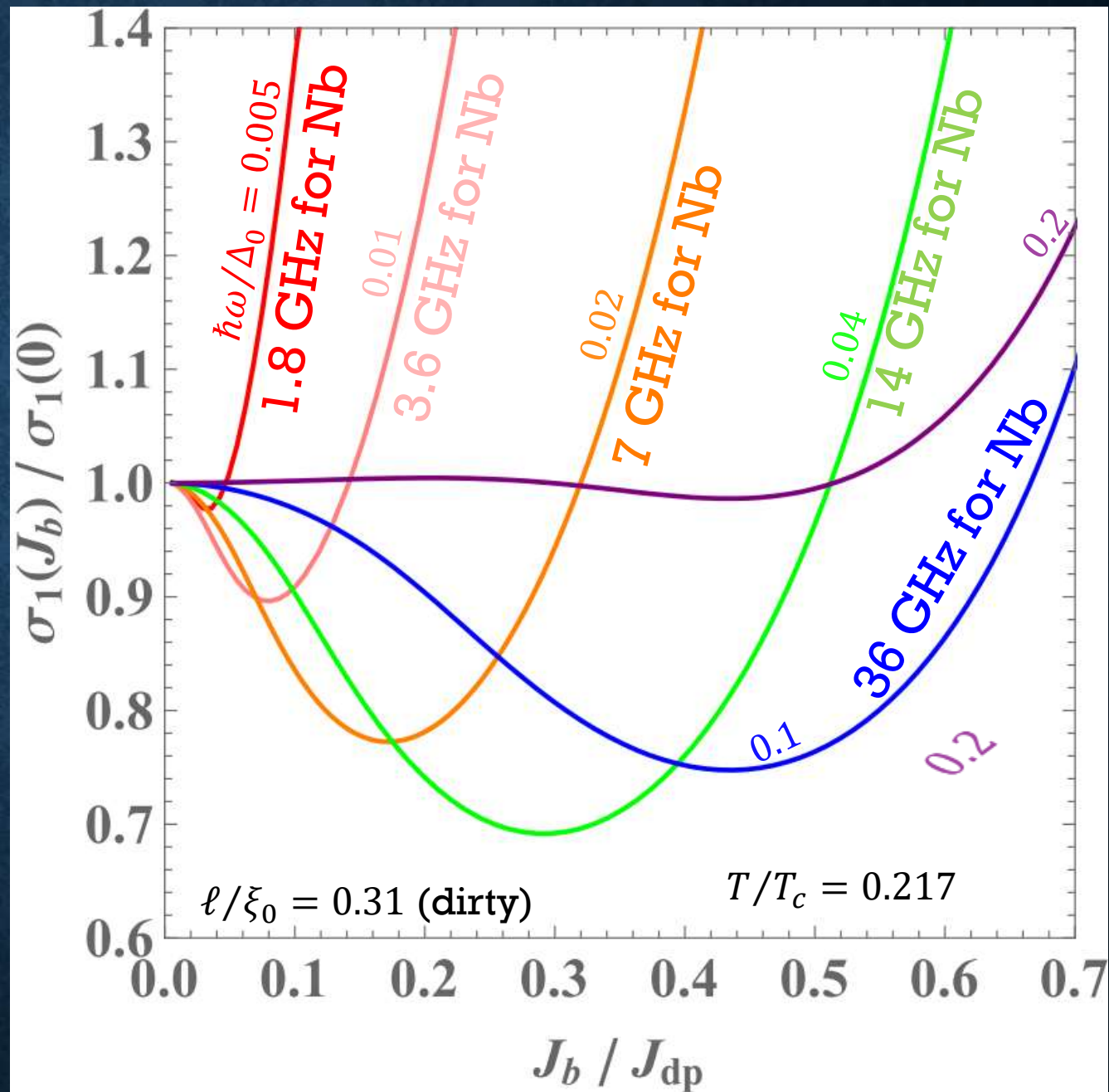
The current-induced reduction of $\sigma_1(J_b)$ is pronounced as the frequency increases

The key player is the direct AAA photon action



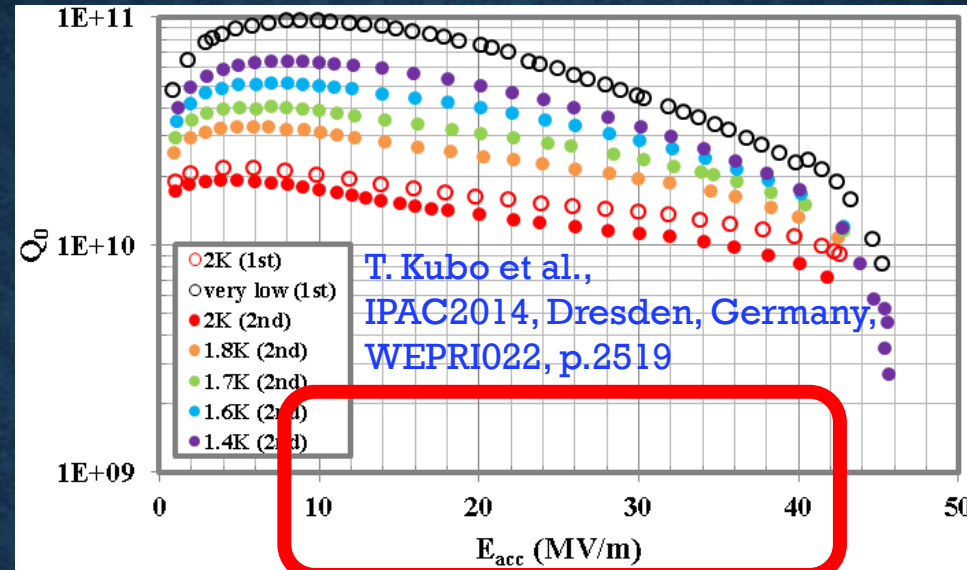
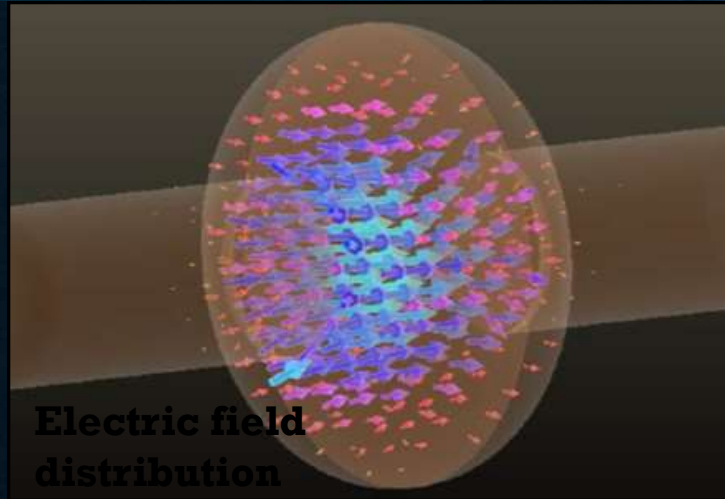
$$\sigma = \sigma^{(0)} + \boxed{\sigma^{(1)}} + \sigma^{(2)}$$

Big clue to understand the pronounced anti Q-slope with increasing frequency

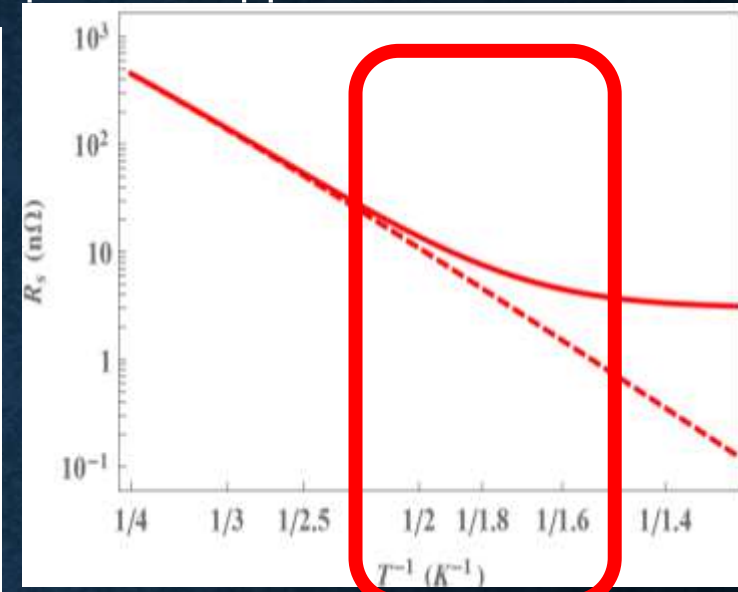


The biggest differences between cavities for accelerators and those for quantum applications

Accelerators



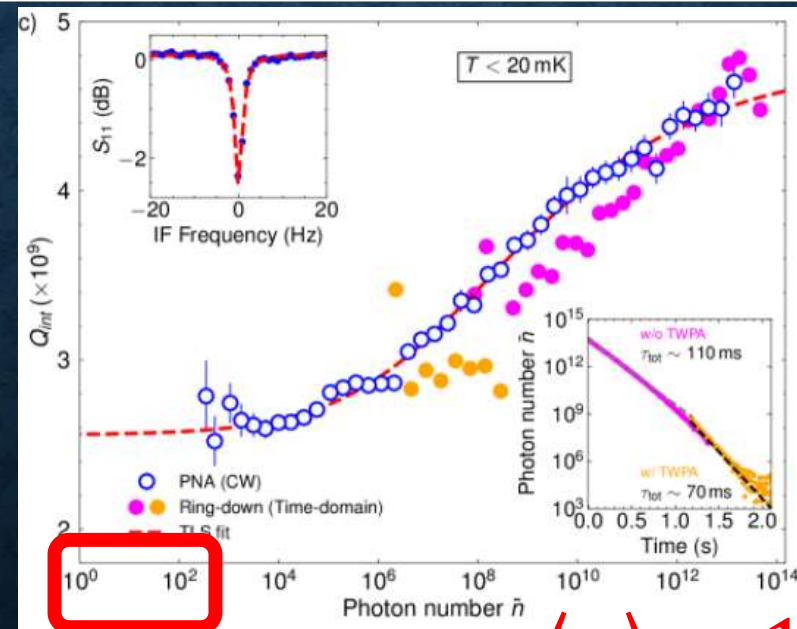
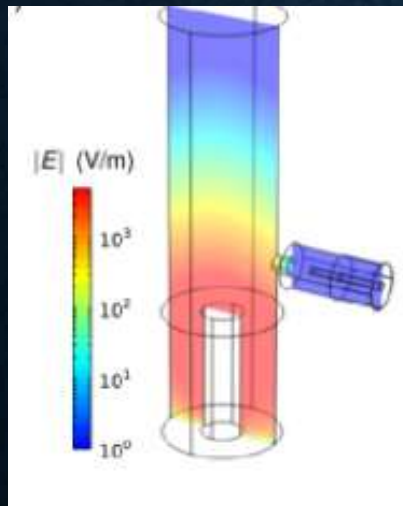
photon number $\langle n \rangle \sim 10^{25}$



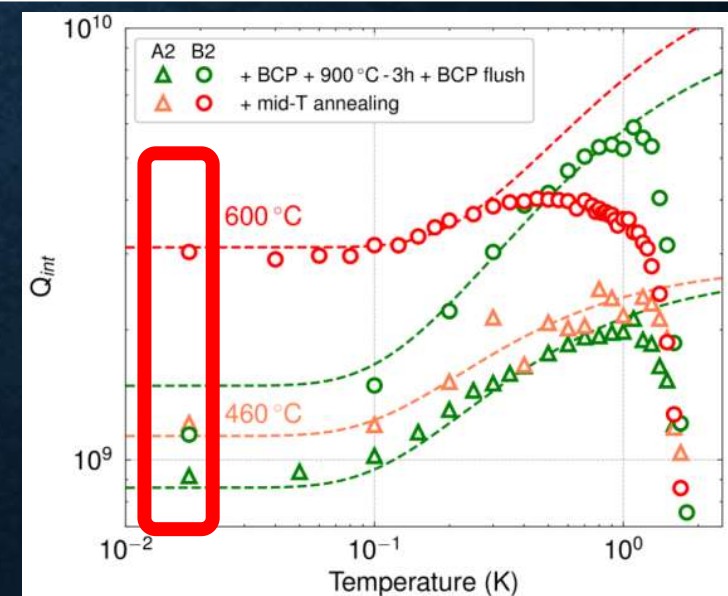
$T \simeq 2$ K

Quantum computing

T. Takenaka, T. Kubo et al.,
arXiv:2510.01819.



photon number $\langle n \rangle \sim 1$



$T \simeq 20$ mK